

Lecture 39

Approximability and Inapproximability of OptTSP

Inapproximability of NP-Hard Problems

Inapproximability of NP-Hard Problems

Polytime approximation algorithms seems like a good solution for NP-Hard problems.

Inapproximability of NP-Hard Problems

Polytime approximation algorithms seems like a good solution for NP-Hard problems.

However,

Inapproximability of NP-Hard Problems

Polytime approximation algorithms seems like a good solution for NP-Hard problems.

However,

- Not all the NP-hard problems have a polytime algorithm with good approximation.

Inapproximability of NP-Hard Problems

Polytime approximation algorithms seems like a good solution for NP-Hard problems.

However,

- Not all the NP-hard problems have a polytime algorithm with good approximation. E.g.,

Inapproximability of NP-Hard Problems

Polytime approximation algorithms seems like a good solution for NP-Hard problems.

However,

- Not all the NP-hard problems have a polytime algorithm with good approximation. E.g.,
Fact: If $P \neq NP$, *OptVertexCover* cannot have a polytime 1.3606-approximation algorithm.

Inapproximability of NP-Hard Problems

Polytime approximation algorithms seems like a good solution for NP-Hard problems.

However,

- Not all the NP-hard problems have a polytime algorithm with good approximation. E.g.,
Fact: If $P \neq NP$, *OptVertexCover* cannot have a polytime 1.3606-approximation algorithm.
- Some NP-hard problems do not have a polytime algorithm with any approximation.

Inapproximability of NP-Hard Problems

Polytime approximation algorithms seems like a good solution for NP-Hard problems.

However,

- Not all the NP-hard problems have a polytime algorithm with good approximation. E.g.,
Fact: If $P \neq NP$, *OptVertexCover* cannot have a polytime 1.3606-approximation algorithm.
- Some NP-hard problems do not have a polytime algorithm with any approximation. E.g.,

Inapproximability of NP-Hard Problems

Polytime approximation algorithms seems like a good solution for NP-Hard problems.

However,

- Not all the NP-hard problems have a polytime algorithm with good approximation. E.g.,
Fact: If $P \neq NP$, *OptVertexCover* cannot have a polytime 1.3606-approximation algorithm.
- Some NP-hard problems do not have a polytime algorithm with any approximation. E.g.,
Fact: If $P \neq NP$, *OptTSP* cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

OptTSP

OptTSP

OptTSP:

OptTSP

OptTSP:

Input: A complete undirected graph G , cost function $c : V \times V \rightarrow \mathbb{N}$.

OptTSP

OptTSP:

Input: A complete undirected graph G , cost function $c : V \times V \rightarrow \mathbb{N}$.

Output: Find the minimum cost TSP tour in G .

OptTSP

OptTSP:

Input: A complete undirected graph G , cost function $c : V \times V \rightarrow \mathbb{N}$.

Output: Find the minimum cost TSP tour in G .

TSP:

OptTSP

OptTSP:

Input: A complete undirected graph G , cost function $c : V \times V \rightarrow \mathbb{N}$.

Output: Find the minimum cost TSP tour in G .

TSP:

Input: A complete undirected graph G , cost function $c : V \times V \rightarrow \mathbb{N}$, and $k \in \mathbb{N}$.

OptTSP

OptTSP:

Input: A complete undirected graph G , cost function $c : V \times V \rightarrow \mathbb{N}$.

Output: Find the minimum cost TSP tour in G .

TSP:

Input: A complete undirected graph G , cost function $c : V \times V \rightarrow \mathbb{N}$, and $k \in \mathbb{N}$.

Output: Find whether G has a TSP tour with cost at most k .

Inapproximability of $OptTSP$

Inapproximability of $OptTSP$

Theorem: If $P \neq NP$, $OptTSP$ cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Inapproximability of $OptTSP$

Theorem: If $P \neq NP$, $OptTSP$ cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof:

Inapproximability of $OptTSP$

Theorem: If $P \neq NP$, $OptTSP$ cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose $OptTSP$ has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Inapproximability of $OptTSP$

Theorem: If $P \neq NP$, $OptTSP$ cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose $OptTSP$ has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

Inapproximability of $OptTSP$

Theorem: If $P \neq NP$, $OptTSP$ cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose $OptTSP$ has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

Inapproximability of $OptTSP$

Theorem: If $P \neq NP$, $OptTSP$ cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose $OptTSP$ has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

We will show now that *Hamcycle* can be solved in polytime using A .

Inapproximability of $OptTSP$

Theorem: If $P \neq NP$, $OptTSP$ cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose $OptTSP$ has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

We will show now that *Hamcycle* can be solved in polytime using A .

Idea: "Reduce" *Hamcycle* to *TSP* (G to $\langle G', c \rangle$) so that:

Inapproximability of $OptTSP$

Theorem: If $P \neq NP$, $OptTSP$ cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose $OptTSP$ has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

We will show now that *Hamcycle* can be solved in polytime using A .

Idea: "Reduce" *Hamcycle* to *TSP* (G to $\langle G', c \rangle$) so that:

- If G has a hamiltonian cycle, G' has a TSP tour of **very small** cost.

Inapproximability of $OptTSP$

Theorem: If $P \neq NP$, $OptTSP$ cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose $OptTSP$ has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

We will show now that *Hamcycle* can be solved in polytime using A .

Idea: "Reduce" *Hamcycle* to *TSP* (G to $\langle G', c \rangle$) so that:

- If G has a hamiltonian cycle, G' has a TSP tour of **very small** cost.
- If G doesn't have a hamiltonian cycle, every TSP tour in G' has a **very large** cost.

Inapproximability of $OptTSP$

Theorem: If $P \neq NP$, $OptTSP$ cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose $OptTSP$ has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

We will show now that *Hamcycle* can be solved in polytime using A .

Idea: "Reduce" *Hamcycle* to *TSP* (G to $\langle G', c \rangle$) so that:

- If G has a hamiltonian cycle, G' has a TSP tour of **very small** cost.
- If G doesn't have a hamiltonian cycle, every TSP tour in G' has a **very large** cost.

The **cost difference** should be so large

Inapproximability of $OptTSP$

Theorem: If $P \neq NP$, $OptTSP$ cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose $OptTSP$ has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

We will show now that *Hamcycle* can be solved in polytime using A .

Idea: "Reduce" *Hamcycle* to *TSP* (G to $\langle G', c \rangle$) so that:

- If G has a hamiltonian cycle, G' has a TSP tour of **very small** cost.
- If G doesn't have a hamiltonian cycle, every TSP tour in G' has a **very large** cost.

The **cost difference** should be so large so that by running A on G' , we should be

Inapproximability of $OptTSP$

Theorem: If $P \neq NP$, $OptTSP$ cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose $OptTSP$ has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

We will show now that *Hamcycle* can be solved in polytime using A .

Idea: "Reduce" *Hamcycle* to *TSP* (G to $\langle G', c \rangle$) so that:

- If G has a hamiltonian cycle, G' has a TSP tour of **very small** cost.
- If G doesn't have a hamiltonian cycle, every TSP tour in G' has a **very large** cost.

The **cost difference** should be so large so that by running A on G' , we should be able to tell whether G has a hamiltonian cycle.

Inapproximability of $OptTSP$

Theorem: If $P \neq NP$, $OptTSP$ cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose $OptTSP$ has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

We will show now that *Hamcycle* can be solved in polytime using A .

Idea: “Reduce” *Hamcycle* to *TSP* (G to $\langle G', c \rangle$) so that:

- If G has a hamiltonian cycle, G' has a TSP tour of **very small** cost.
- If G doesn't have a hamiltonian cycle, every TSP tour in G' has a **very large** cost.

The **cost difference** should be so large so that by running A on G' , we should be able to tell whether G has a hamiltonian cycle.

Inapproximability of $OptTSP$

Theorem: If $P \neq NP$, $OptTSP$ cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose $OptTSP$ has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

We will show now that *Hamcycle* can be solved in polytime using A .

Inapproximability of $OptTSP$

Theorem: If $P \neq NP$, $OptTSP$ cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose $OptTSP$ has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

We will show now that *Hamcycle* can be solved in polytime using A .

Consider the following “reduction” from *Hamcycle* to *TSP*:

Inapproximability of $OptTSP$

Theorem: If $P \neq NP$, $OptTSP$ cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose $OptTSP$ has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

We will show now that *Hamcycle* can be solved in polytime using A .

Consider the following “reduction” from *Hamcycle* to *TSP*:

$\langle G \rangle \rightarrow \langle G', c \rangle$:

Inapproximability of *OptTSP*

Theorem: If $P \neq NP$, *OptTSP* cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose *OptTSP* has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

We will show now that *Hamcycle* can be solved in polytime using A .

Consider the following “reduction” from *Hamcycle* to *TSP*:

$\langle G \rangle \rightarrow \langle G', c \rangle$:

- G' is a complete graph on vertex set V of G .

Inapproximability of *OptTSP*

Theorem: If $P \neq NP$, *OptTSP* cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose *OptTSP* has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

We will show now that *Hamcycle* can be solved in polytime using A .

Consider the following “reduction” from *Hamcycle* to *TSP*:

$\langle G \rangle \rightarrow \langle G', c \rangle$:

- G' is a complete graph on vertex set V of G .
- $c(\{u, v\}) =$

Inapproximability of *OptTSP*

Theorem: If $P \neq NP$, *OptTSP* cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose *OptTSP* has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

We will show now that *Hamcycle* can be solved in polytime using A .

Consider the following “reduction” from *Hamcycle* to *TSP*:

$\langle G \rangle \rightarrow \langle G', c \rangle$:

- G' is a complete graph on vertex set V of G .

- $c(\{u, v\}) = \begin{cases} 1 & \text{if } (u, v) \in E \\ 0 & \text{otherwise} \end{cases}$

Inapproximability of *OptTSP*

Theorem: If $P \neq NP$, *OptTSP* cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose *OptTSP* has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

We will show now that *Hamcycle* can be solved in polytime using A .

Consider the following “reduction” from *Hamcycle* to *TSP*:

$\langle G \rangle \rightarrow \langle G', c \rangle$:

- G' is a complete graph on vertex set V of G .

- $c(\{u, v\}) = \begin{cases} 1, & \text{if } \{u, v\} \in E(G) \end{cases}$

Inapproximability of $OptTSP$

Theorem: If $P \neq NP$, $OptTSP$ cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose $OptTSP$ has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

We will show now that *Hamcycle* can be solved in polytime using A .

Consider the following “reduction” from *Hamcycle* to *TSP*:

$\langle G \rangle \rightarrow \langle G', c \rangle$:

- G' is a complete graph on vertex set V of G .

- $c(\{u, v\}) = \begin{cases} 1, & \text{if } \{u, v\} \in E(G) \\ \rho |V| + 1, & \text{if } \{u, v\} \notin E(G) \end{cases}$

Inapproximability of *OptTSP*

Theorem: If $P \neq NP$, *OptTSP* cannot have a polytime ρ -approximation algorithm for any $\rho \geq 1$.

Proof: Suppose *OptTSP* has a polytime ρ -approximation algorithm A for some $\rho \geq 1$.

Hence, on any input $\langle G', c \rangle$, if cost of a minimum cost TSP tour is C , then

$$C \leq A(\langle G', c \rangle) \leq \rho \cdot C$$

We will show now that *Hamcycle* can be solved in polytime using A .

Consider the following “reduction” from *Hamcycle* to *TSP*:

$\langle G \rangle \rightarrow \langle G', c \rangle$:

- G' is a complete graph on vertex set V of G .

- $c(\{u, v\}) = \begin{cases} 1, & \text{if } \{u, v\} \in E(G) \\ \rho |V| + 1, & \text{if } \{u, v\} \notin E(G) \end{cases}$

Inapproximability of $OptTSP$

Proof: $\langle G \rangle \rightarrow \langle G', c \rangle$:

- G' is a complete graph on vertex set V of G .
- $c(\{u, v\}) = \begin{cases} 1, & \text{if } \{u, v\} \in E(G) \\ \rho |V| + 1, & \text{if } \{u, v\} \notin E(G) \end{cases}$

Inapproximability of $OptTSP$

Proof: $\langle G \rangle \rightarrow \langle G', c \rangle$:

- G' is a complete graph on vertex set V of G .
- $c(\{u, v\}) = \begin{cases} 1, & \text{if } \{u, v\} \in E(G) \\ \rho |V| + 1, & \text{if } \{u, v\} \notin E(G) \end{cases}$

If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

Inapproximability of $OptTSP$

Proof: $\langle G \rangle \rightarrow \langle G', c \rangle$:

- G' is a complete graph on vertex set V of G .
- $c(\{u, v\}) = \begin{cases} 1, & \text{if } \{u, v\} \in E(G) \\ \rho |V| + 1, & \text{if } \{u, v\} \notin E(G) \end{cases}$

If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost at least

Inapproximability of $OptTSP$

Proof: $\langle G \rangle \rightarrow \langle G', c \rangle$:

- G' is a complete graph on vertex set V of G .
- $c(\{u, v\}) = \begin{cases} 1, & \text{if } \{u, v\} \in E(G) \\ \rho |V| + 1, & \text{if } \{u, v\} \notin E(G) \end{cases}$

If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost at least

$$(\rho |V| + 1) + (|V| - 1)$$

Inapproximability of $OptTSP$

Proof: $\langle G \rangle \rightarrow \langle G', c \rangle$:

- G' is a complete graph on vertex set V of G .
- $c(\{u, v\}) = \begin{cases} 1, & \text{if } \{u, v\} \in E(G) \\ \rho |V| + 1, & \text{if } \{u, v\} \notin E(G) \end{cases}$

If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost at least

$$\underbrace{(\rho |V| + 1) + (|V| - 1)}$$


Because any TSP tour in G' contains at least one edge which is not in G .

Inapproximability of $OptTSP$

Proof: $\langle G \rangle \rightarrow \langle G', c \rangle$:

- G' is a complete graph on vertex set V of G .
- $c(\{u, v\}) = \begin{cases} 1, & \text{if } \{u, v\} \in E(G) \\ \rho |V| + 1, & \text{if } \{u, v\} \notin E(G) \end{cases}$

If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost at least

$$\underbrace{(\rho |V| + 1) + (|V| - 1)}_{\nearrow} > \rho |V|$$

Because any TSP tour in G' contains at least one edge which is not in G .

Inapproximability of $OptTSP$

Proof: $\langle G \rangle \rightarrow \langle G', c \rangle$:

- G' is a complete graph on vertex set V of G .
- $c(\{u, v\}) = \begin{cases} 1, & \text{if } \{u, v\} \in E(G) \\ \rho |V| + 1, & \text{if } \{u, v\} \notin E(G) \end{cases}$

If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost at least

$$\underbrace{(\rho |V| + 1) + (|V| - 1)}_{\nearrow} > \rho |V|$$

Because any TSP tour in G' contains at least one edge which is not in G .

Inapproximability of $OptTSP$

Proof: $\langle G \rangle \rightarrow \langle G', c \rangle$:

- G' is a complete graph on vertex set V of G .
- $c(\{u, v\}) = \begin{cases} 1, & \text{if } \{u, v\} \in E(G) \\ \rho |V| + 1, & \text{if } \{u, v\} \notin E(G) \end{cases}$

If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Inapproximability of $OptTSP$

Proof: $\langle G \rangle \rightarrow \langle G', c \rangle$:

- G' is a complete graph on vertex set V of G .
- $c(\{u, v\}) = \begin{cases} 1, & \text{if } \{u, v\} \in E(G) \\ \rho |V| + 1, & \text{if } \{u, v\} \notin E(G) \end{cases}$

If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Inapproximability of $OptTSP$

Proof: If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Inapproximability of $OptTSP$

Proof: If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Recall that A is a polytime ρ -approximation algorithm for $OptTSP$.

Inapproximability of $OptTSP$

Proof: If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Recall that A is a polytime ρ -approximation algorithm for $OptTSP$.

Polytime algorithm for *Hamcycle* will work in the following way:

Inapproximability of $OptTSP$

Proof: If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Recall that A is a polytime ρ -approximation algorithm for $OptTSP$.

Polytime algorithm for *Hamcycle* will work in the following way:

G

Inapproximability of $OptTSP$

Proof: If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Recall that A is a polytime ρ -approximation algorithm for $OptTSP$.

Polytime algorithm for *Hamcycle* will work in the following way:

$$G \longrightarrow \langle G', c \rangle$$

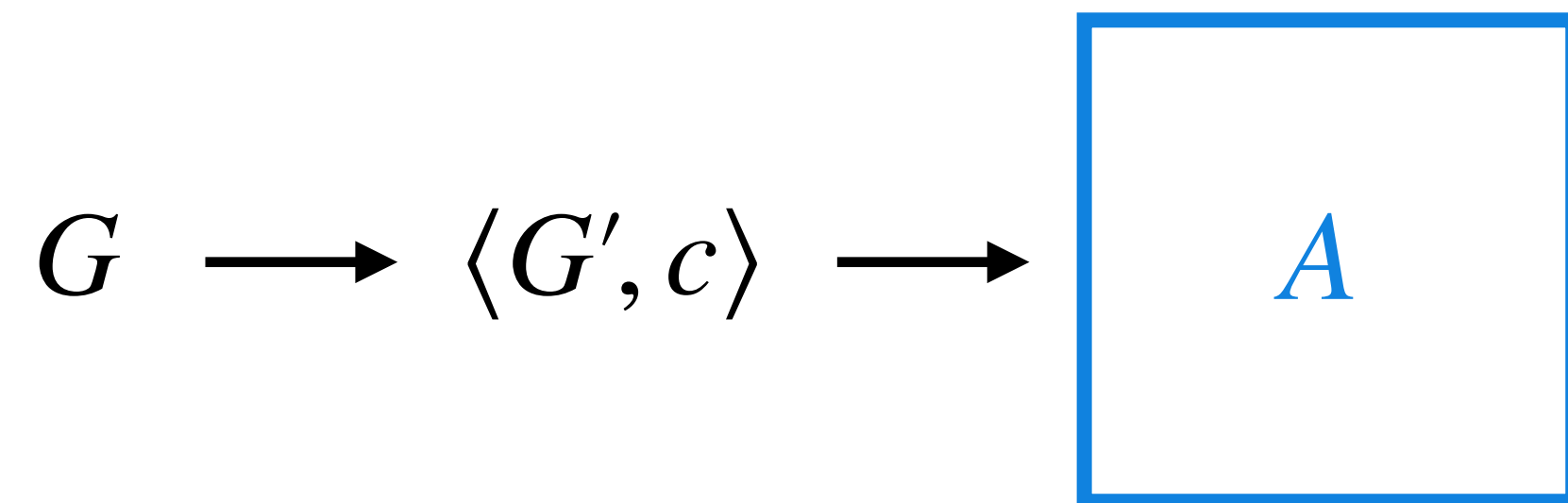
Inapproximability of $OptTSP$

Proof: If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Recall that A is a polytime ρ -approximation algorithm for $OptTSP$.

Polytime algorithm for *Hamcycle* will work in the following way:



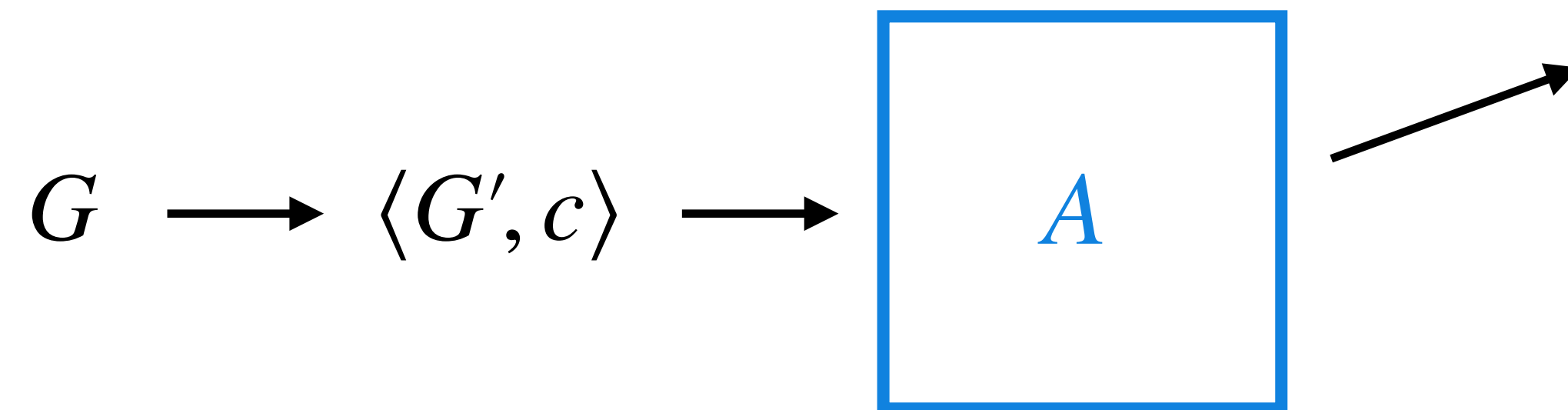
Inapproximability of $OptTSP$

Proof: If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Recall that A is a polytime ρ -approximation algorithm for $OptTSP$.

Polytime algorithm for *Hamcycle* will work in the following way:



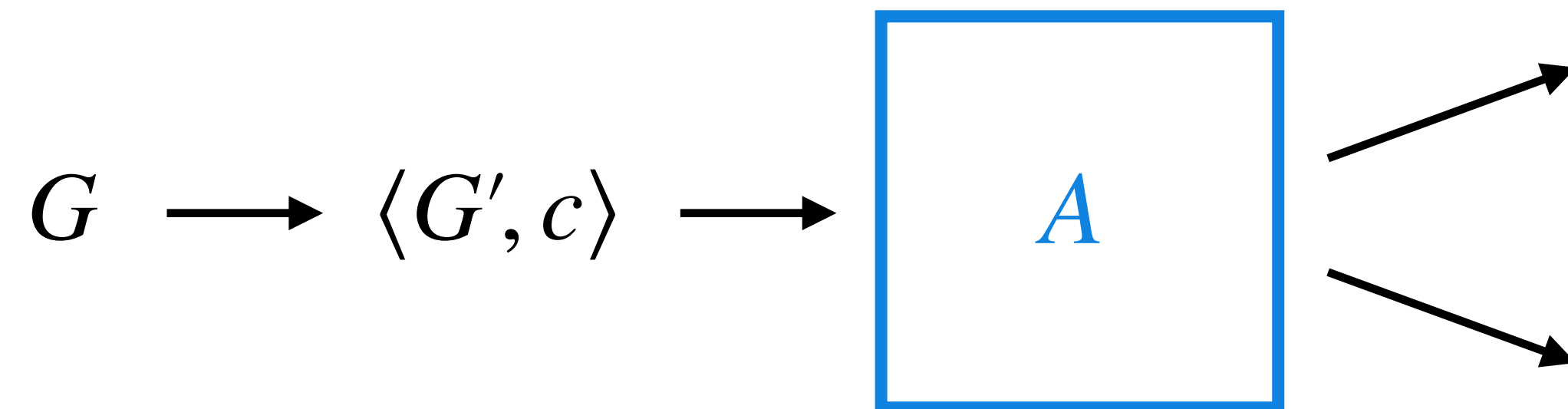
Inapproximability of $OptTSP$

Proof: If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Recall that A is a polytime ρ -approximation algorithm for $OptTSP$.

Polytime algorithm for *Hamcycle* will work in the following way:



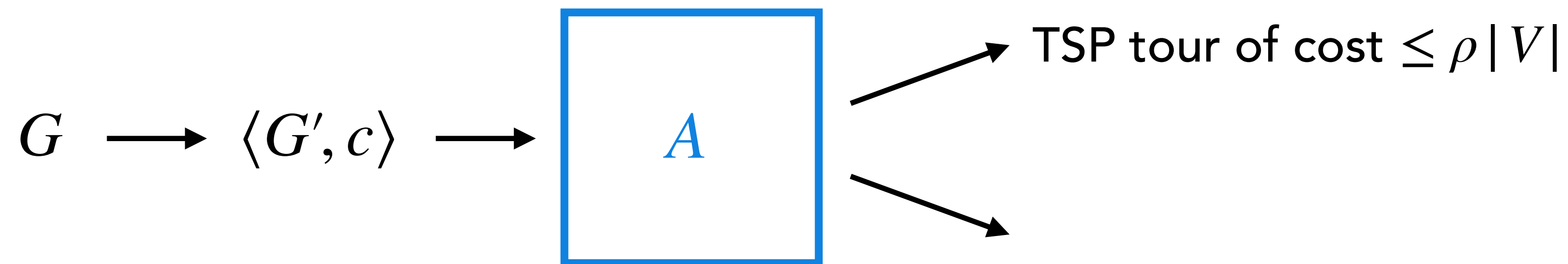
Inapproximability of $OptTSP$

Proof: If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Recall that A is a polytime ρ -approximation algorithm for $OptTSP$.

Polytime algorithm for *Hamcycle* will work in the following way:



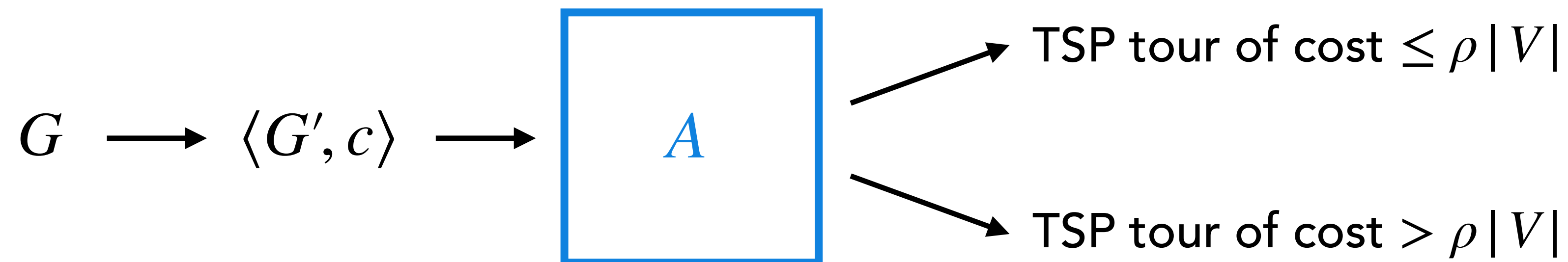
Inapproximability of $OptTSP$

Proof: If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Recall that A is a polytime ρ -approximation algorithm for $OptTSP$.

Polytime algorithm for *Hamcycle* will work in the following way:



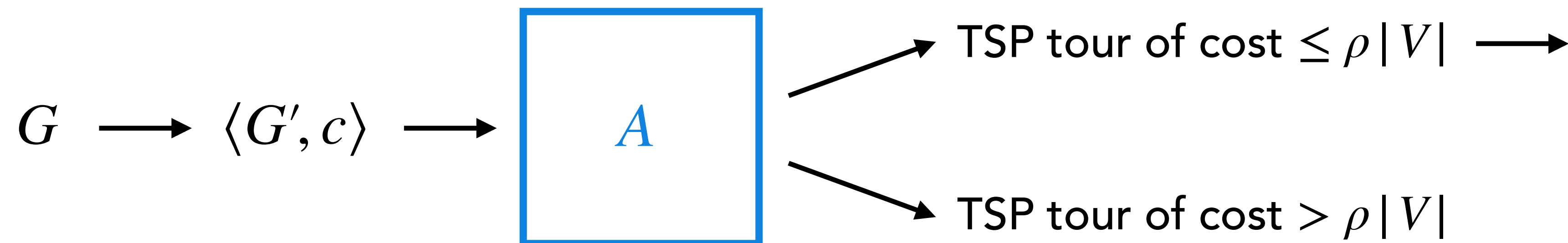
Inapproximability of $OptTSP$

Proof: If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Recall that A is a polytime ρ -approximation algorithm for $OptTSP$.

Polytime algorithm for *Hamcycle* will work in the following way:



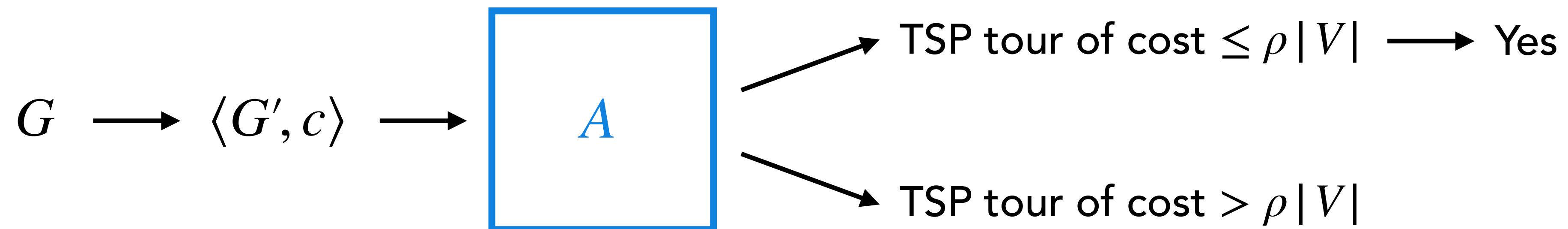
Inapproximability of $OptTSP$

Proof: If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Recall that A is a polytime ρ -approximation algorithm for $OptTSP$.

Polytime algorithm for *Hamcycle* will work in the following way:



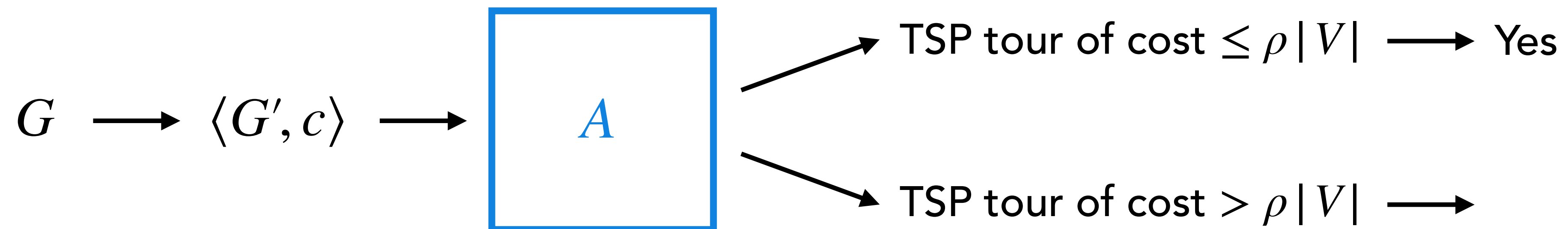
Inapproximability of $OptTSP$

Proof: If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Recall that A is a polytime ρ -approximation algorithm for $OptTSP$.

Polytime algorithm for *Hamcycle* will work in the following way:



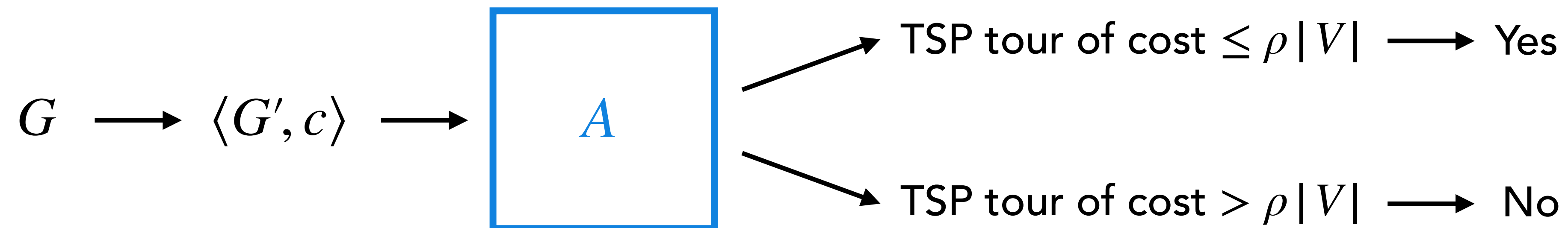
Inapproximability of $OptTSP$

Proof: If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Recall that A is a polytime ρ -approximation algorithm for $OptTSP$.

Polytime algorithm for *Hamcycle* will work in the following way:



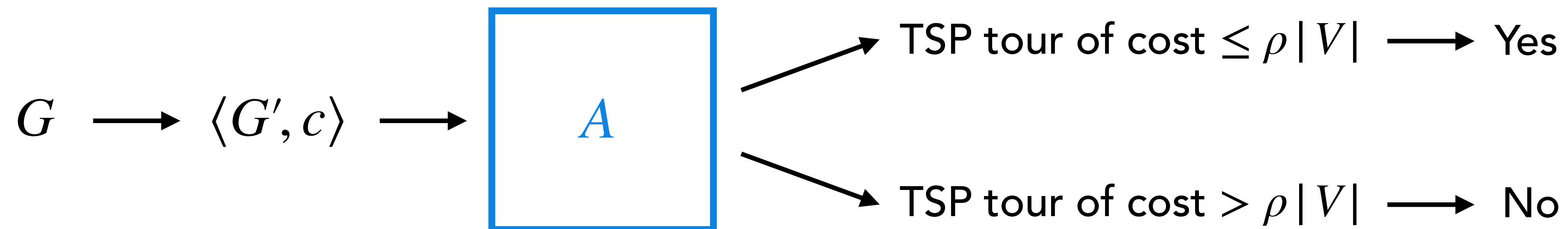
Inapproximability of $OptTSP$

Proof: If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Recall that A is a polytime ρ -approximation algorithm for $OptTSP$.

Polytime algorithm for *Hamcycle* will work in the following way:



$Hamcycle \in P \implies P = NP$.

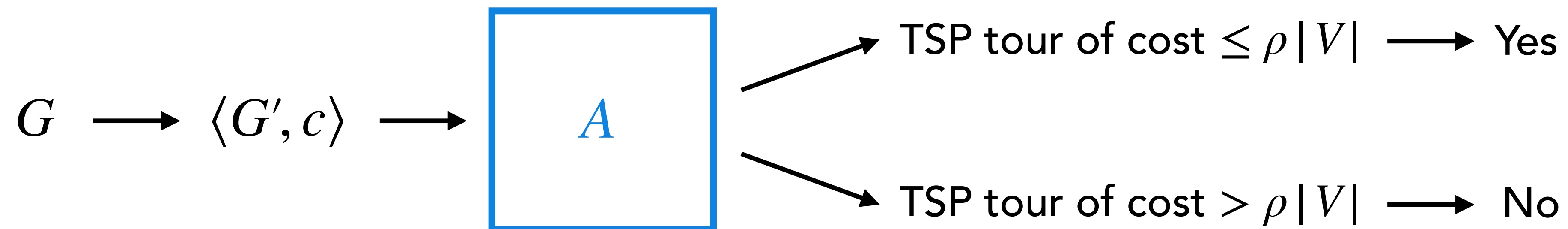
Inapproximability of $OptTSP$

Proof: If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Recall that A is a polytime ρ -approximation algorithm for $OptTSP$.

Polytime algorithm for *Hamcycle* will work in the following way:



$Hamcycle \in P \implies P = NP$. **Contradiction!**

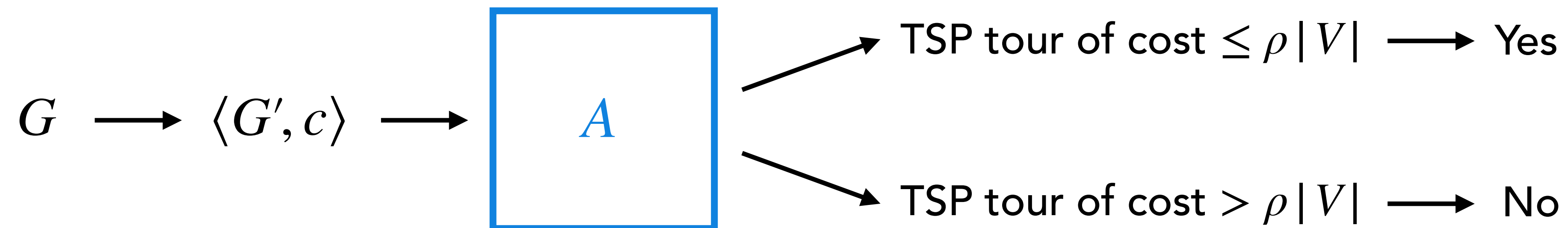
Inapproximability of $OptTSP$

Proof: If G has a hamiltonian cycle, then G' has a TSP tour of cost $|V|$.

If G doesn't have a hamiltonian cycle, then any TSP tour in G' has cost $> \rho |V|$

Recall that A is a polytime ρ -approximation algorithm for $OptTSP$.

Polytime algorithm for *Hamcycle* will work in the following way:



$Hamcycle \in P \implies P = NP$. **Contradiction!**



2-Approx. of $OptTSP$ Under Triangle Inequality

2-Approx. of $OptTSP$ Under Triangle Inequality

We will now see a polytime 2-approximation algorithm for $OptTSP$

2-Approx. of $OptTSP$ Under Triangle Inequality

We will now see a polytime **2-approximation** algorithm for $OptTSP$ if the graph G satisfies

2-Approx. of $OptTSP$ Under Triangle Inequality

We will now see a polytime **2-approximation** algorithm for $OptTSP$ if the graph G satisfies the **triangle inequality**

2-Approx. of $OptTSP$ Under Triangle Inequality

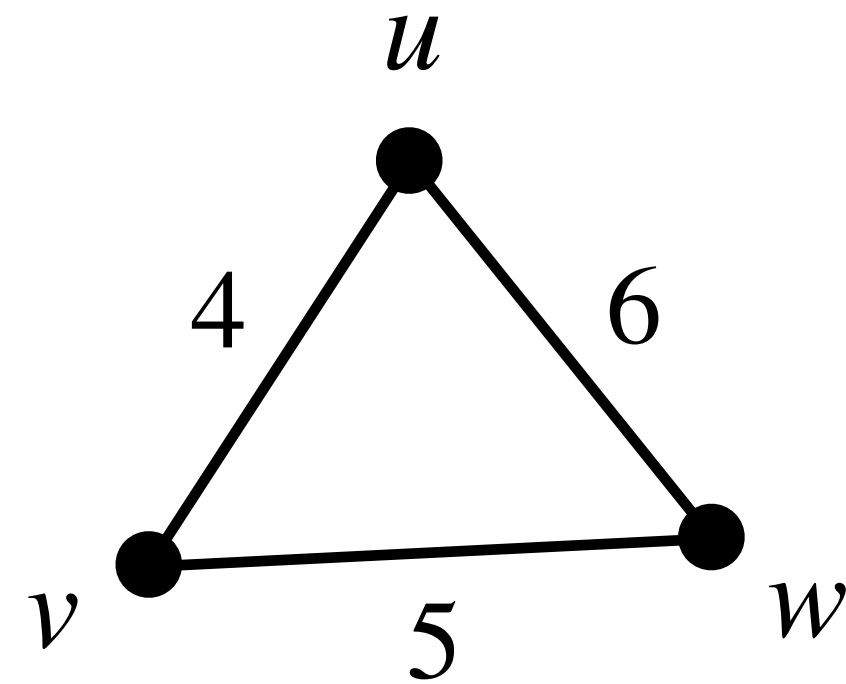
We will now see a polytime **2-approximation** algorithm for $OptTSP$ if the graph G satisfies the **triangle inequality**, that is, for any three vertices $u, v, w \in G$,

2-Approx. of $OptTSP$ Under Triangle Inequality

We will now see a polytime **2-approximation** algorithm for $OptTSP$ if the graph G satisfies the **triangle inequality**, that is, for any three vertices $u, v, w \in G$, $c(u, w) \leq c(u, v) + c(v, w)$.

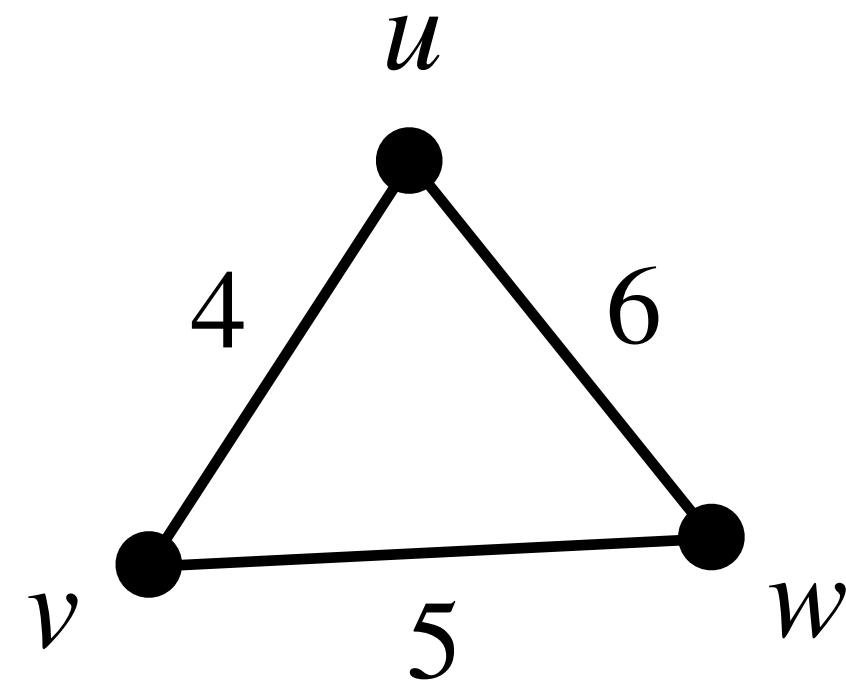
2-Approx. of $OptTSP$ Under Triangle Inequality

We will now see a polytime **2-approximation** algorithm for $OptTSP$ if the graph G satisfies the **triangle inequality**, that is, for any three vertices $u, v, w \in G$, $c(u, w) \leq c(u, v) + c(v, w)$.



2-Approx. of $OptTSP$ Under Triangle Inequality

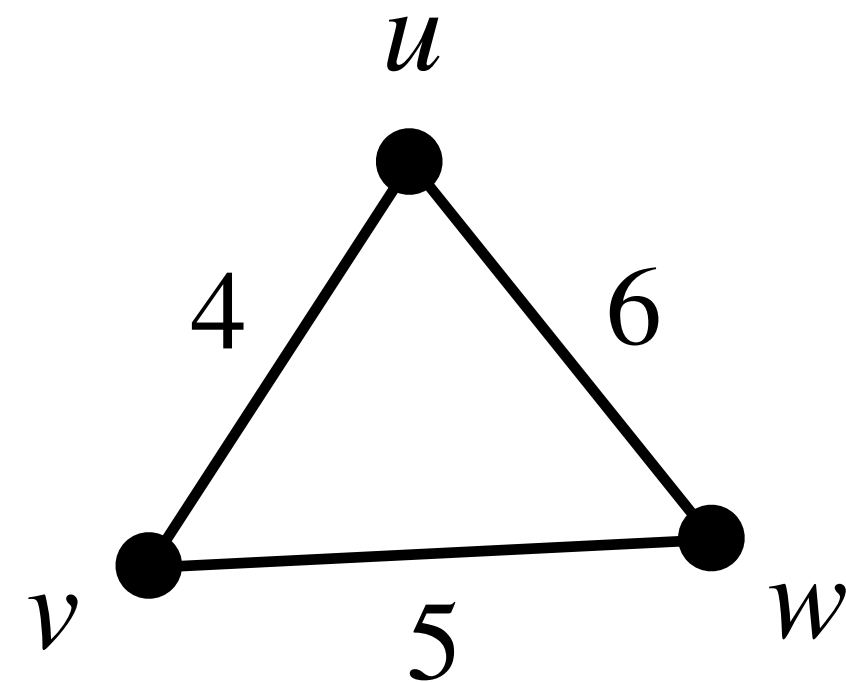
We will now see a polytime **2-approximation** algorithm for $OptTSP$ if the graph G satisfies the **triangle inequality**, that is, for any three vertices $u, v, w \in G$, $c(u, w) \leq c(u, v) + c(v, w)$.



$$6 \leq 4 + 5$$

2-Approx. of $OptTSP$ Under Triangle Inequality

We will now see a polytime **2-approximation** algorithm for $OptTSP$ if the graph G satisfies the **triangle inequality**, that is, for any three vertices $u, v, w \in G$, $c(u, w) \leq c(u, v) + c(v, w)$.

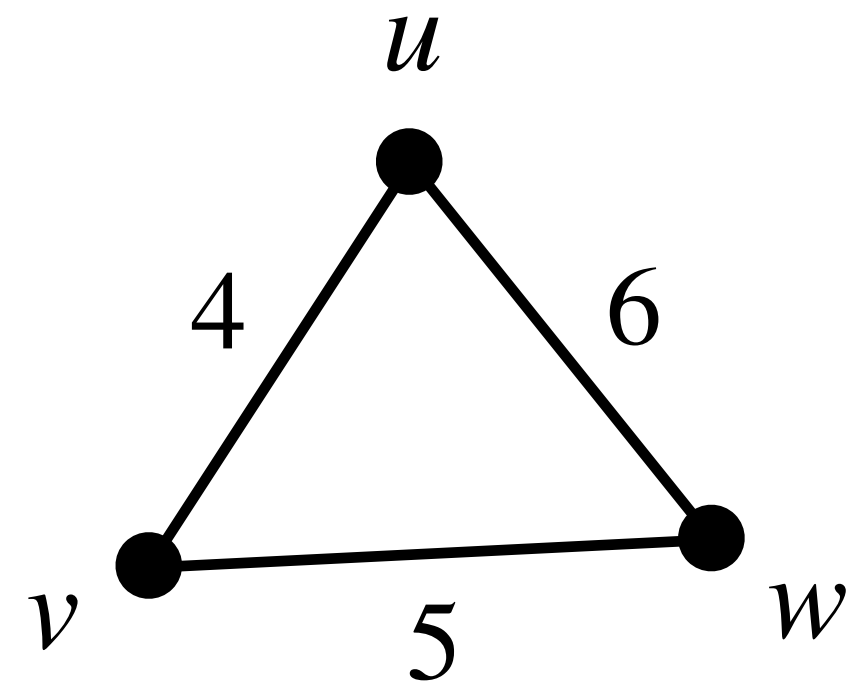


$$6 \leq 4 + 5$$

$$4 \leq 6 + 5$$

2-Approx. of $OptTSP$ Under Triangle Inequality

We will now see a polytime **2-approximation** algorithm for $OptTSP$ if the graph G satisfies the **triangle inequality**, that is, for any three vertices $u, v, w \in G$, $c(u, w) \leq c(u, v) + c(v, w)$.



$$6 \leq 4 + 5$$

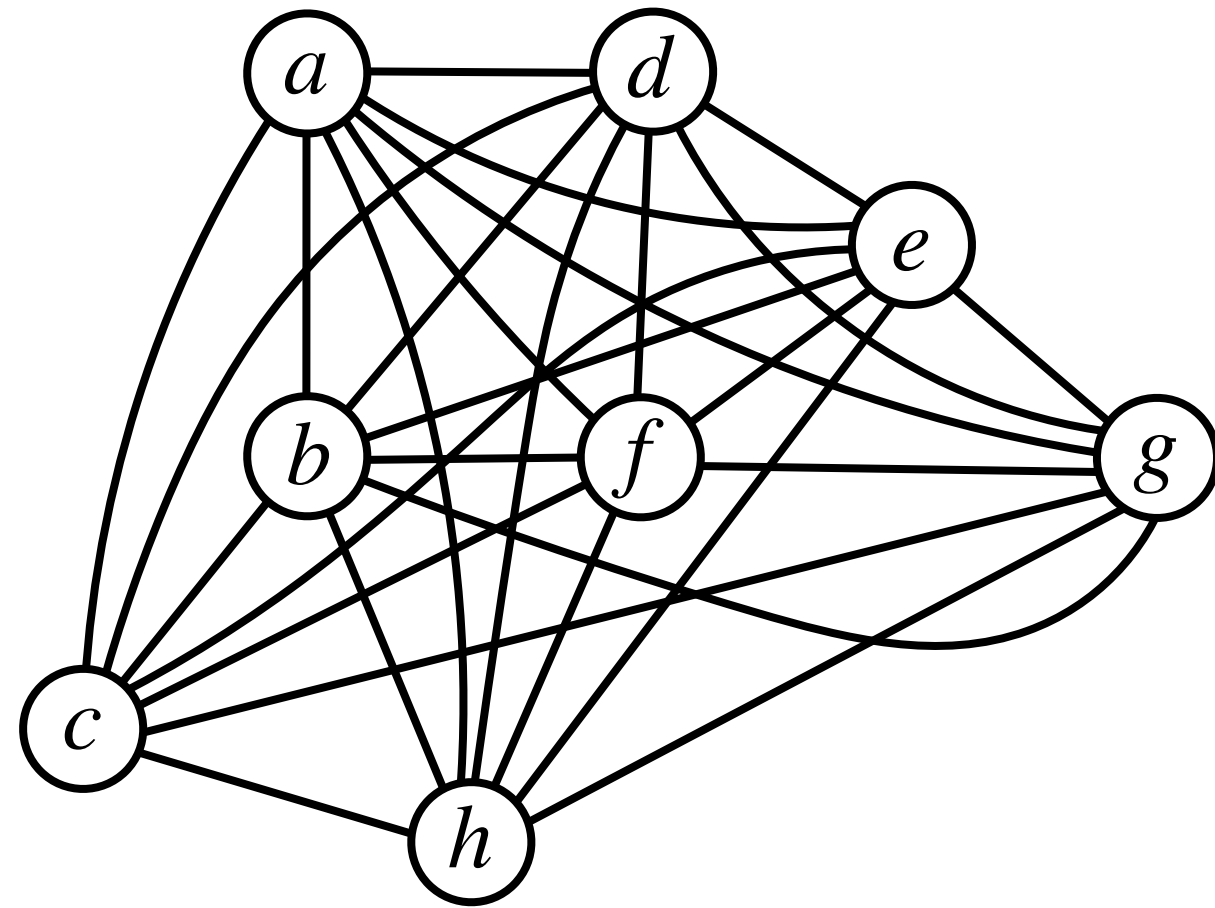
$$4 \leq 6 + 5$$

$$5 \leq 4 + 6$$

2-Approx. of $OptTSP$ Under Triangle Inequality

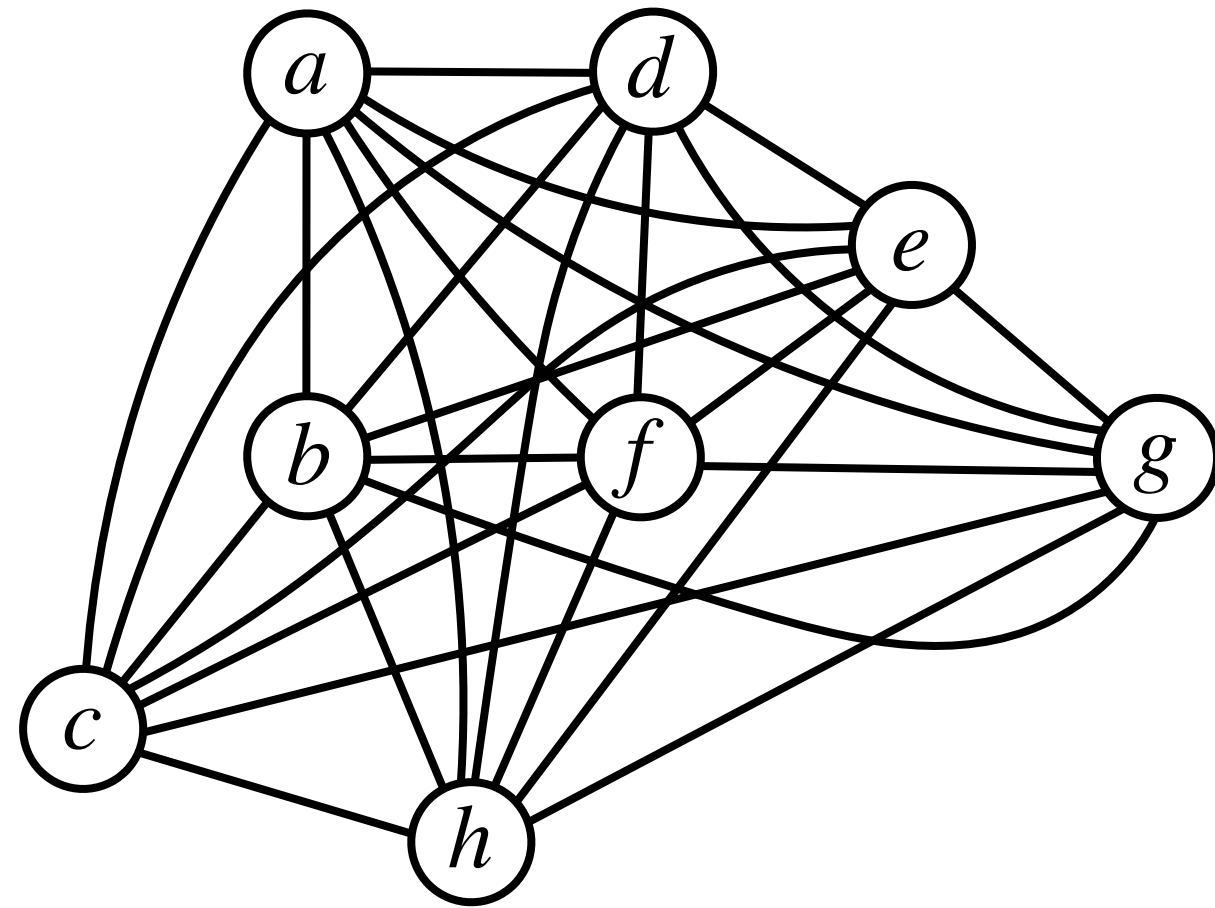
2-Approx. of $OptTSP$ Under Triangle Inequality

G with cost c

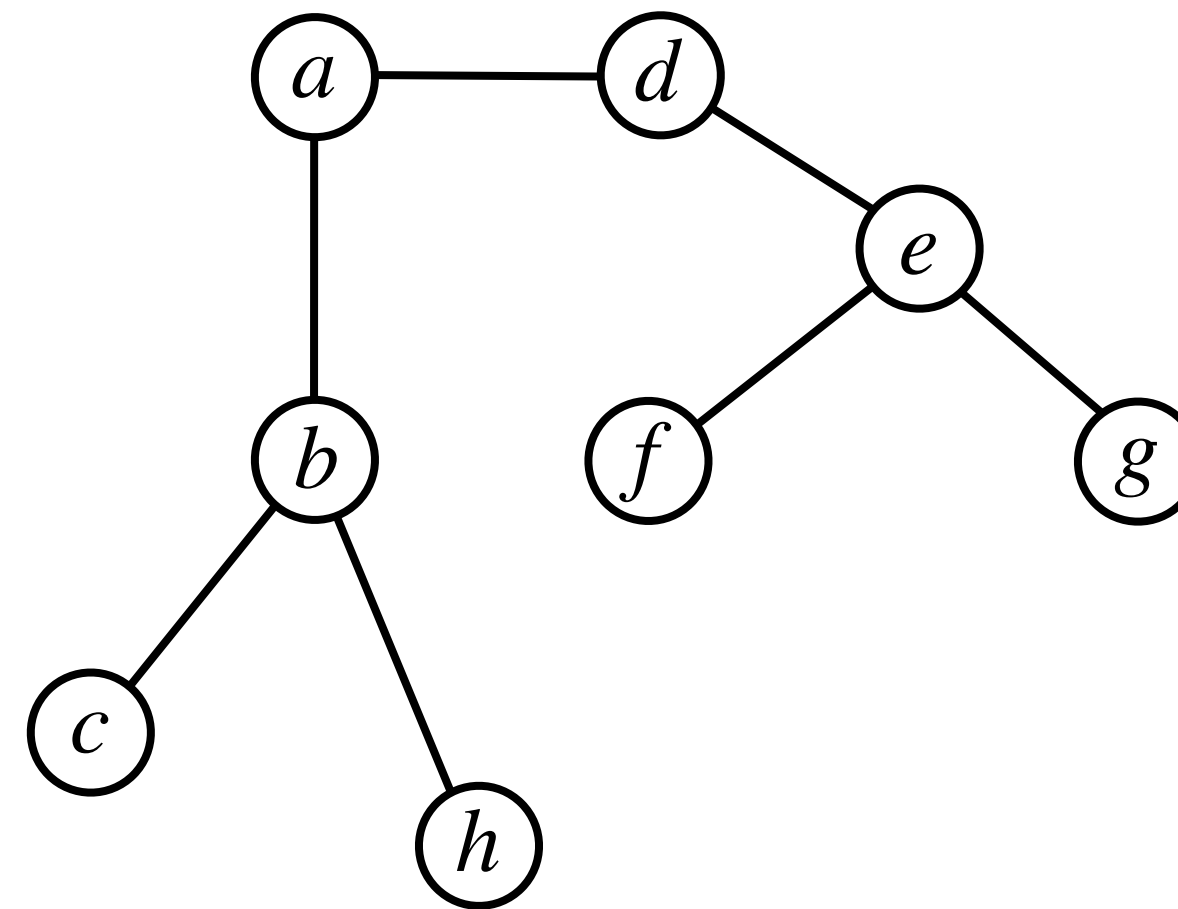


2-Approx. of $OptTSP$ Under Triangle Inequality

G with cost c

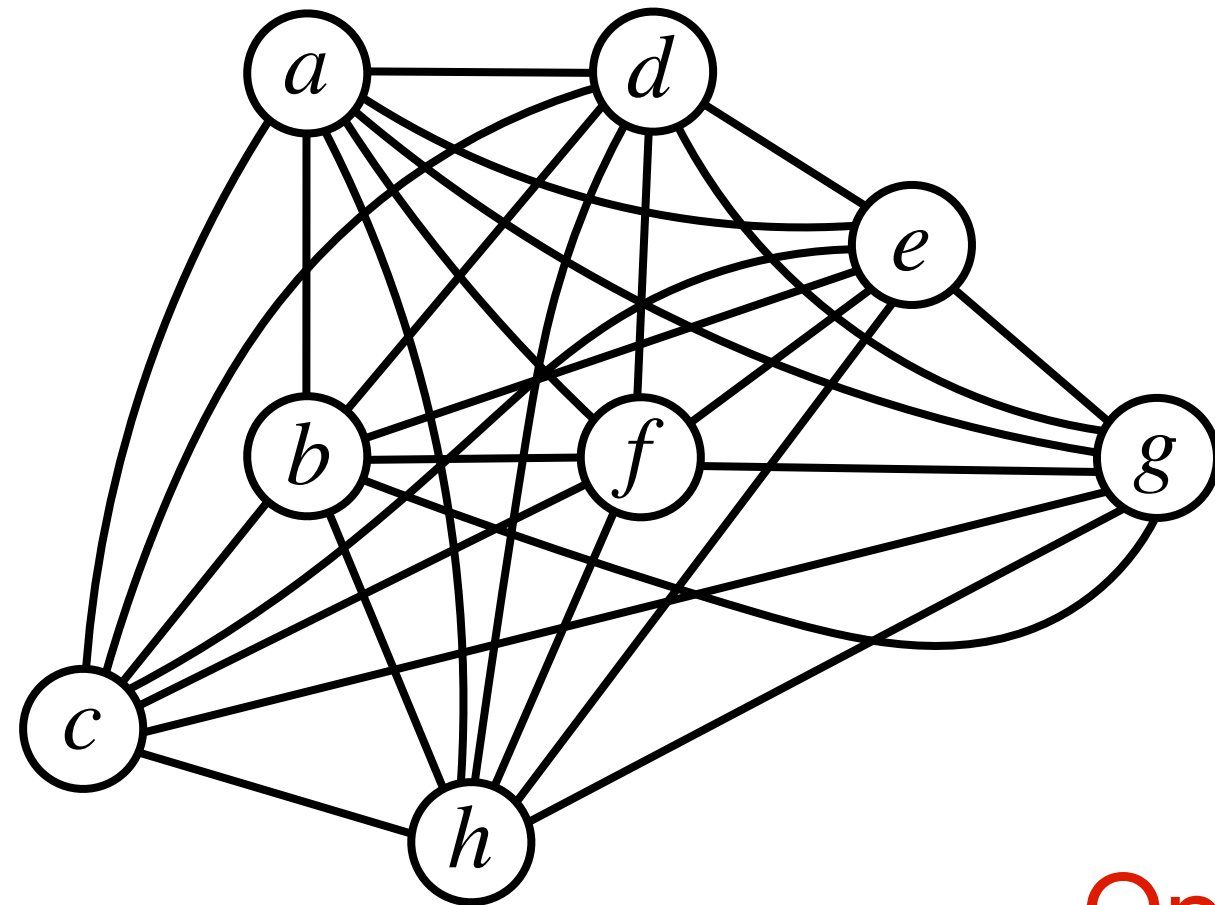


MST T of G

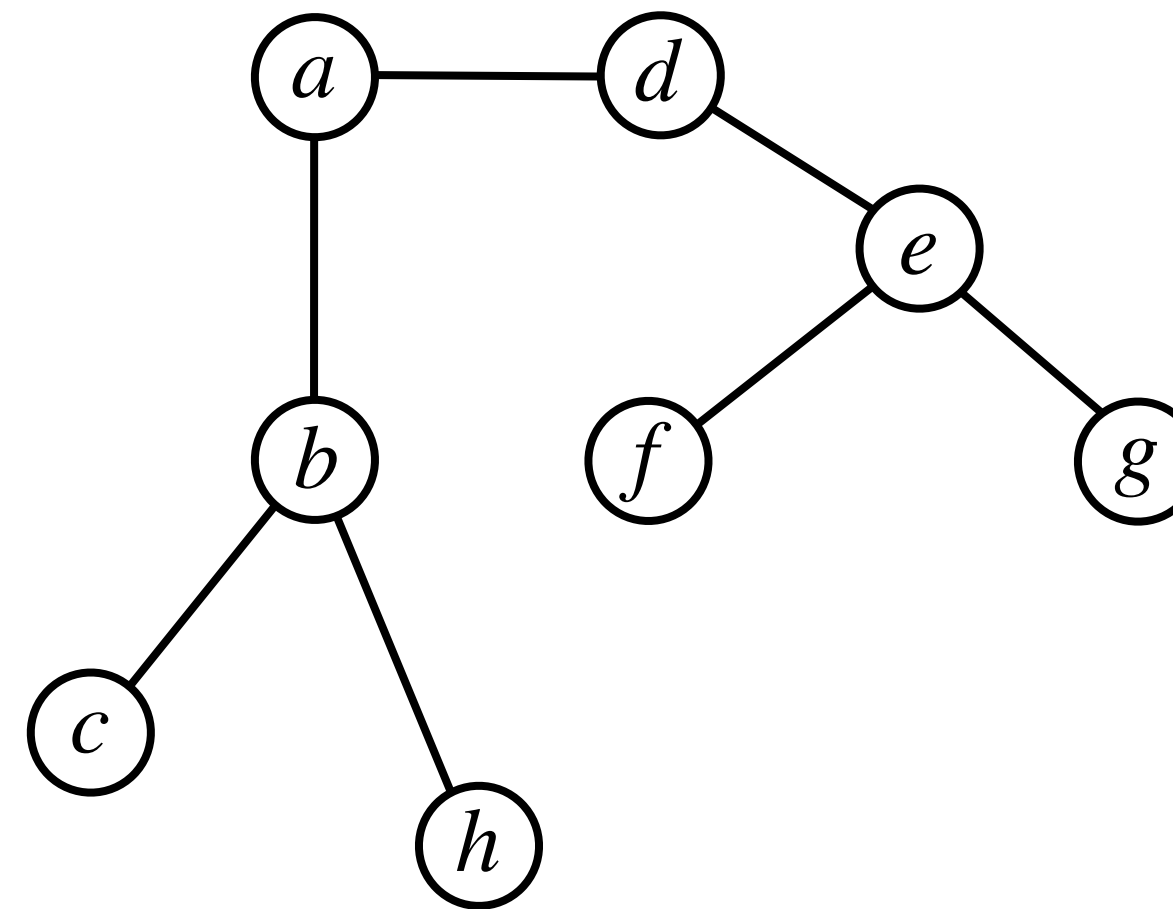


2-Approx. of $OptTSP$ Under Triangle Inequality

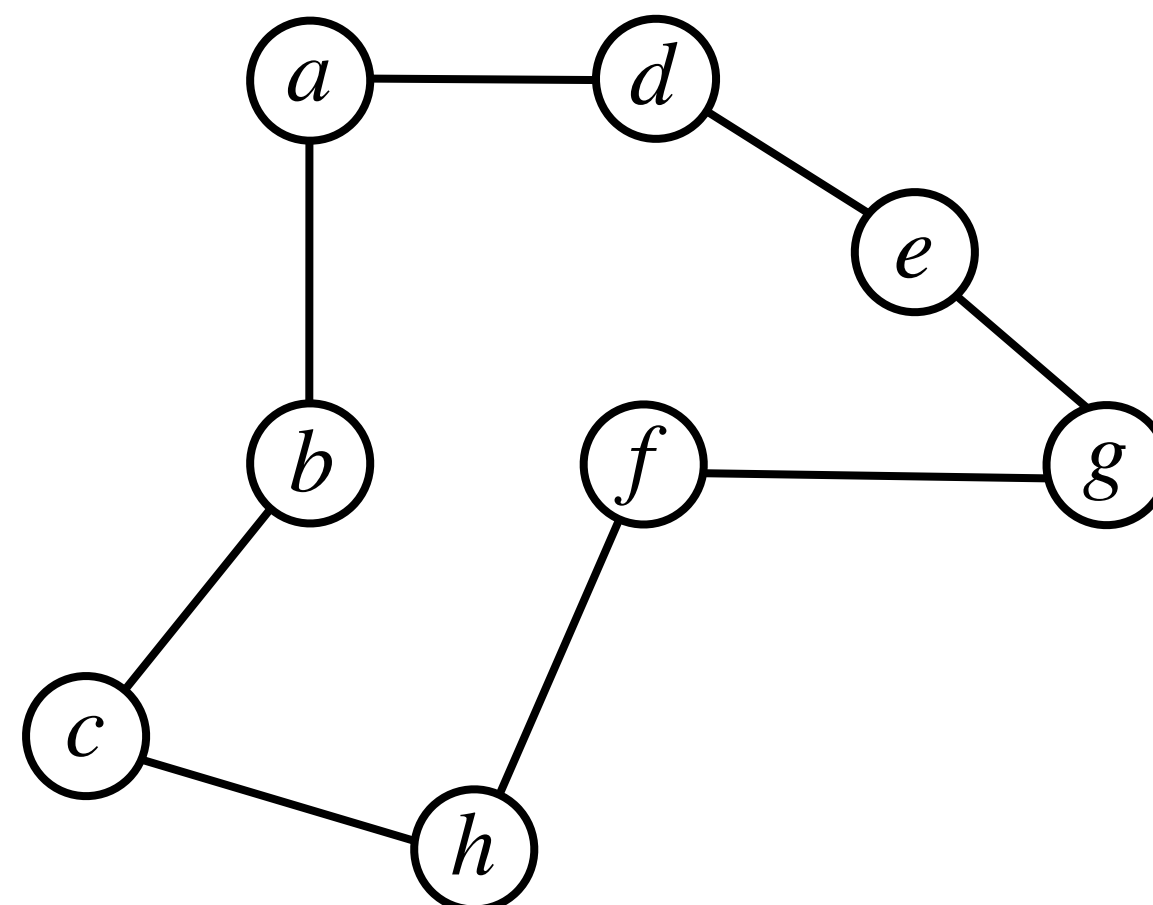
G with cost c



MST T of G

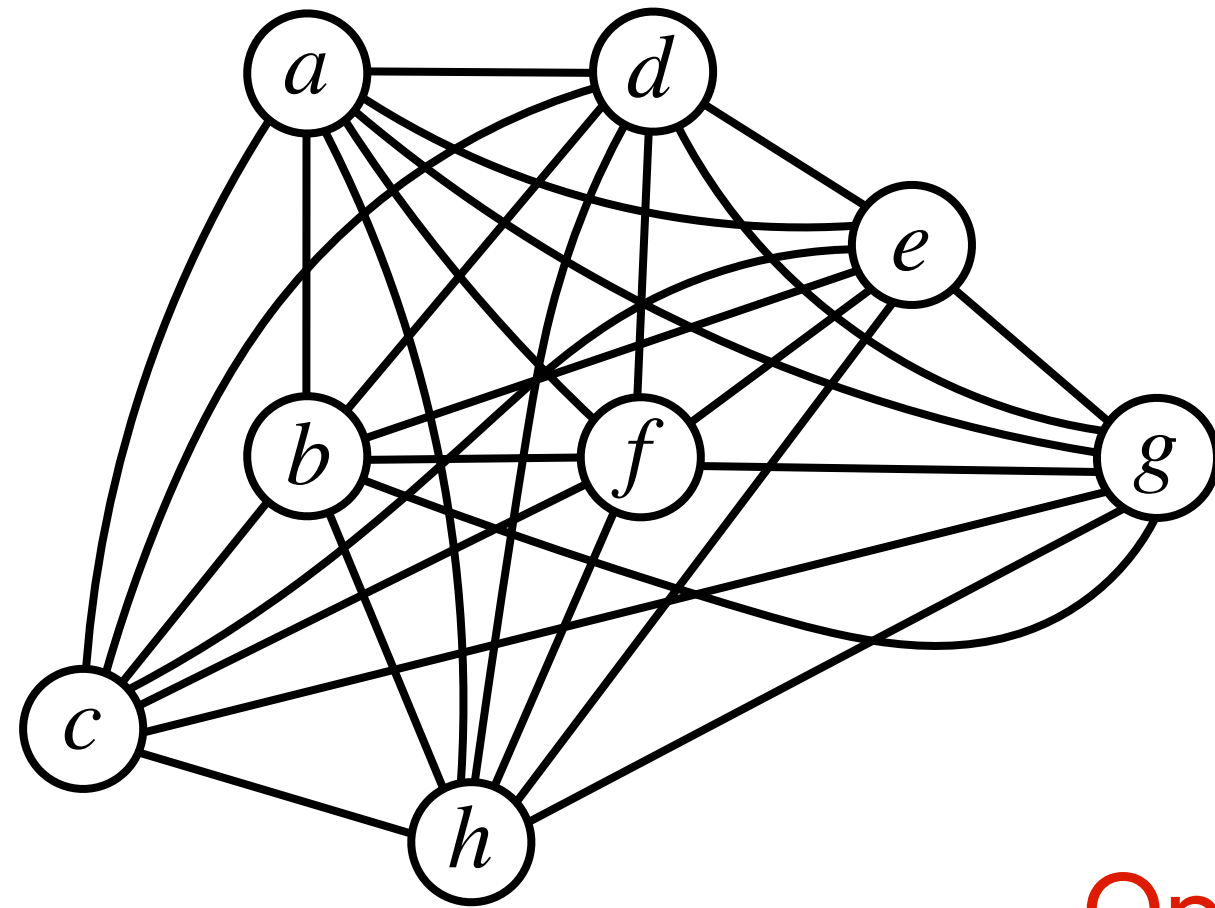


OptTSP tour H^* in G

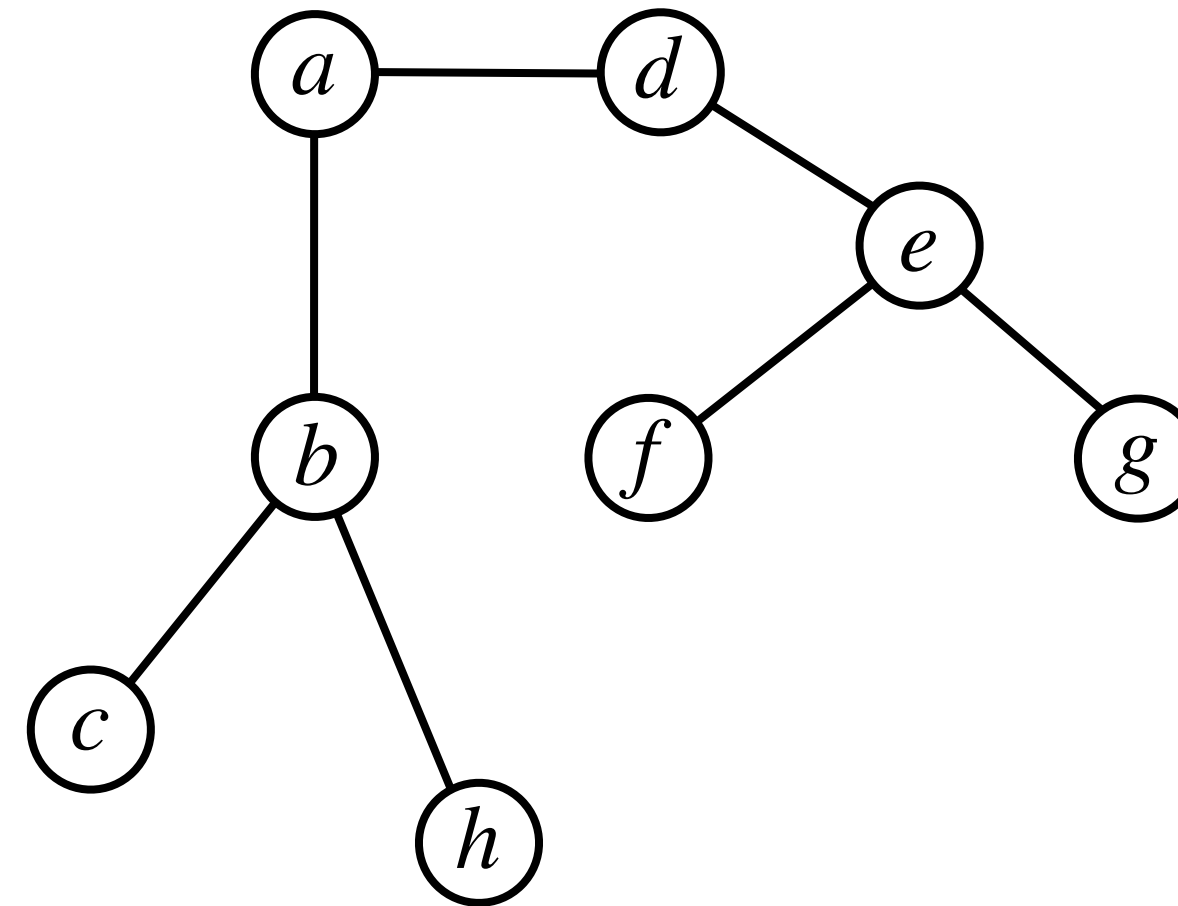


2-Approx. of $OptTSP$ Under Triangle Inequality

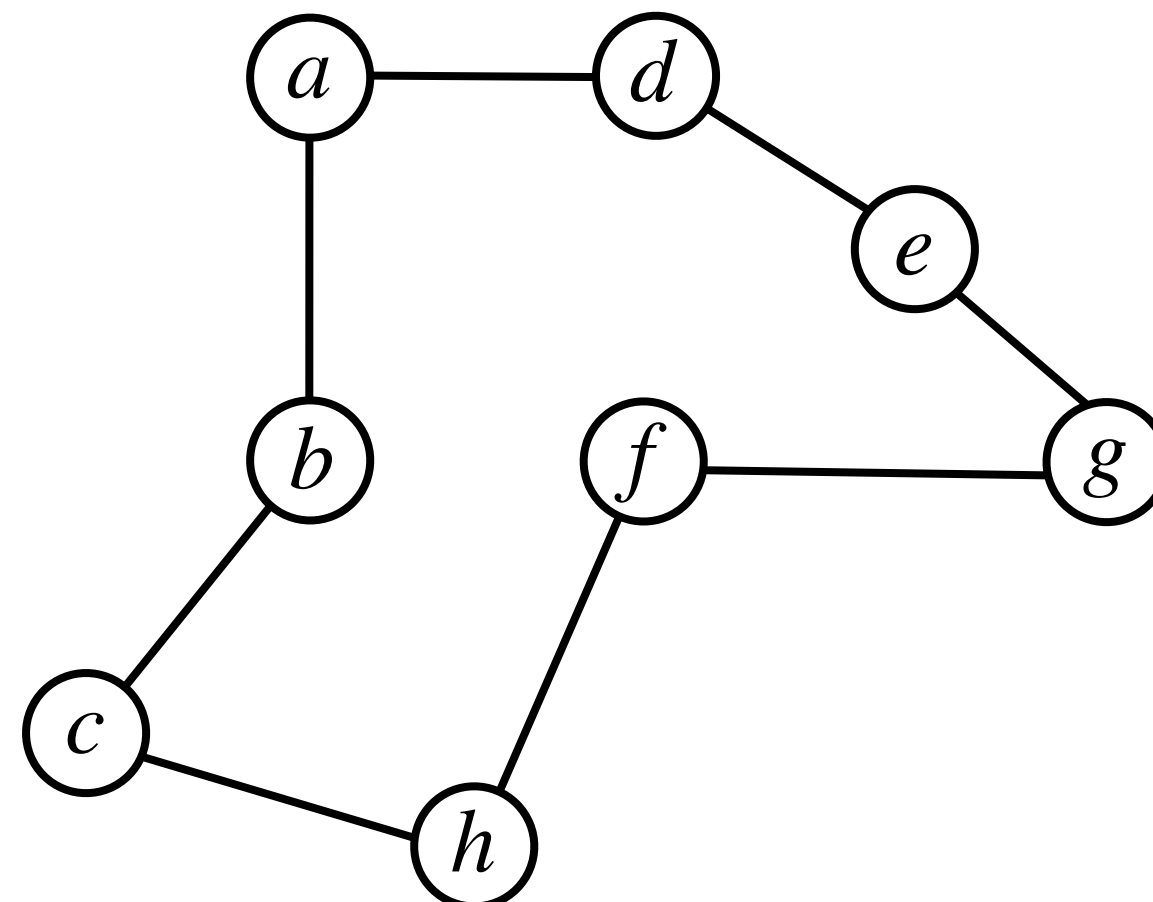
G with cost c



MST T of G



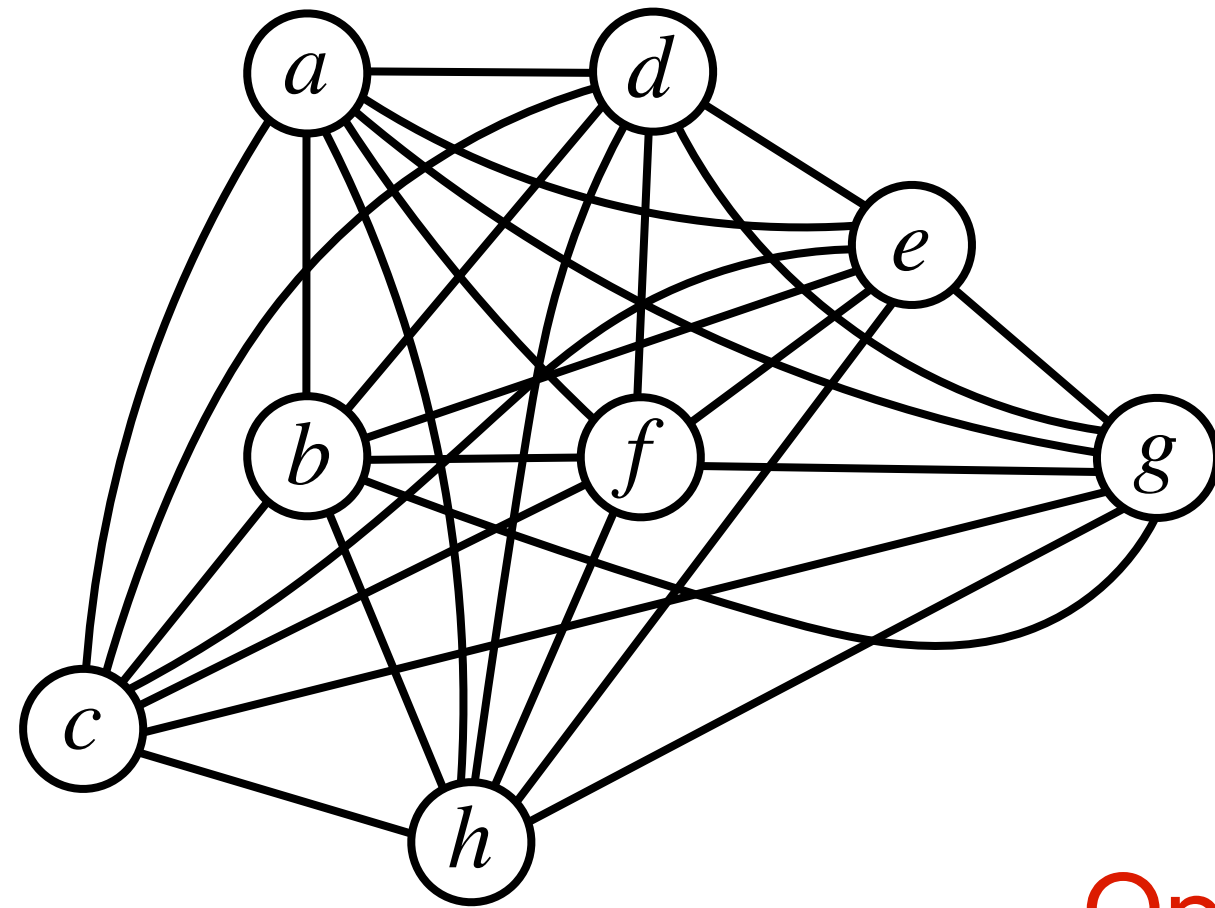
OptTSP tour H^* in G



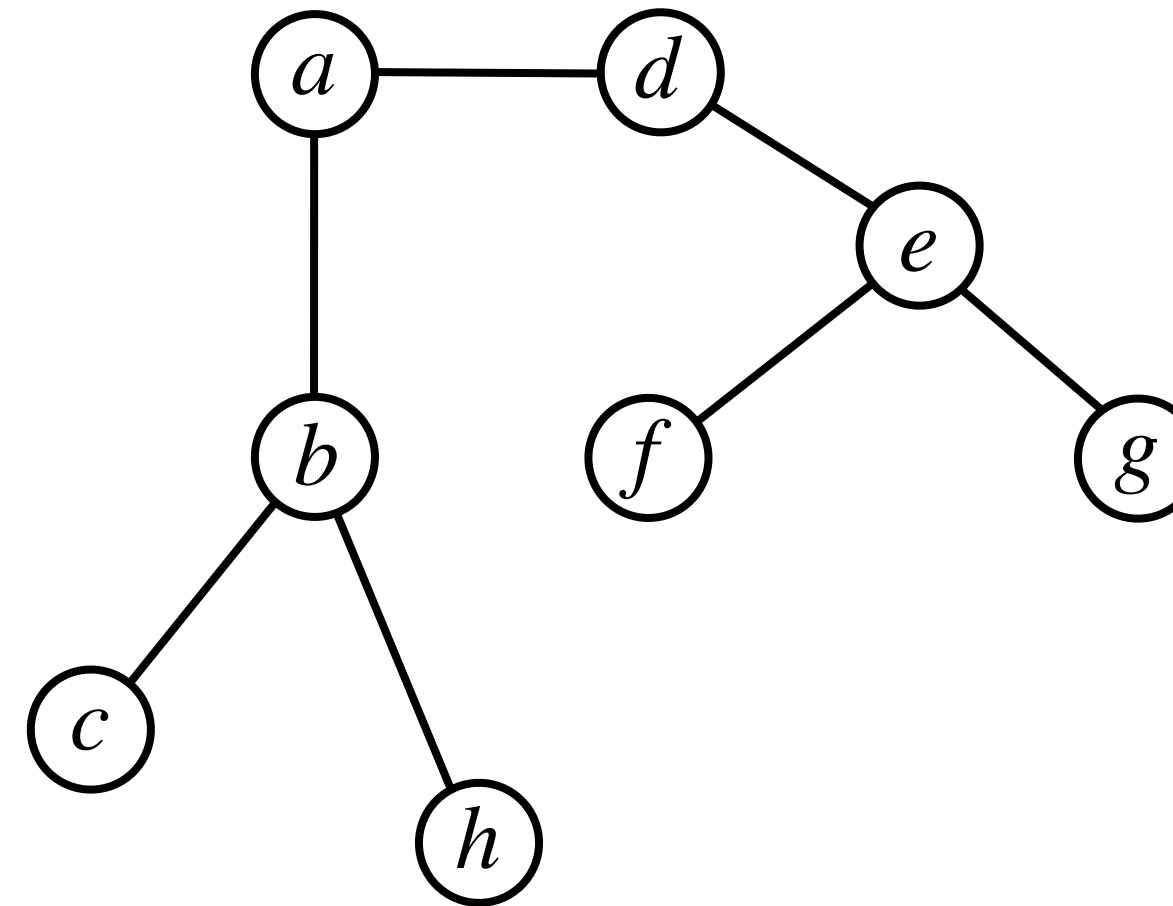
$c(H^*)$

2-Approx. of $OptTSP$ Under Triangle Inequality

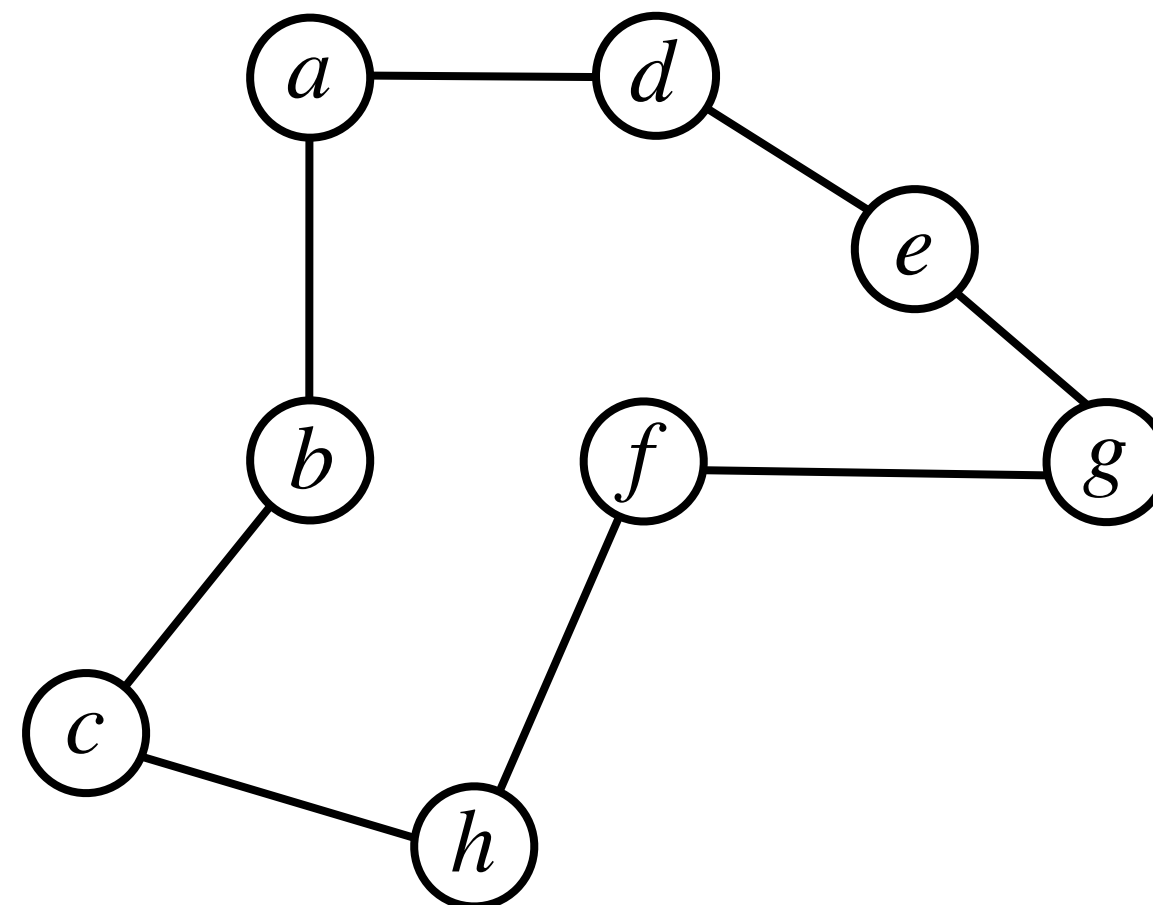
G with cost c



MST T of G



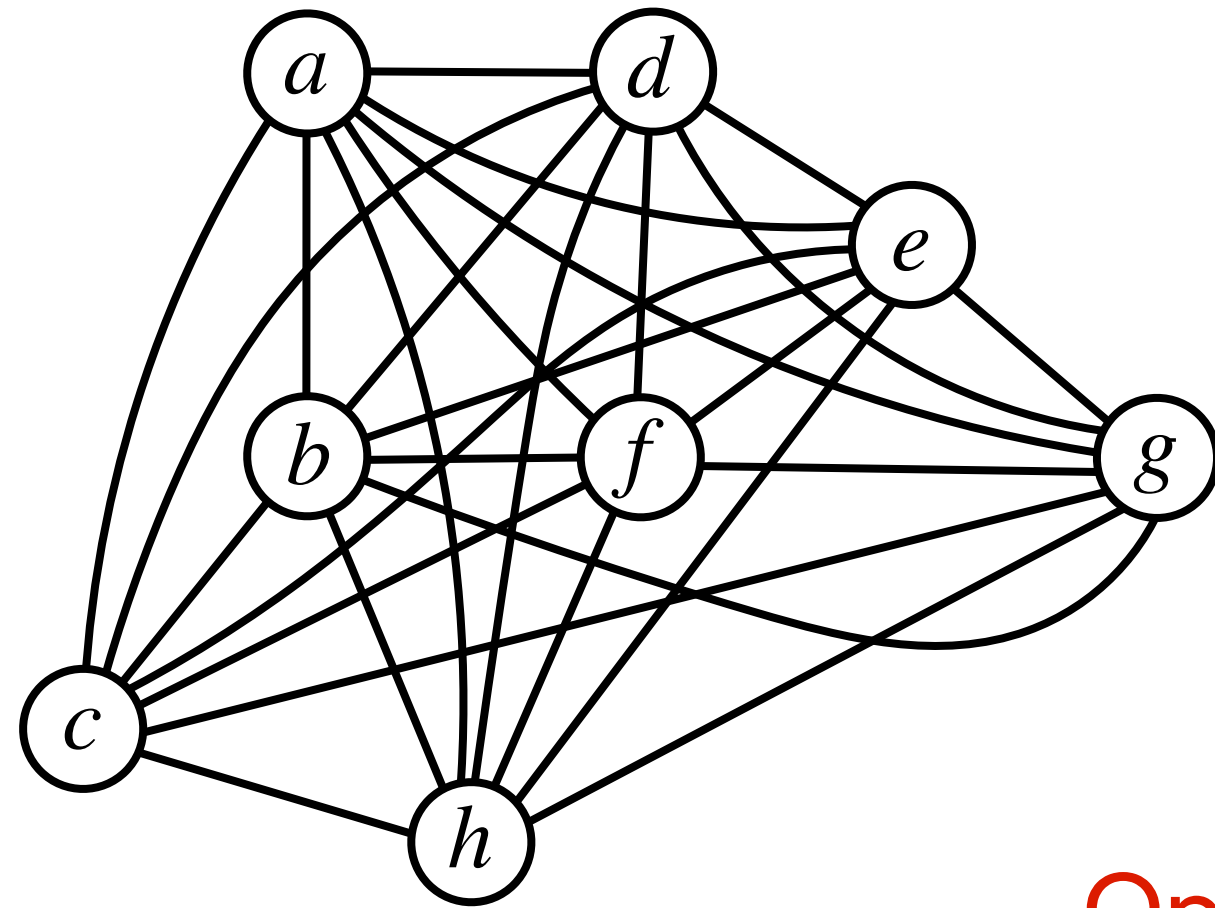
OptTSP tour H^* in G



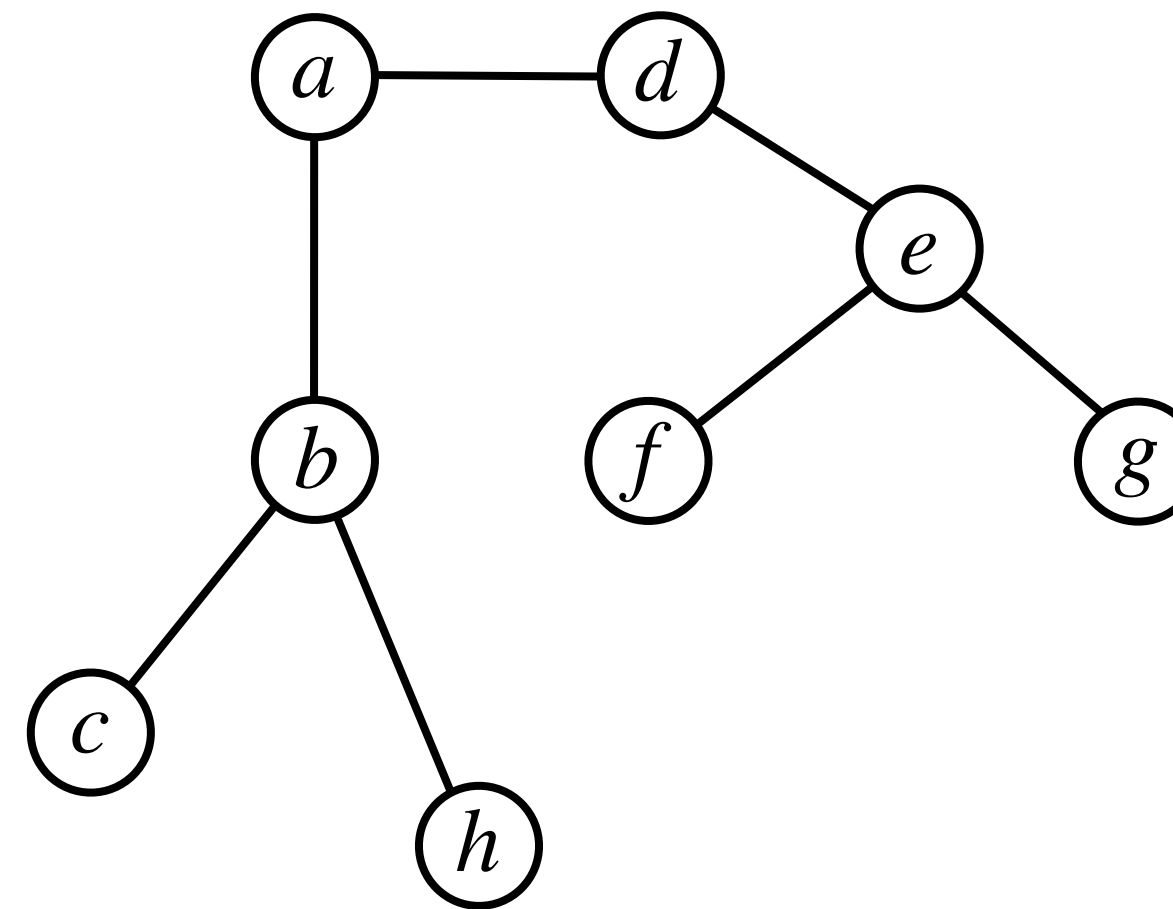
$$c(H^*) \geq c(H^* - \{f, h\})$$

2-Approx. of $OptTSP$ Under Triangle Inequality

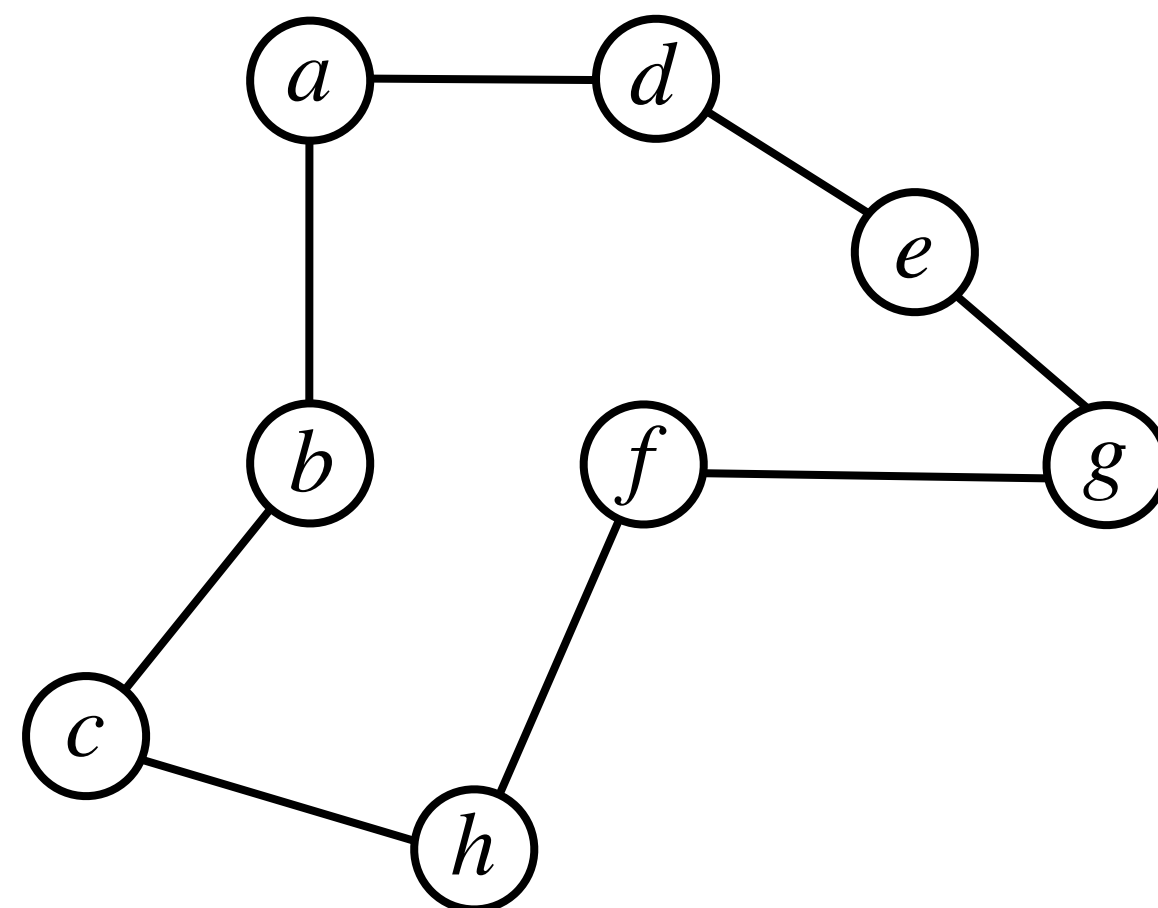
G with cost c



MST T of G



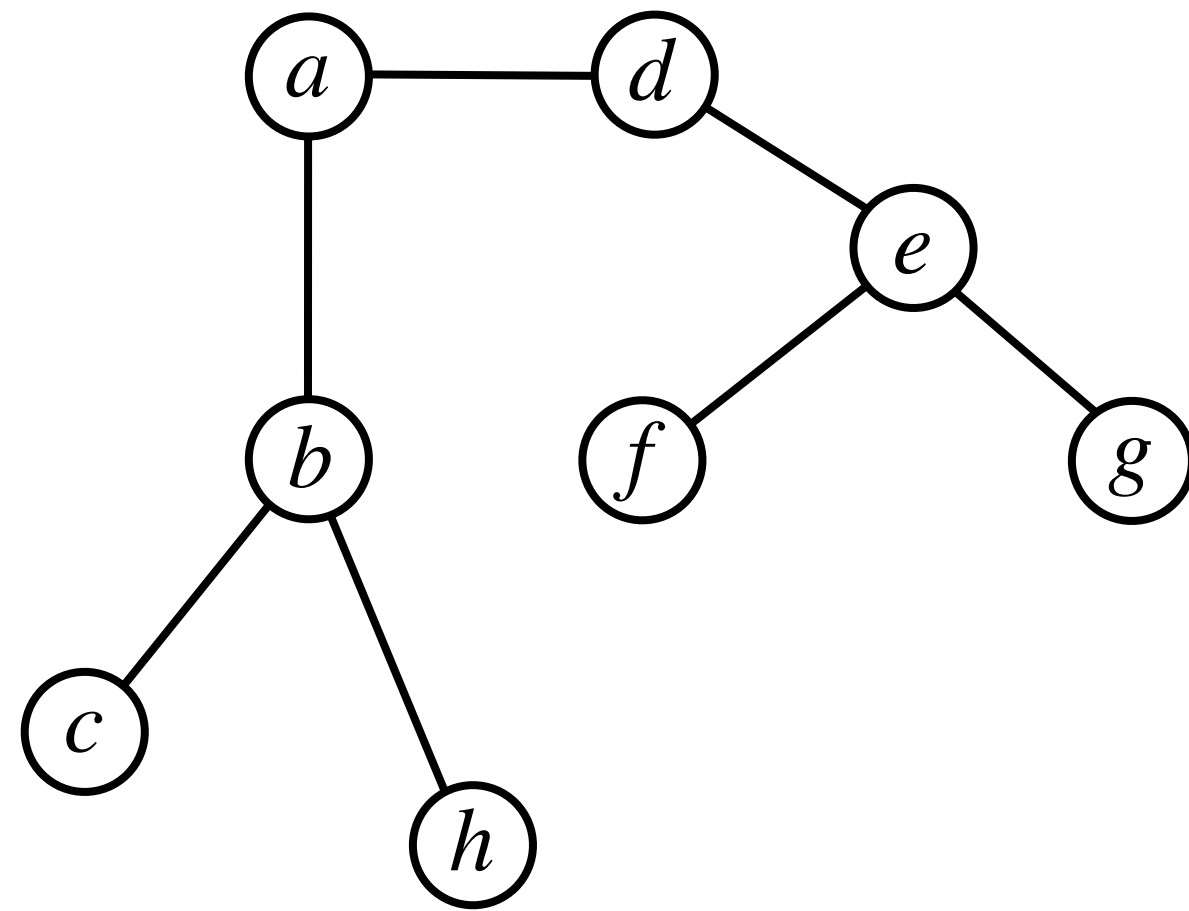
OptTSP tour H^* in G



$$c(H^*) \geq c(H^* - \{f, h\}) \geq c(T)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

MST T of G

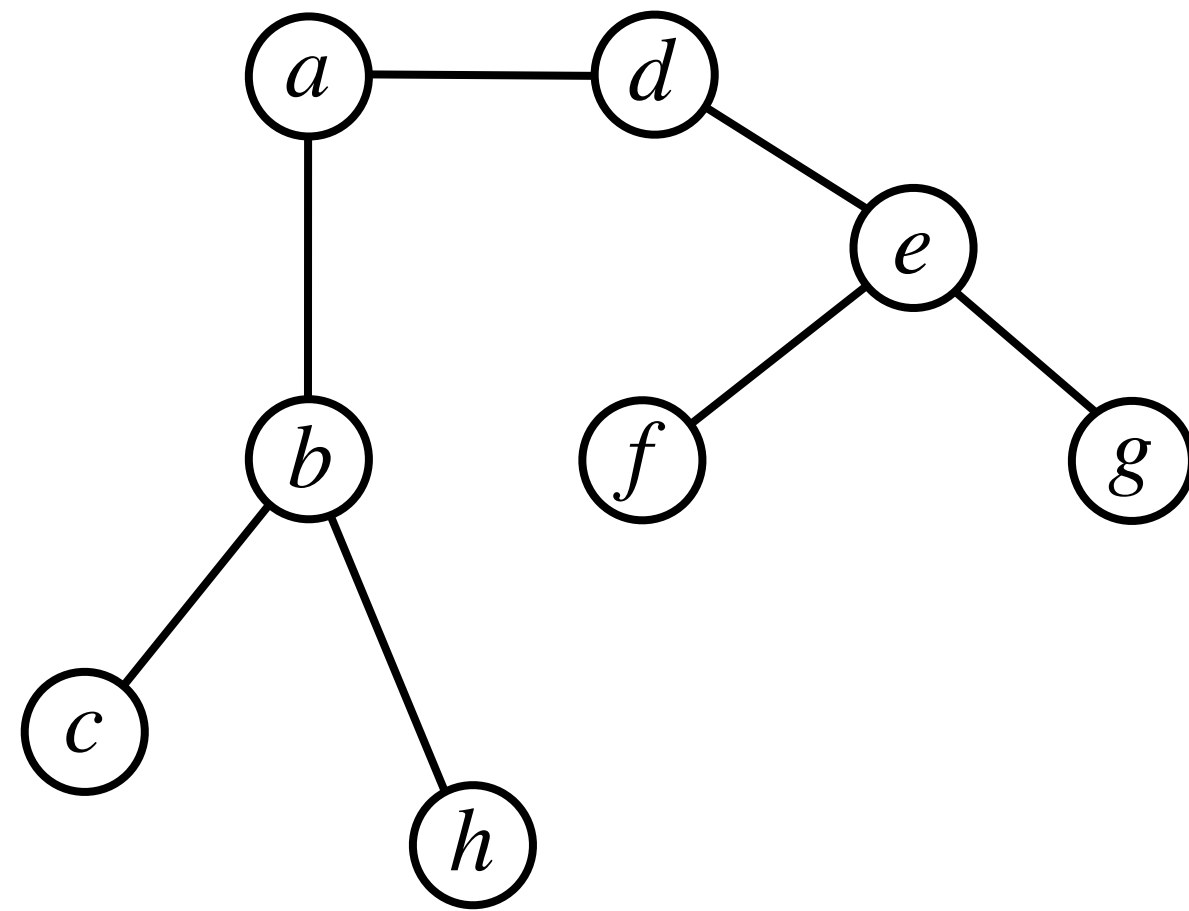


$$c(H^*) \geq c(T)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Consider the tour formed by walking around T .

MST T of G

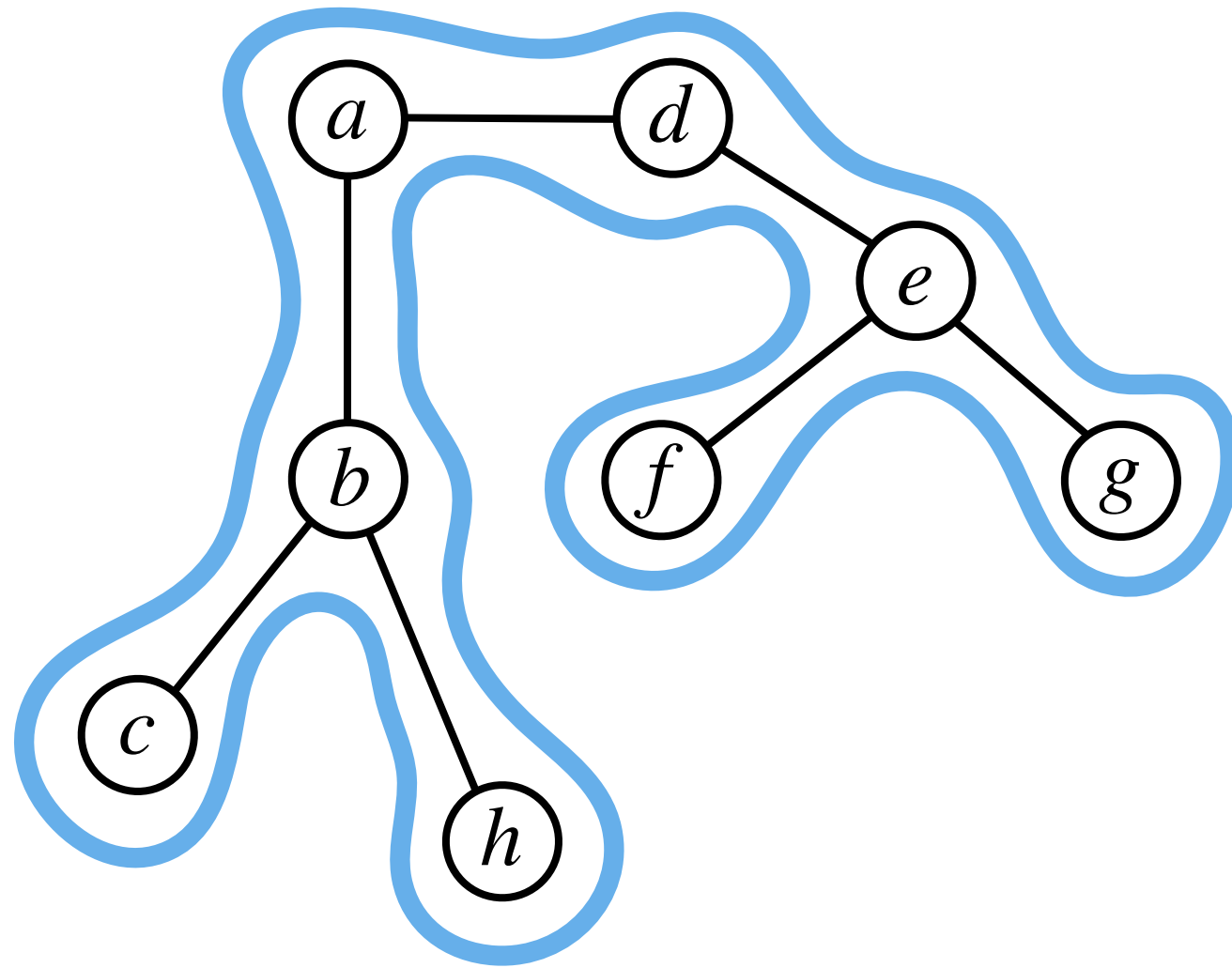


$$c(H^*) \geq c(T)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Consider the tour formed by walking around T .

Tour around MST T

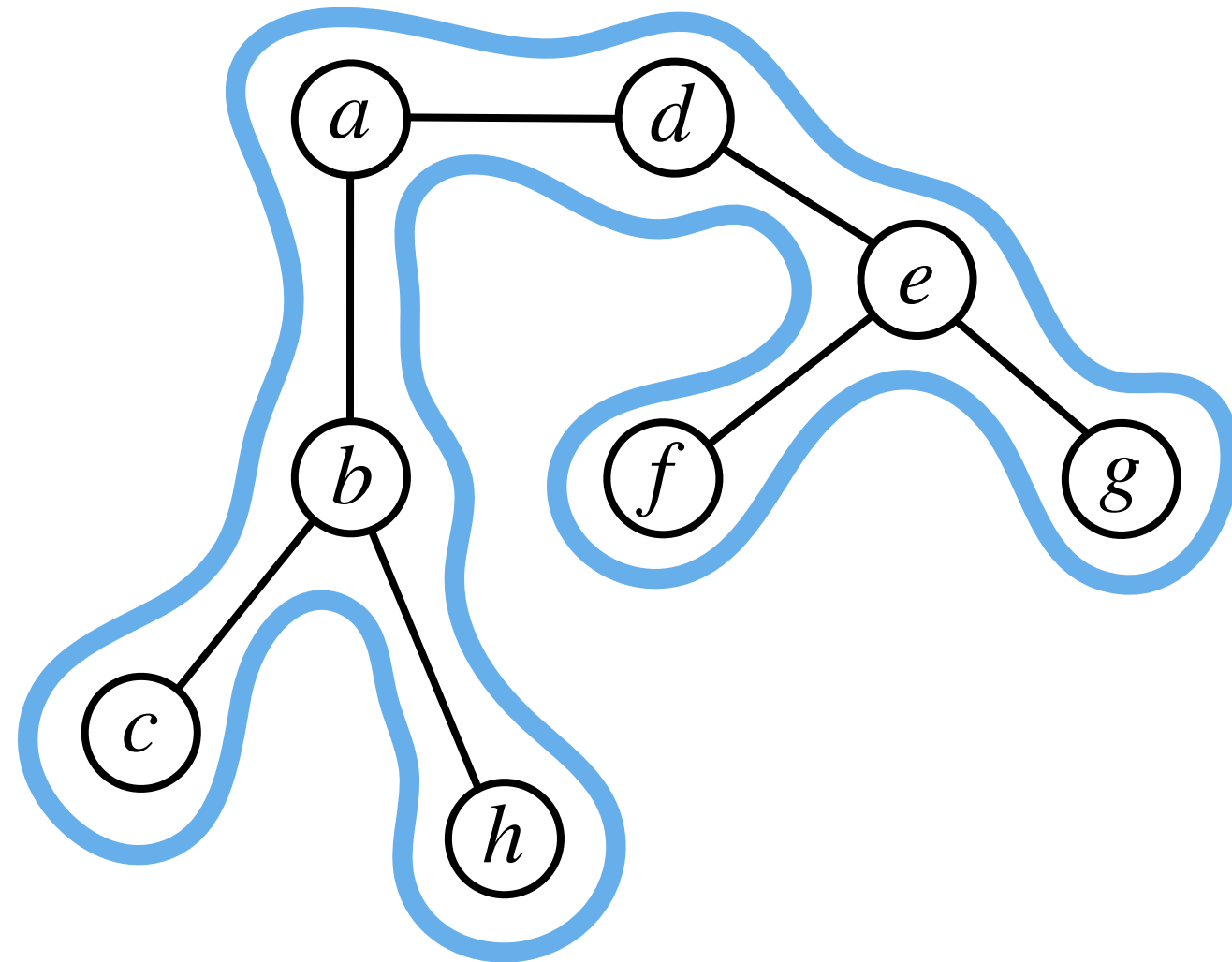


$$c(H^*) \geq c(T)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Consider the tour formed by walking around T .

Tour around MST T



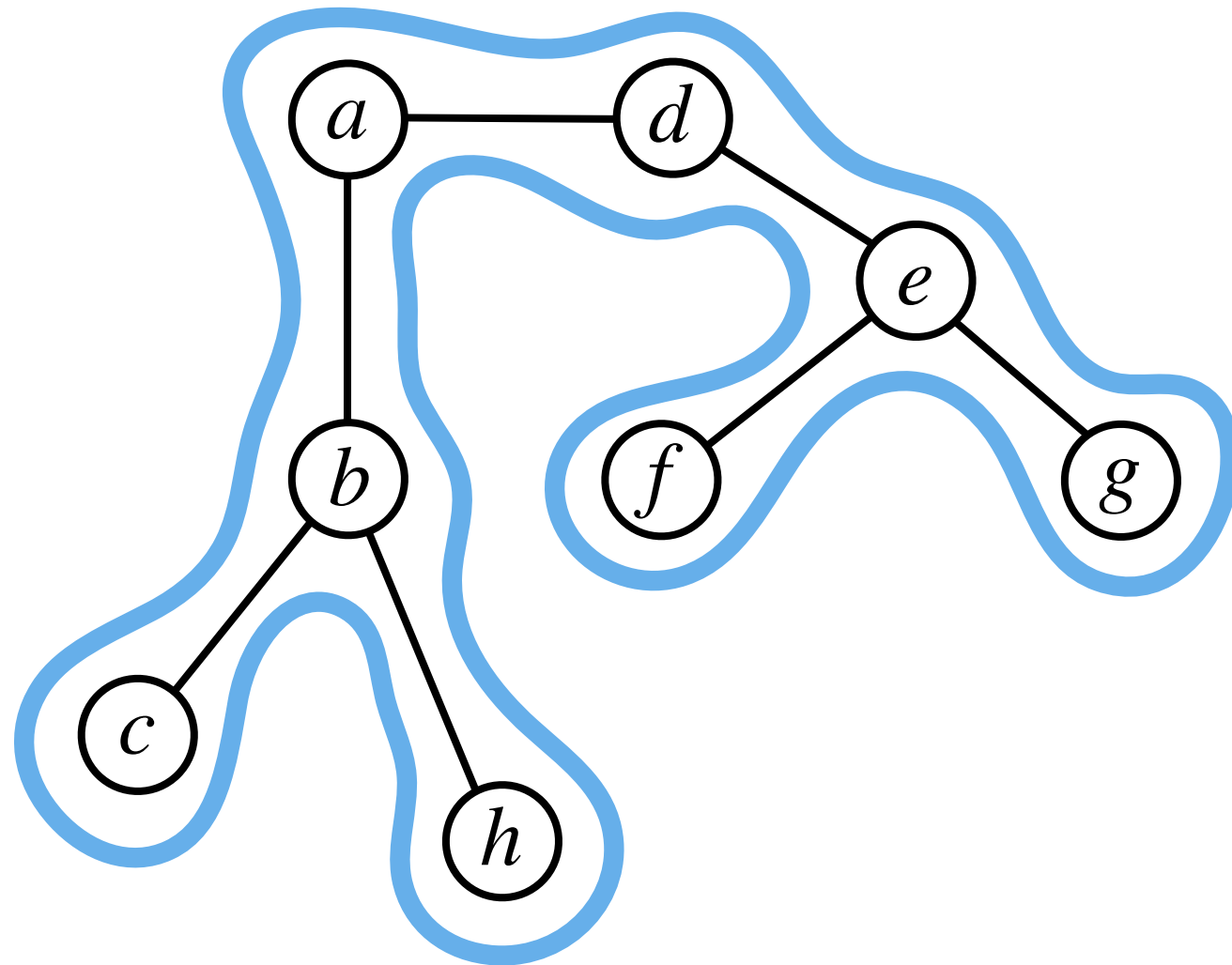
$$c(H^*) \geq c(T)$$

$$H = (a, b, c, b, h, b, a, d, e, f, e, g, e, d, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Consider the tour formed by walking around T .

Tour around MST T



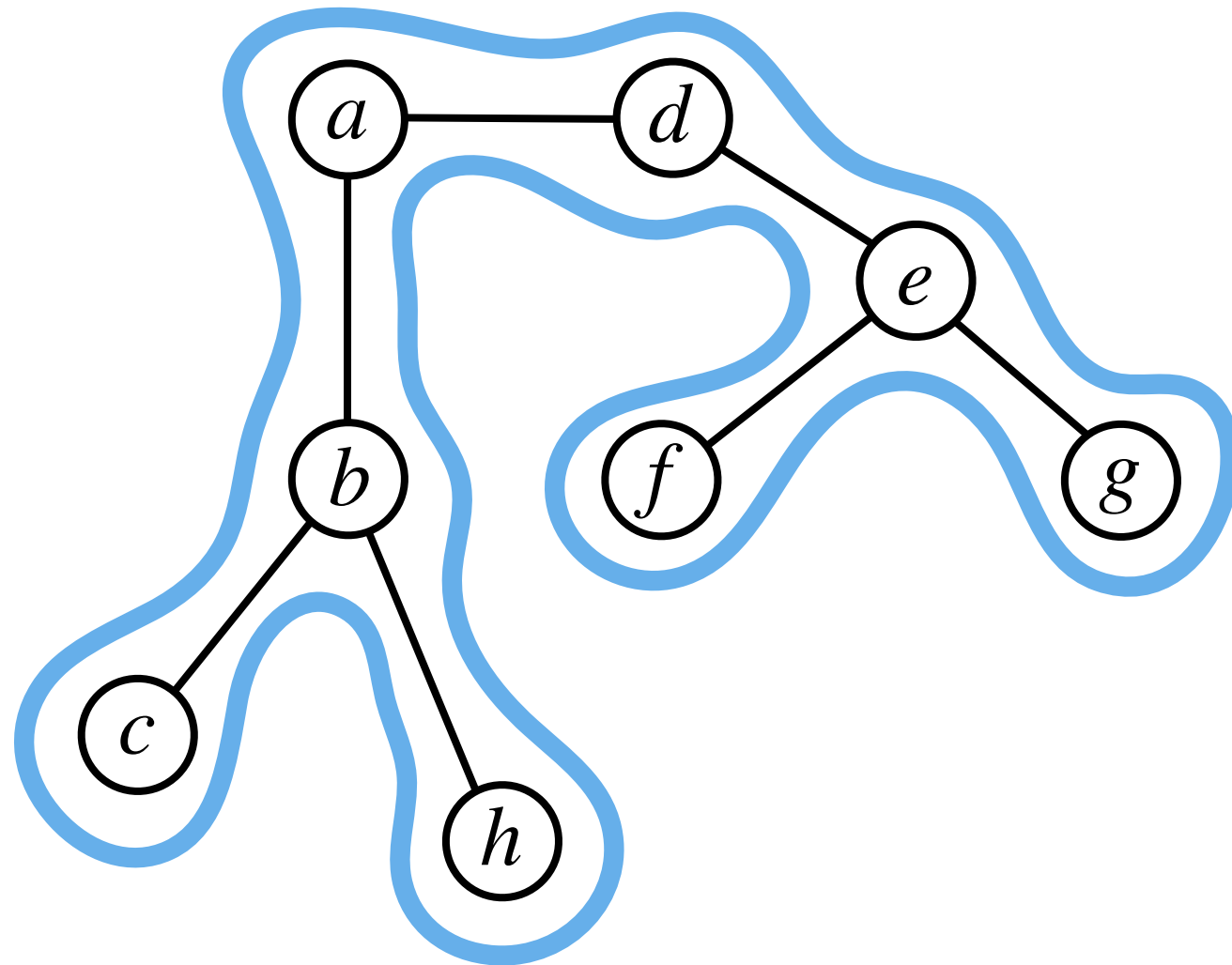
$$2.c(H^*) \geq 2.c(T)$$

$$H = (a, b, c, b, h, b, a, d, e, f, e, g, e, d, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Consider the tour formed by walking around T .

Tour around MST T



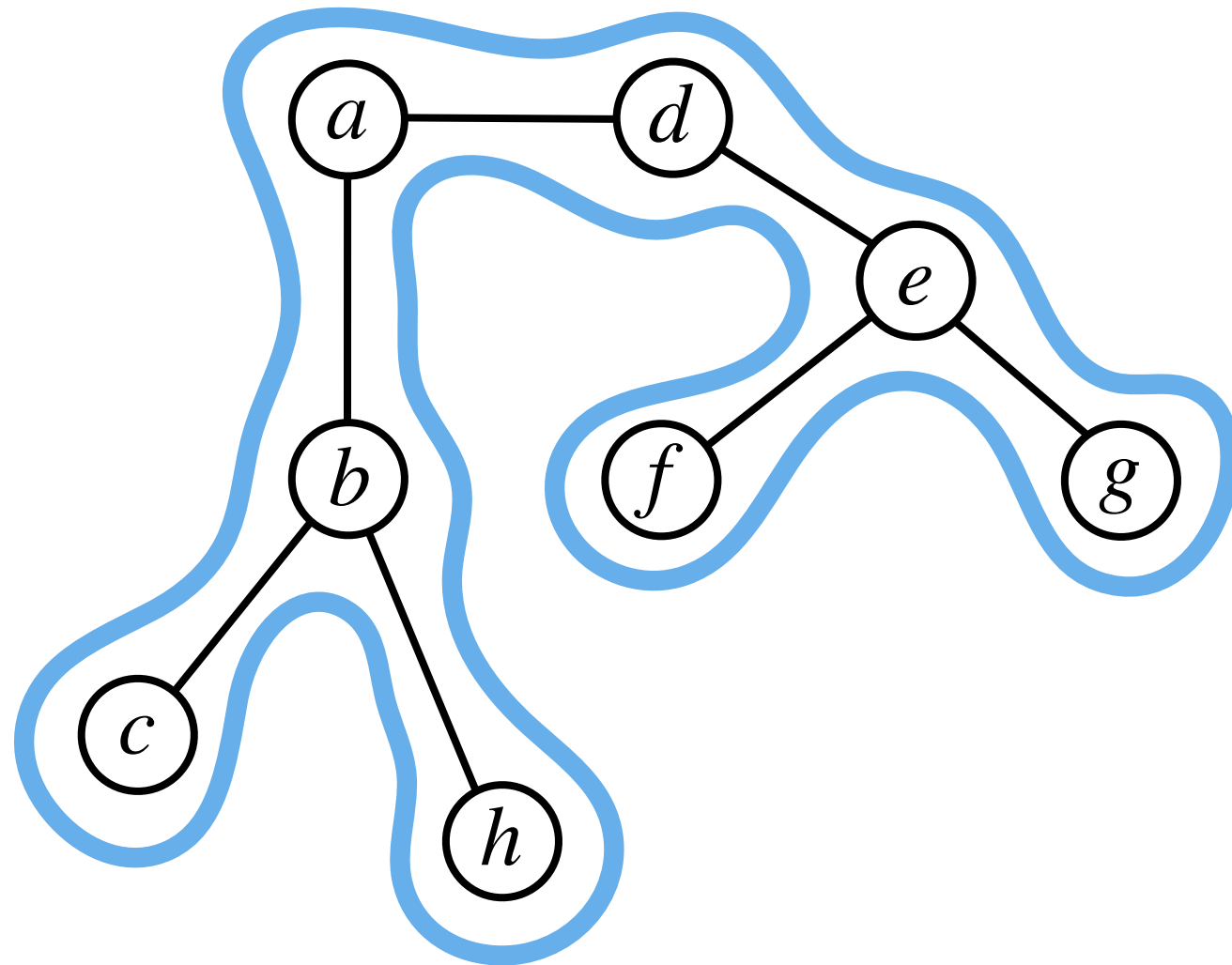
$$2 \cdot c(H^*) \geq 2 \cdot c(T) = c(H)$$

$$H = (a, b, c, b, h, b, a, d, e, f, e, g, e, d, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Consider the tour formed by walking around T .

Tour around MST T



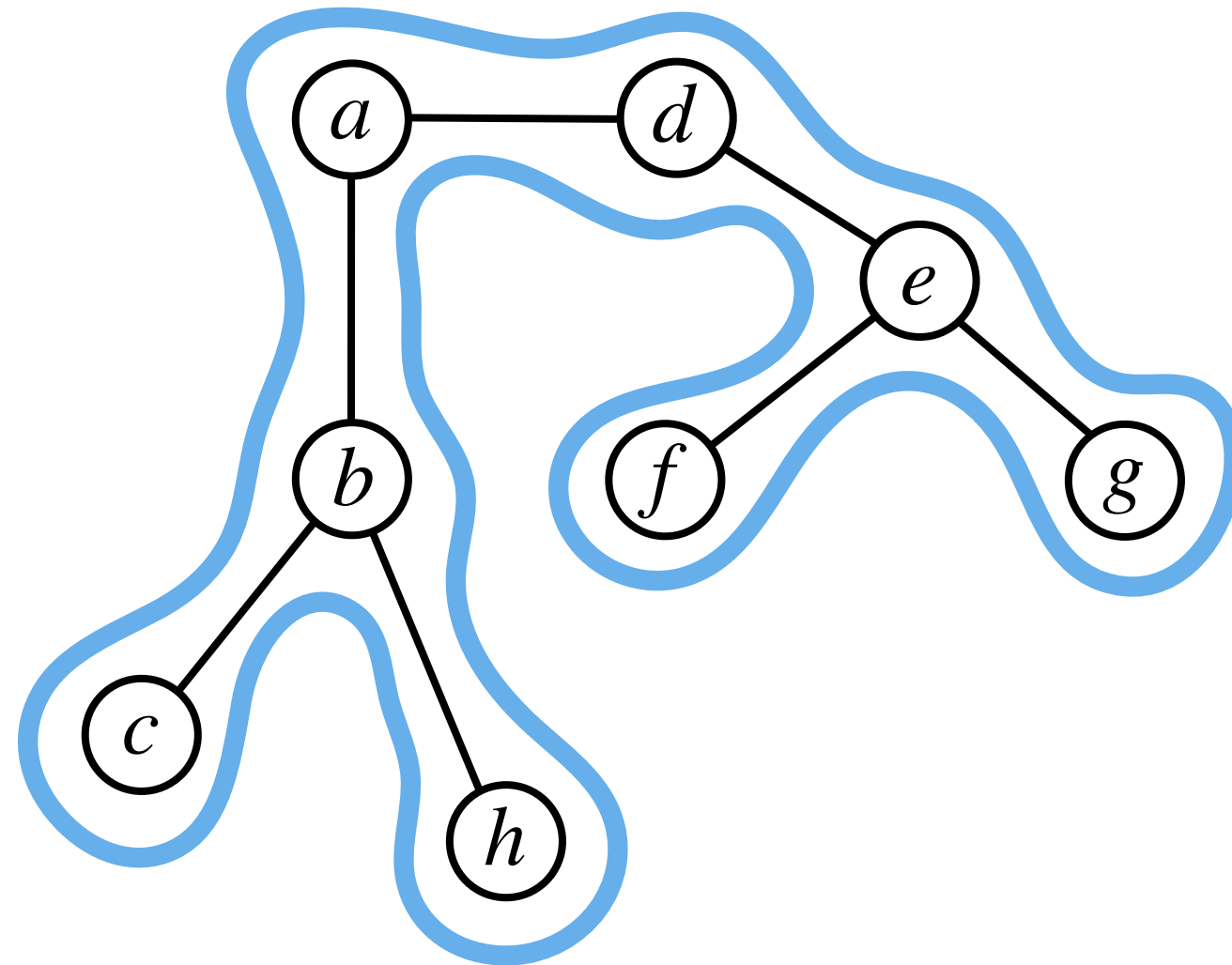
$$2 \cdot c(H^*) \geq c(H)$$

$$H = (a, b, c, b, h, b, a, d, e, f, e, g, e, d, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Consider the tour formed by walking around T .

Tour around MST T



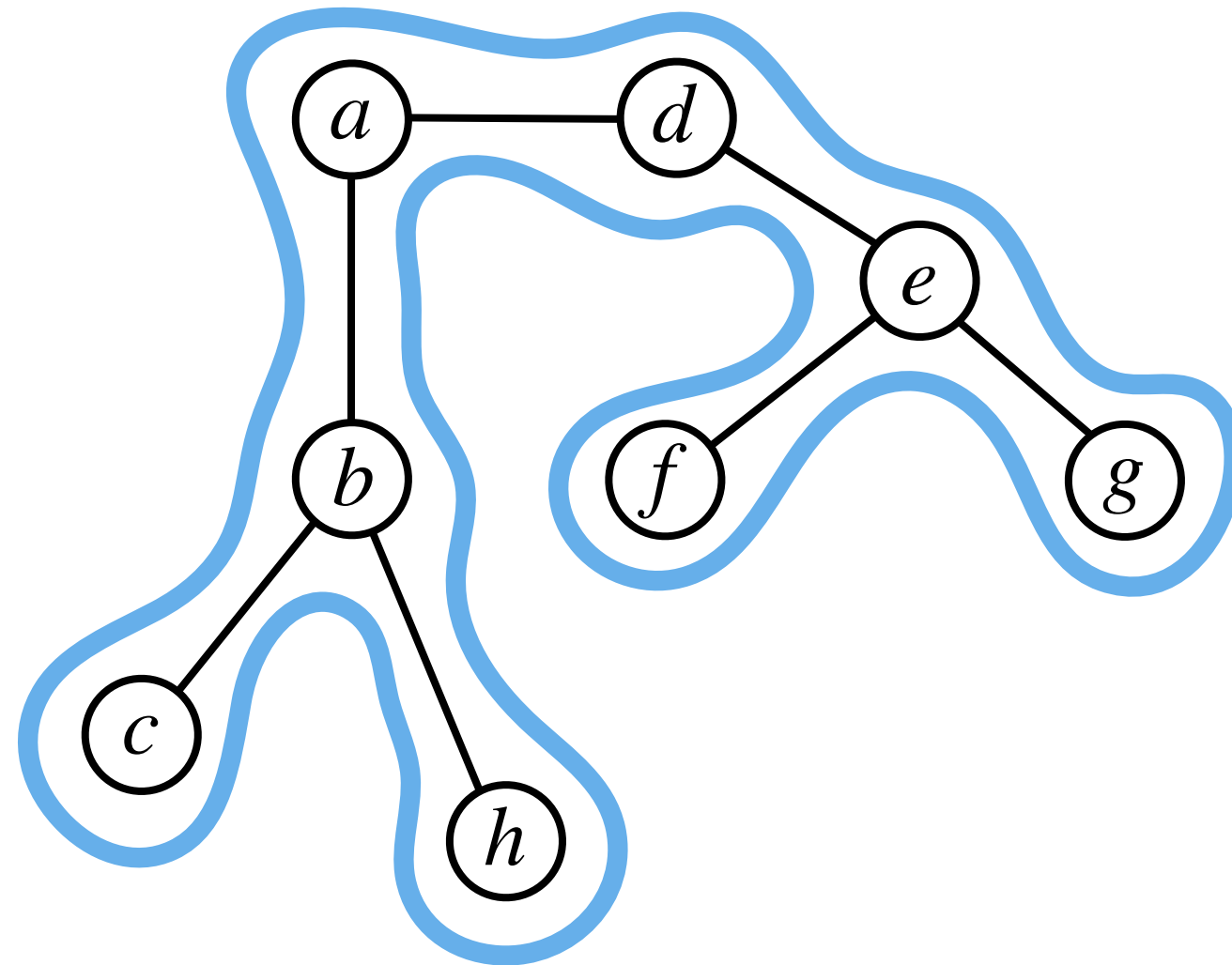
$$2 \cdot c(H^*) \geq c(H)$$

$H = (a, b, c, b, h, b, a, d, e, f, e, g, e, d, a)$ ← But H is not a TSP tour.

2-Approx. of $OptTSP$ Under Triangle Inequality

Consider the tour formed by walking around T .

Tour around MST T



$$2 \cdot c(H^*) \geq c(H)$$

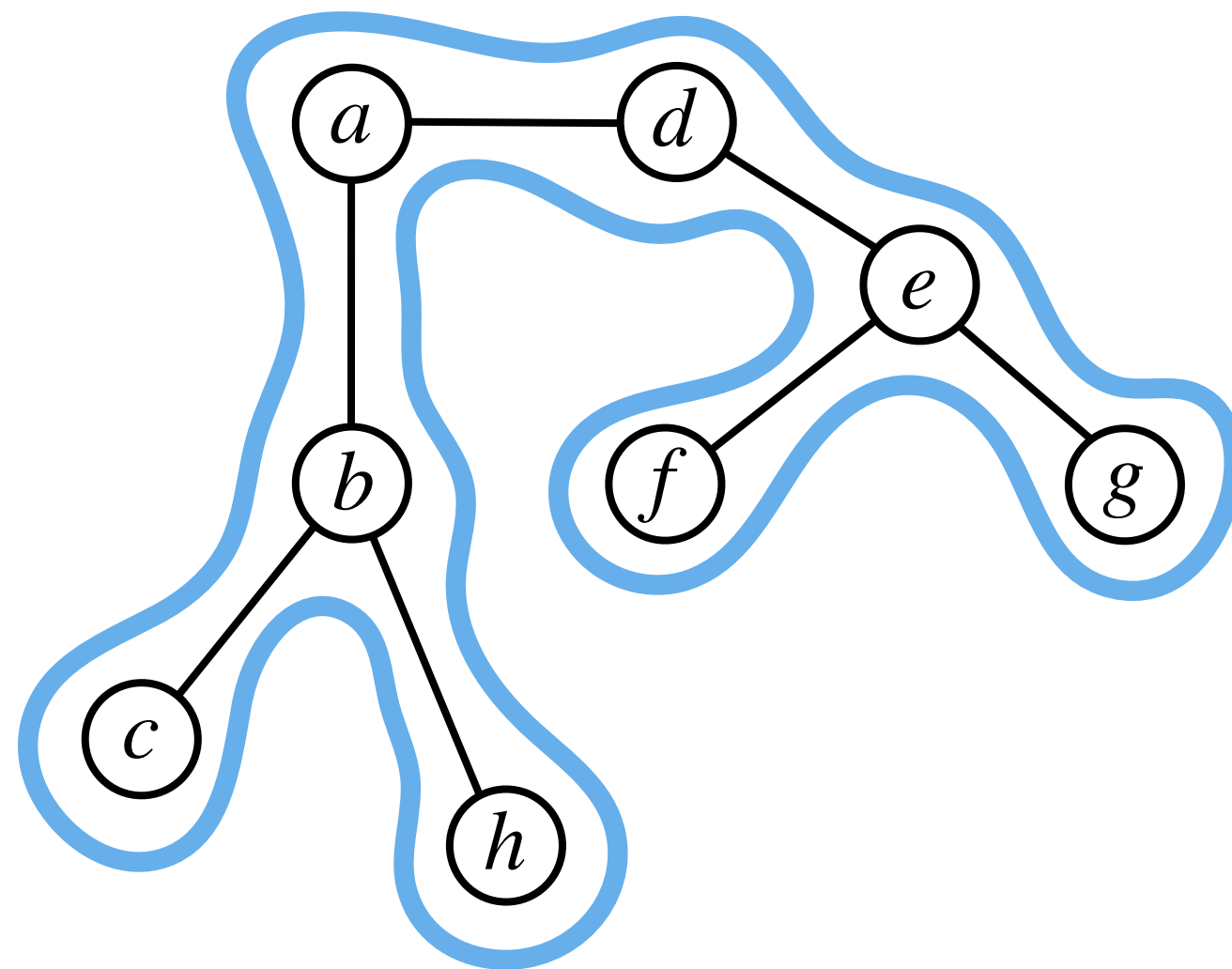
$H = (a, b, c, b, h, b, a, d, e, f, e, g, e, d, a)$ ← But H is not a TSP tour.

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



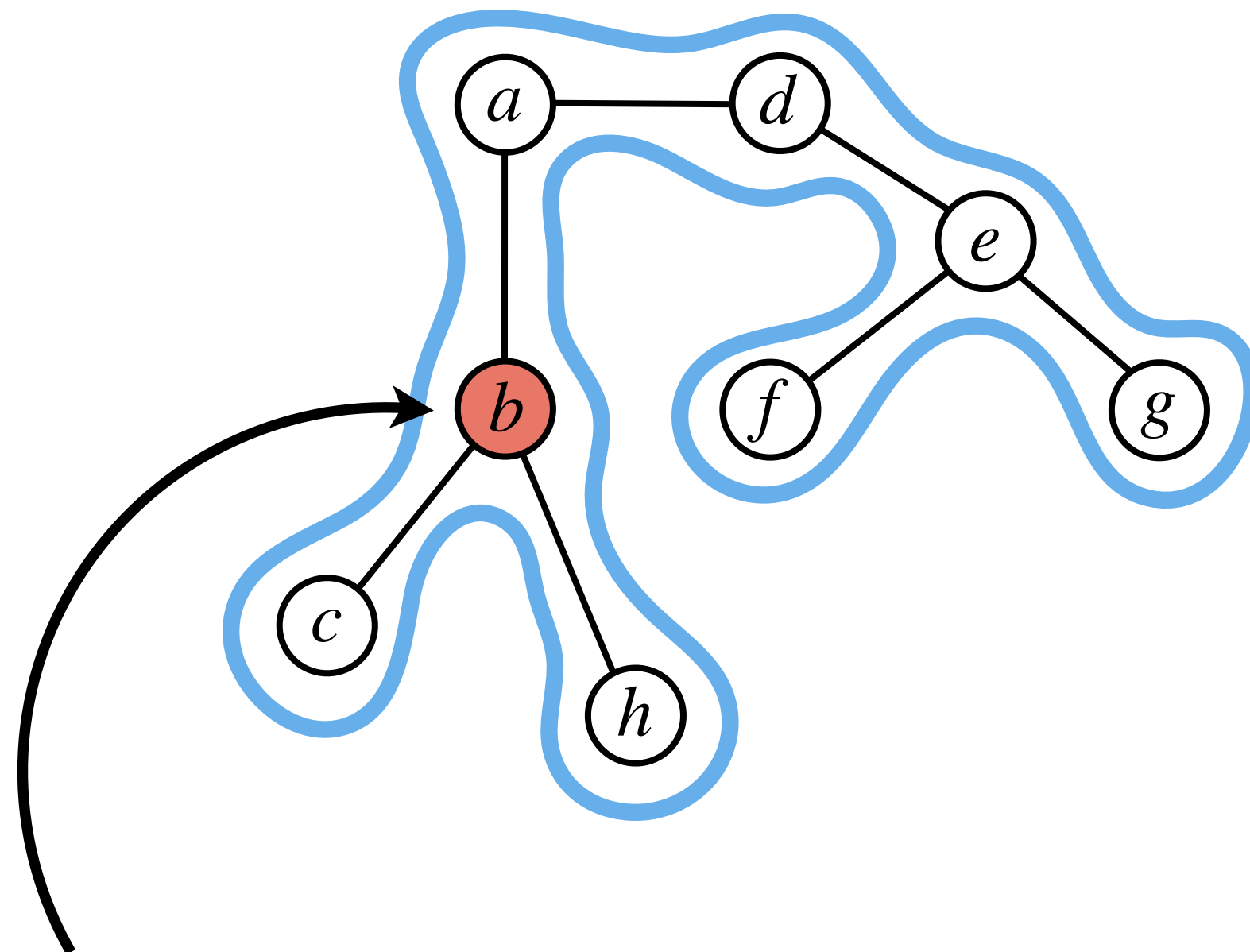
$$2 \cdot c(H^*) \geq c(H)$$

$$H = (a, b, c, b, h, b, a, d, e, f, e, g, e, d, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



b is getting repeated.

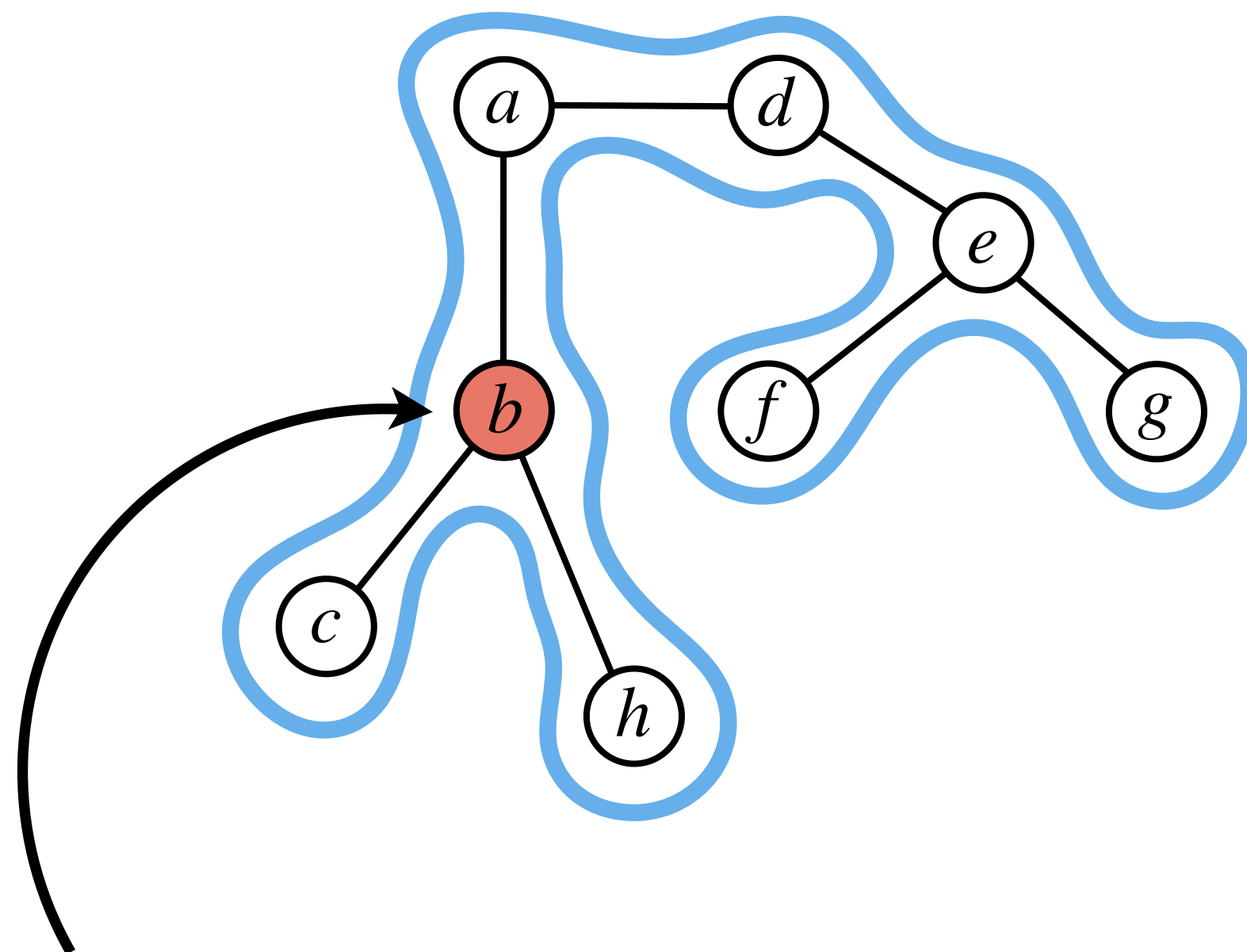
$$2 \cdot c(H^*) \geq c(H)$$

$$H = (a, b, c, b, h, b, a, d, e, f, e, g, e, d, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



$$2 \cdot c(H^*) \geq c(H)$$

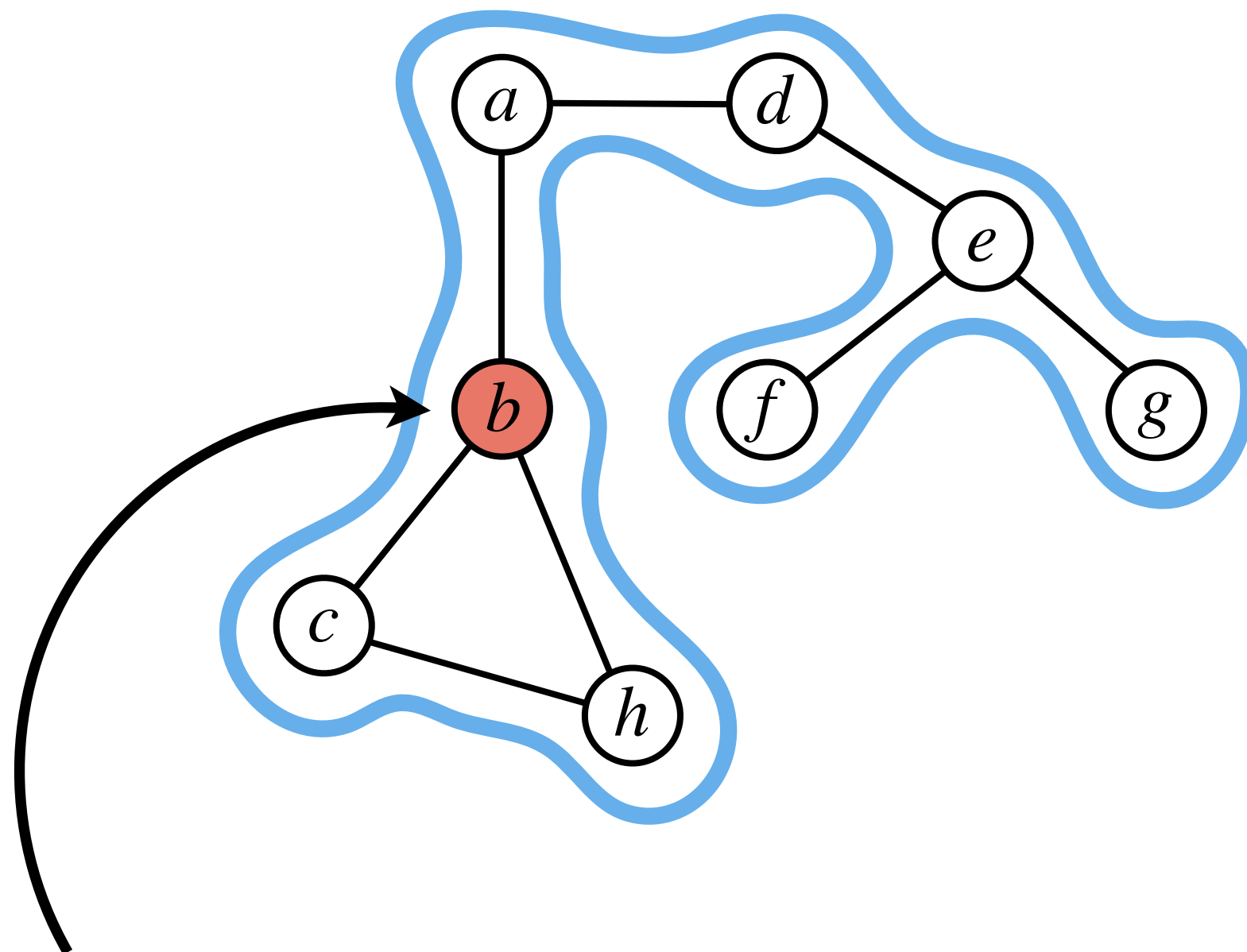
$$H = (a, b, c, b, h, b, a, d, e, f, e, g, e, d, a)$$

What will happen if we skip b and go directly to h via edge $\{c, h\}$?

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



$$2 \cdot c(H^*) \geq c(H)$$

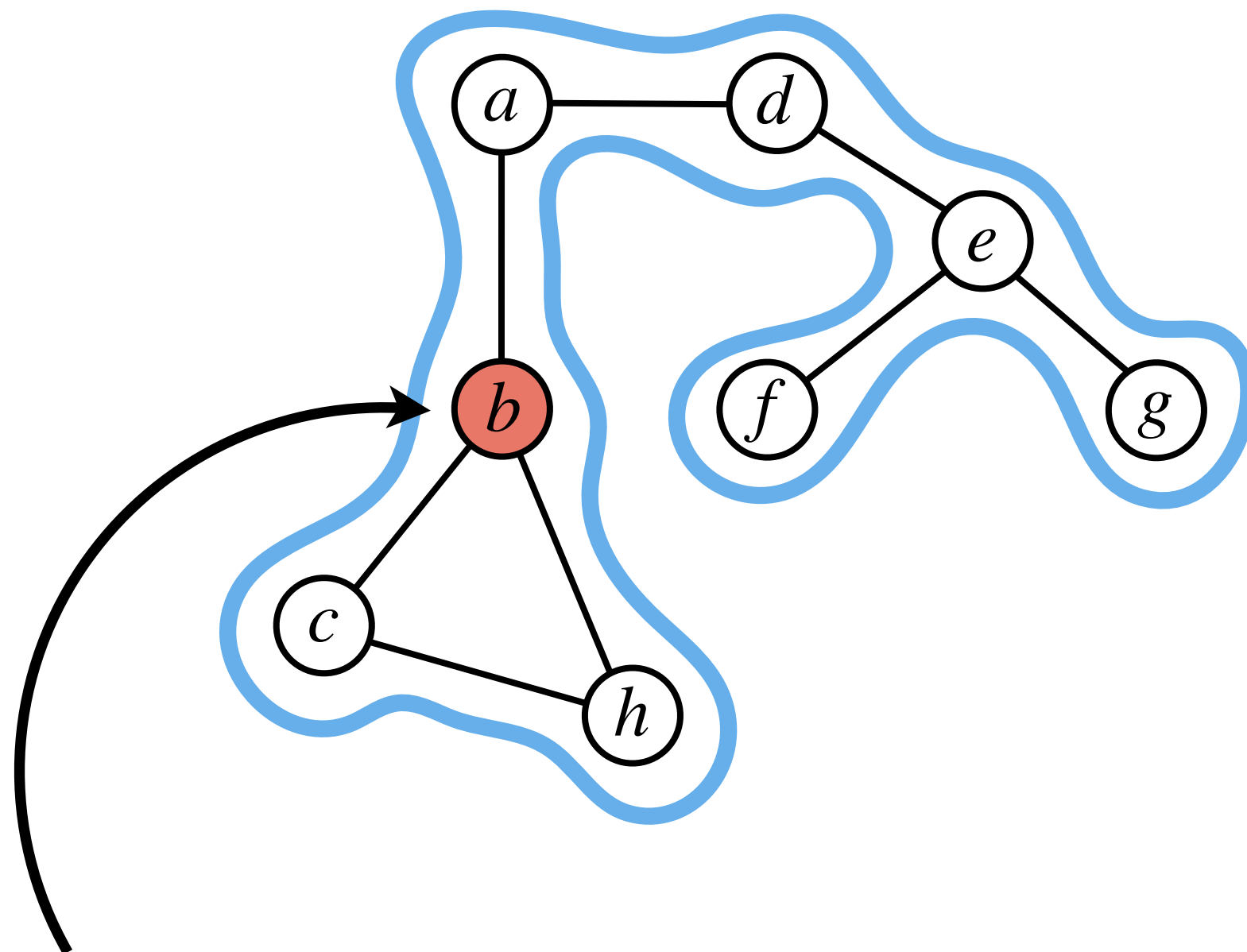
$$H = (a, b, c, b, h, b, a, d, e, f, e, g, e, d, a)$$

What will happen if we skip b and go directly to h via edge $\{c, h\}$?

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



$$2 \cdot c(H^*) \geq c(H)$$

$$H = (a, b, c, b, h, b, a, d, e, f, e, g, e, d, a)$$

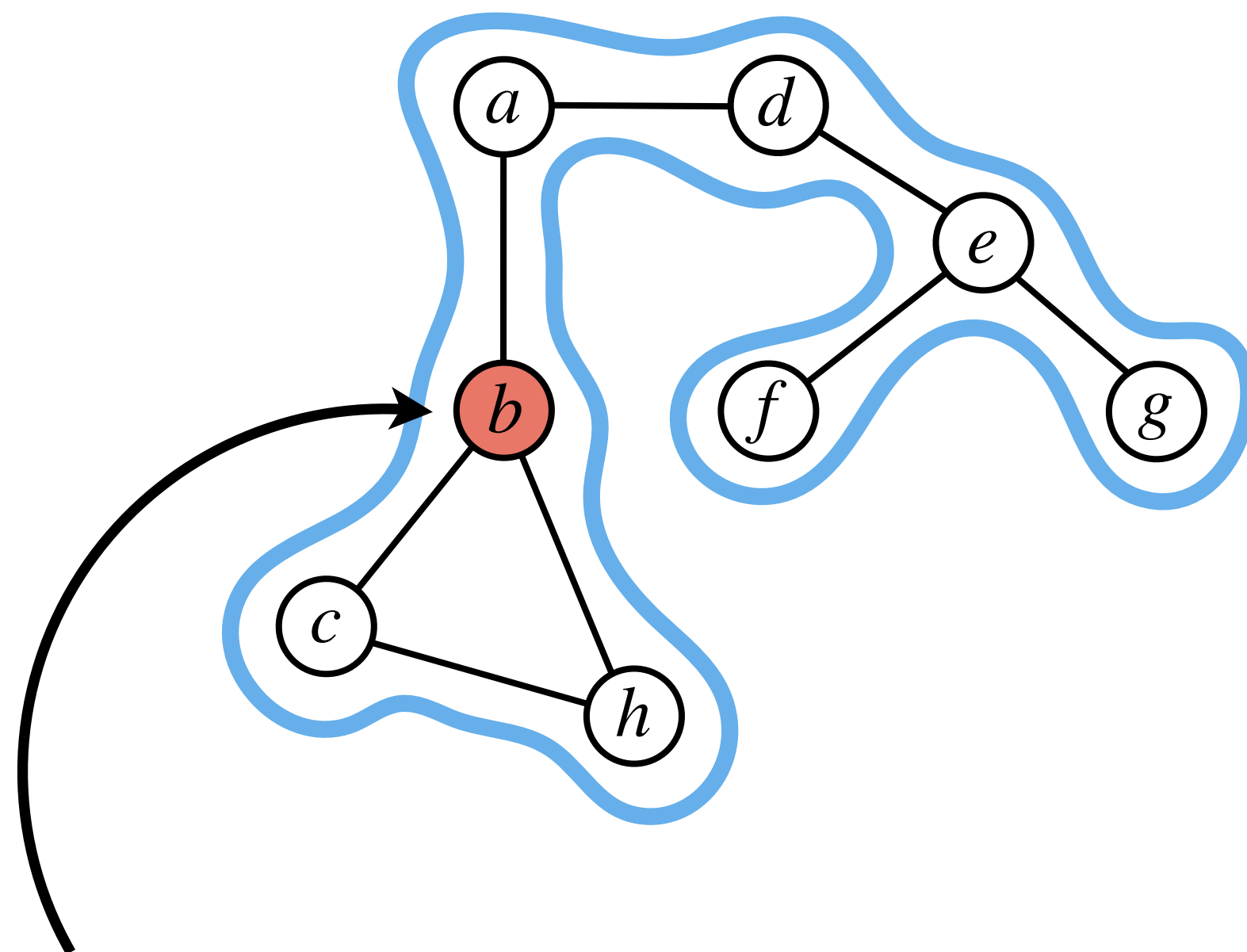
What will happen if we skip b and go directly to h via edge $\{c, h\}$?

Cost of tour will only decrease or be the same as $c(\{c, h\}) \leq c(\{c, b\}) + c(\{b, h\})$.

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



$$2 \cdot c(H^*) \geq c(H)$$

$$H = (a, b, c, \cancel{b}, h, b, a, d, e, f, e, g, e, d, a)$$

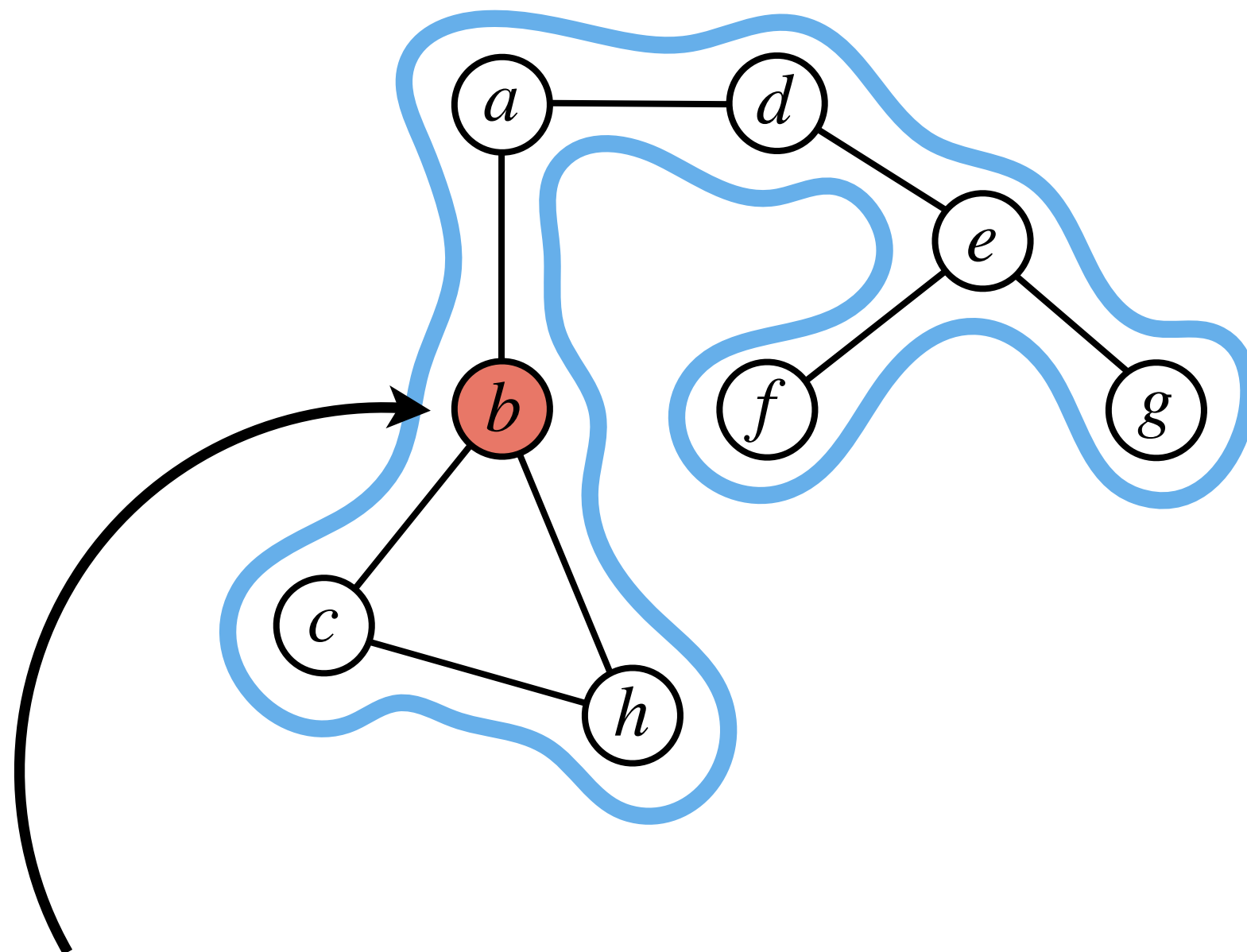
What will happen if we skip b and go directly to h via edge $\{c, h\}$?

Cost of tour will only decrease or be the same as $c(\{c, h\}) \leq c(\{c, b\}) + c(\{b, h\})$.

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



b is getting repeated again after h .

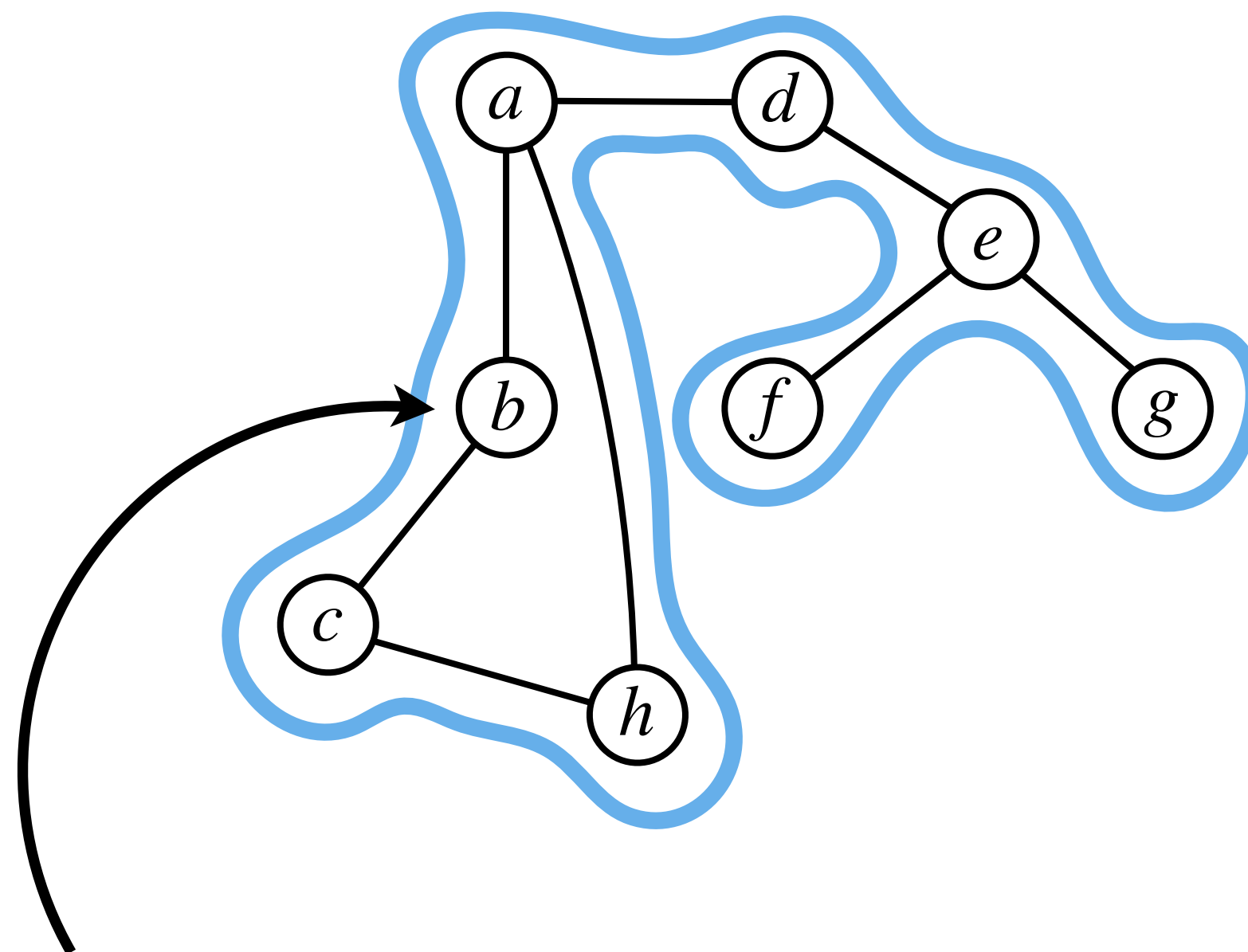
$$2 \cdot c(H^*) \geq c(H)$$

$$H = (a, b, c, \cancel{b}, h, b, a, d, e, f, e, g, e, d, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a TSP tour so that every change makes the cost **smaller or same**.

Tour around MST T



Skip b and go directly to a via edge $\{h, a\}$

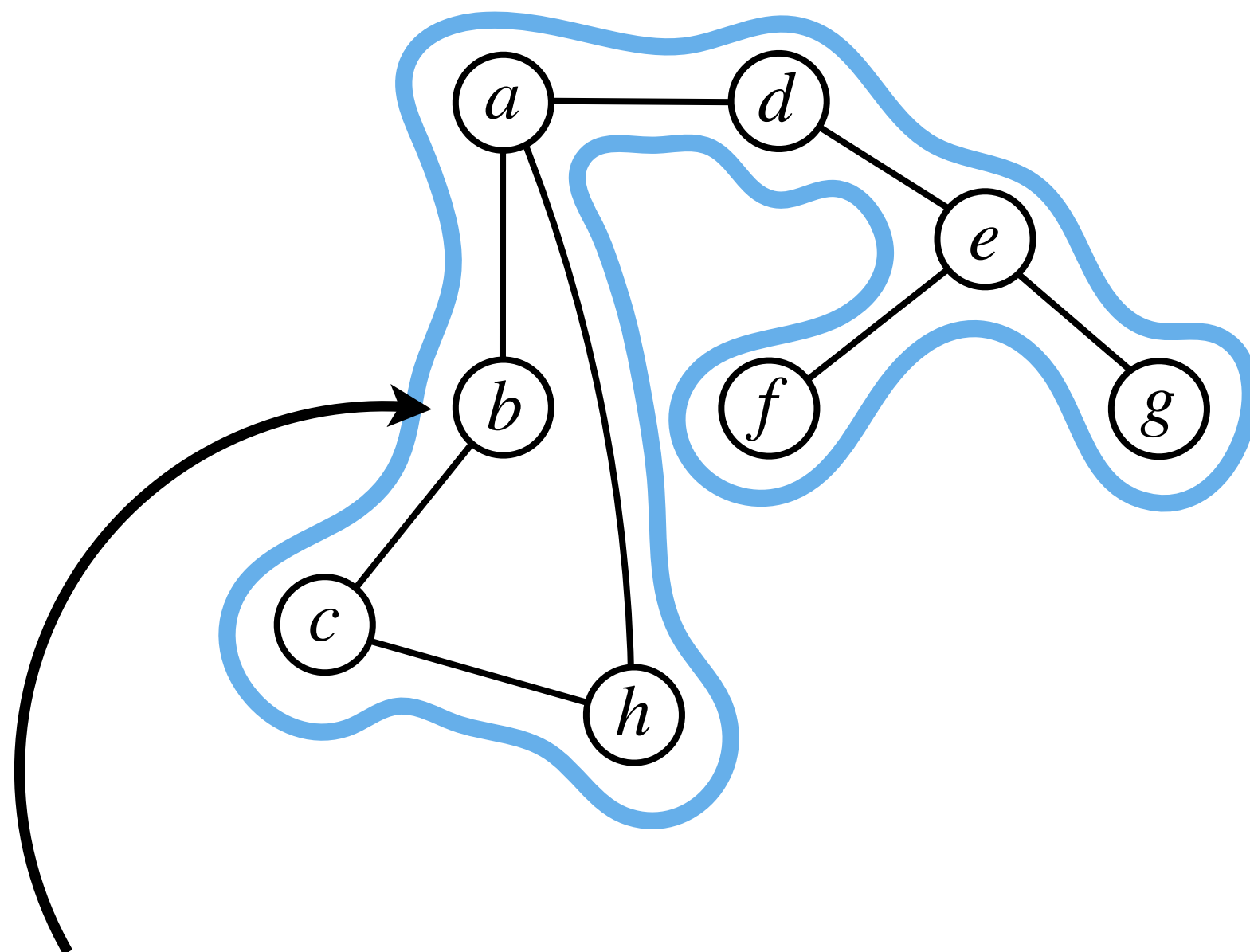
$$2 \cdot c(H^*) \geq c(H)$$

$$H = (a, b, c, \cancel{b}, h, b, a, d, e, f, e, g, e, d, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a TSP tour so that every change makes the cost **smaller or same**.

Tour around MST T



Skip b and go directly to a via edge $\{h, a\}$

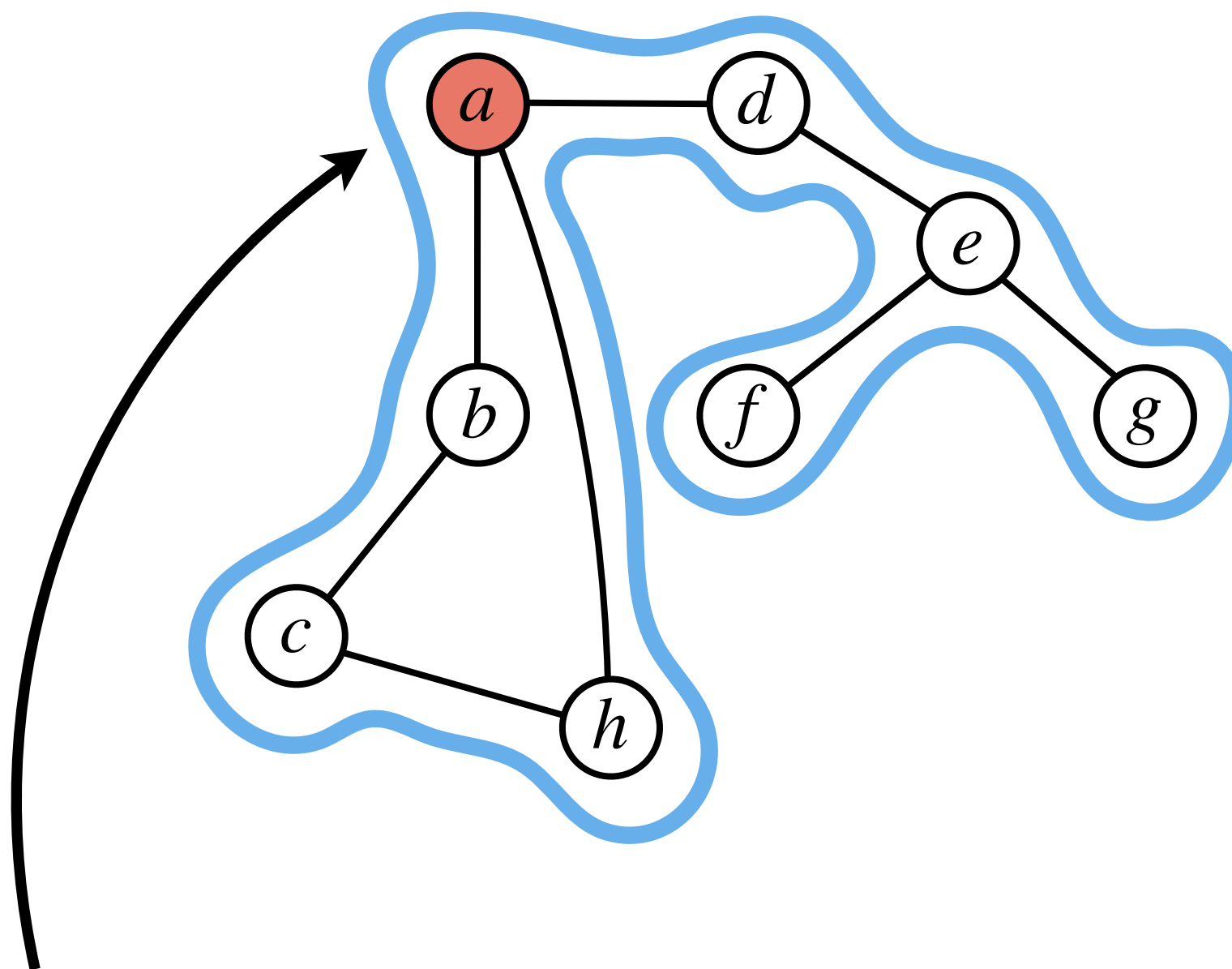
$$2 \cdot c(H^*) \geq c(H)$$

$$H = (a, b, c, \cancel{b}, h, \cancel{b}, a, d, e, f, e, g, e, d, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



a is getting repeated.

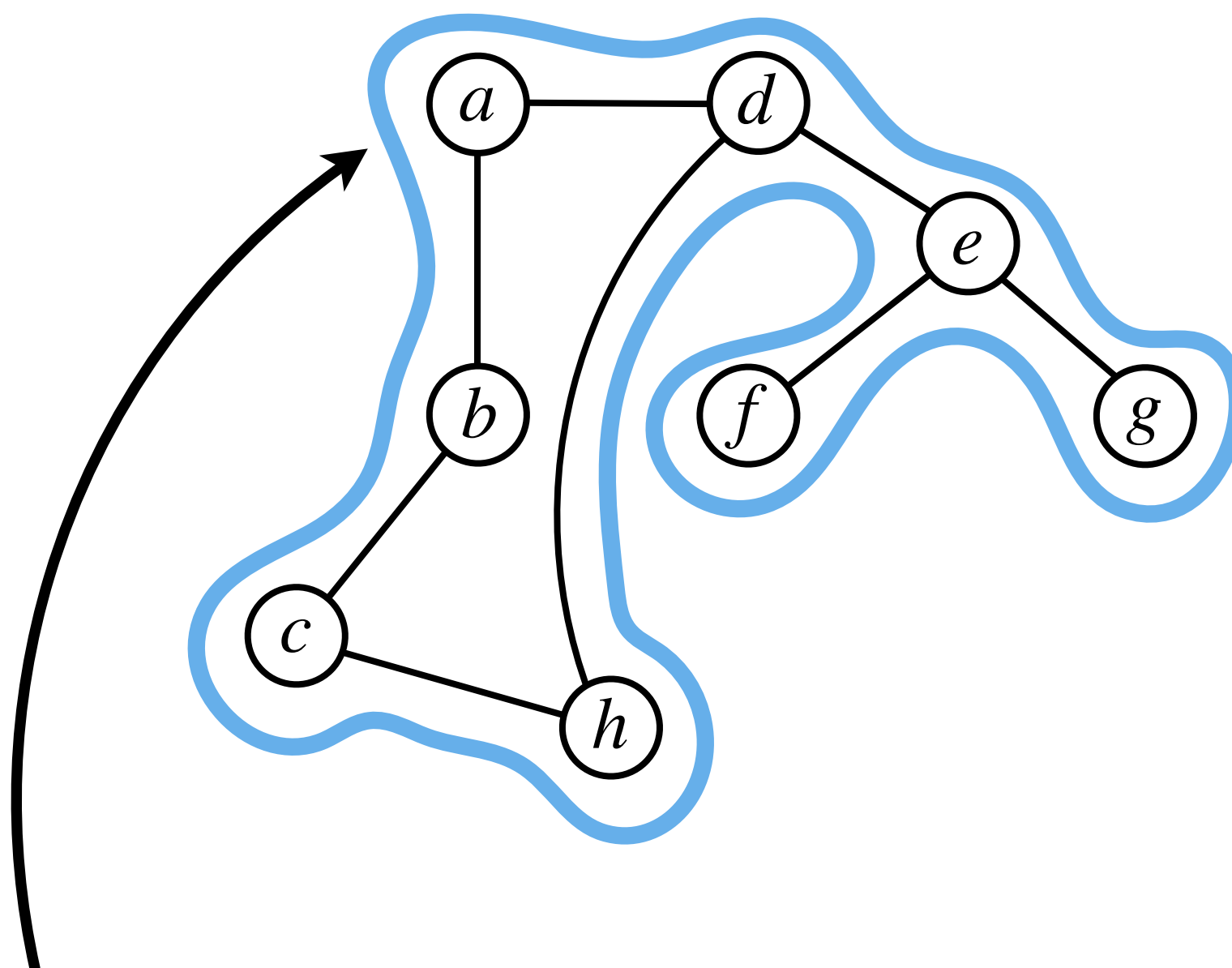
$$2 \cdot c(H^*) \geq c(H)$$

$$H = (a, b, c, \cancel{b}, h, \cancel{b}, a, d, e, f, e, g, e, d, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



Skip a and go directly to d via edge $\{h, d\}$

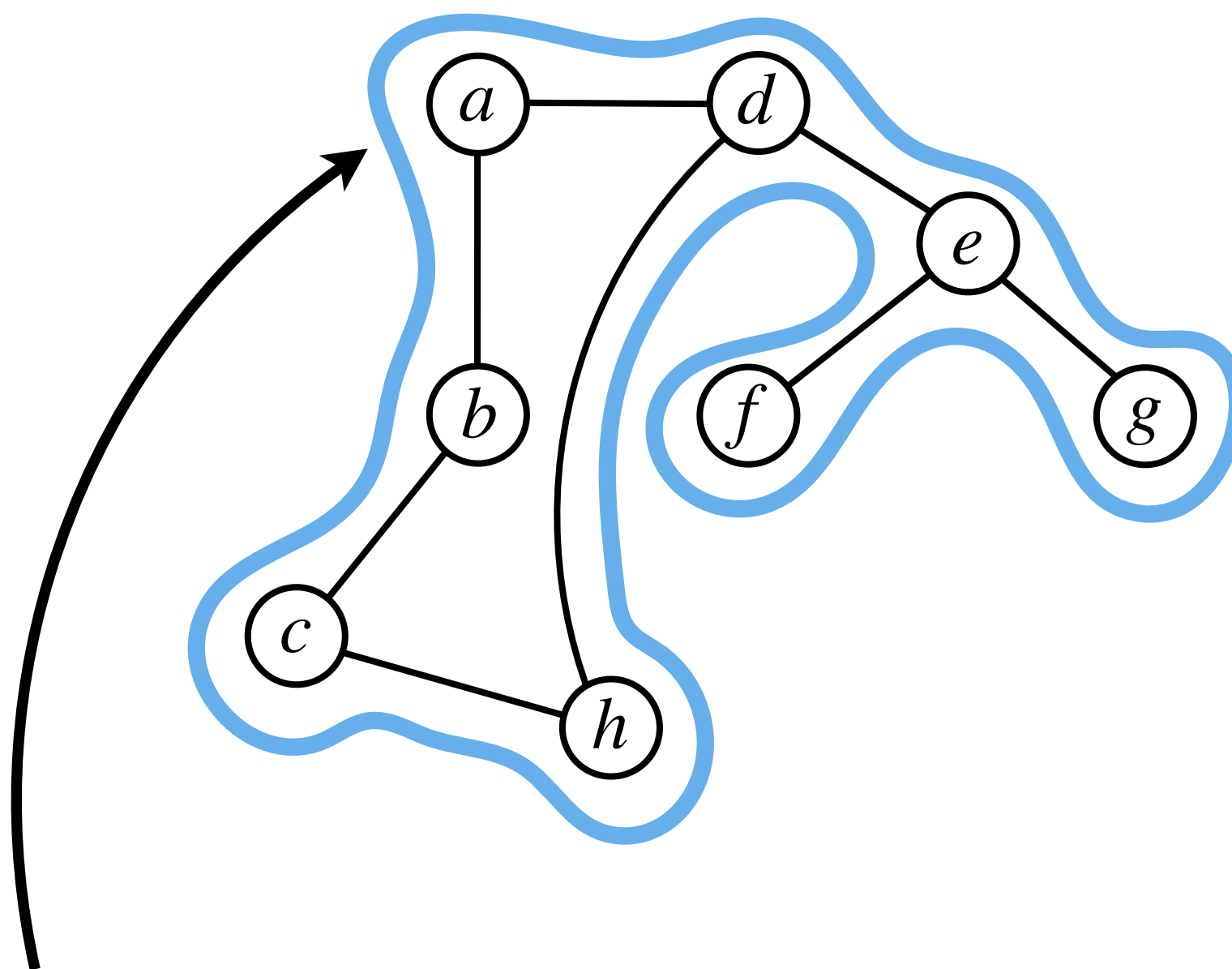
$$2 \cdot c(H^*) \geq c(H)$$

$$H = (a, b, c, \cancel{b}, h, \cancel{b}, a, d, e, f, e, g, e, d, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



Skip a and go directly to d via edge $\{h, d\}$

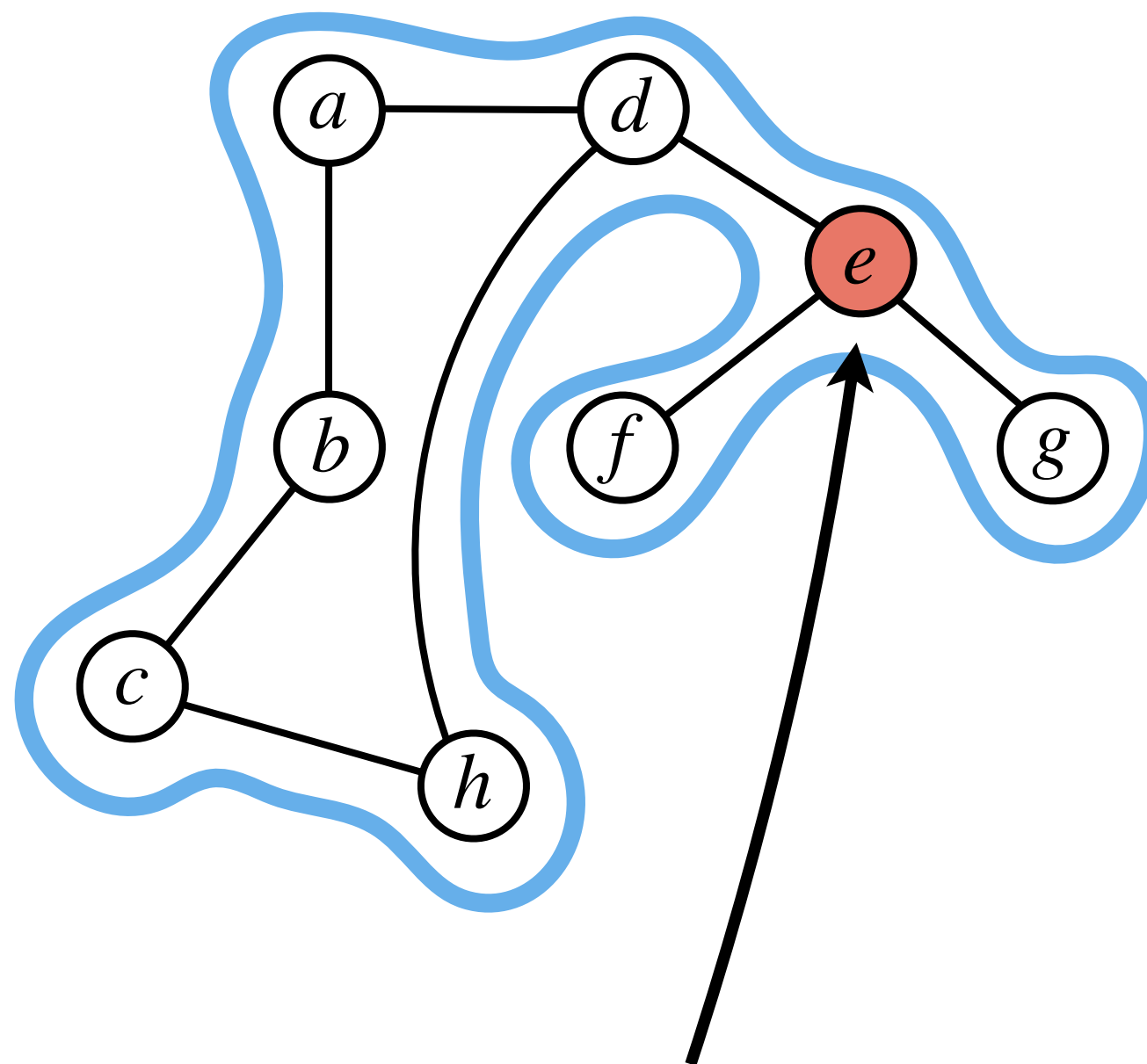
$$2 \cdot c(H^*) \geq c(H)$$

$$H = (a, b, c, \cancel{b}, h, \cancel{b}, \cancel{d}, d, e, f, e, g, e, d, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



e is getting repeated.

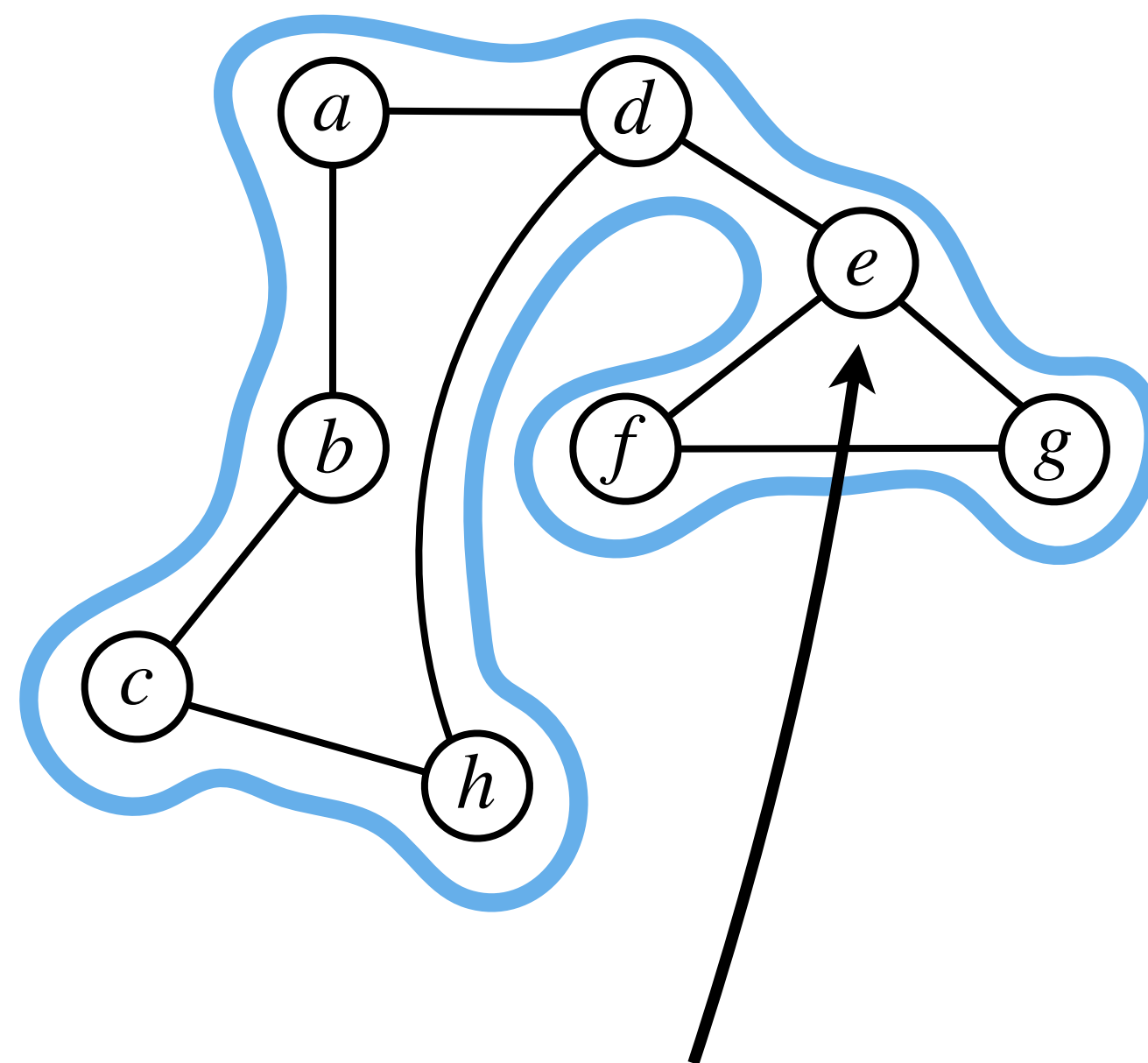
$$2 \cdot c(H^*) \geq c(H)$$

$$H = (a, b, c, \cancel{b}, h, \cancel{b}, \cancel{d}, d, e, f, e, g, e, d, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



$$2 \cdot c(H^*) \geq c(H)$$

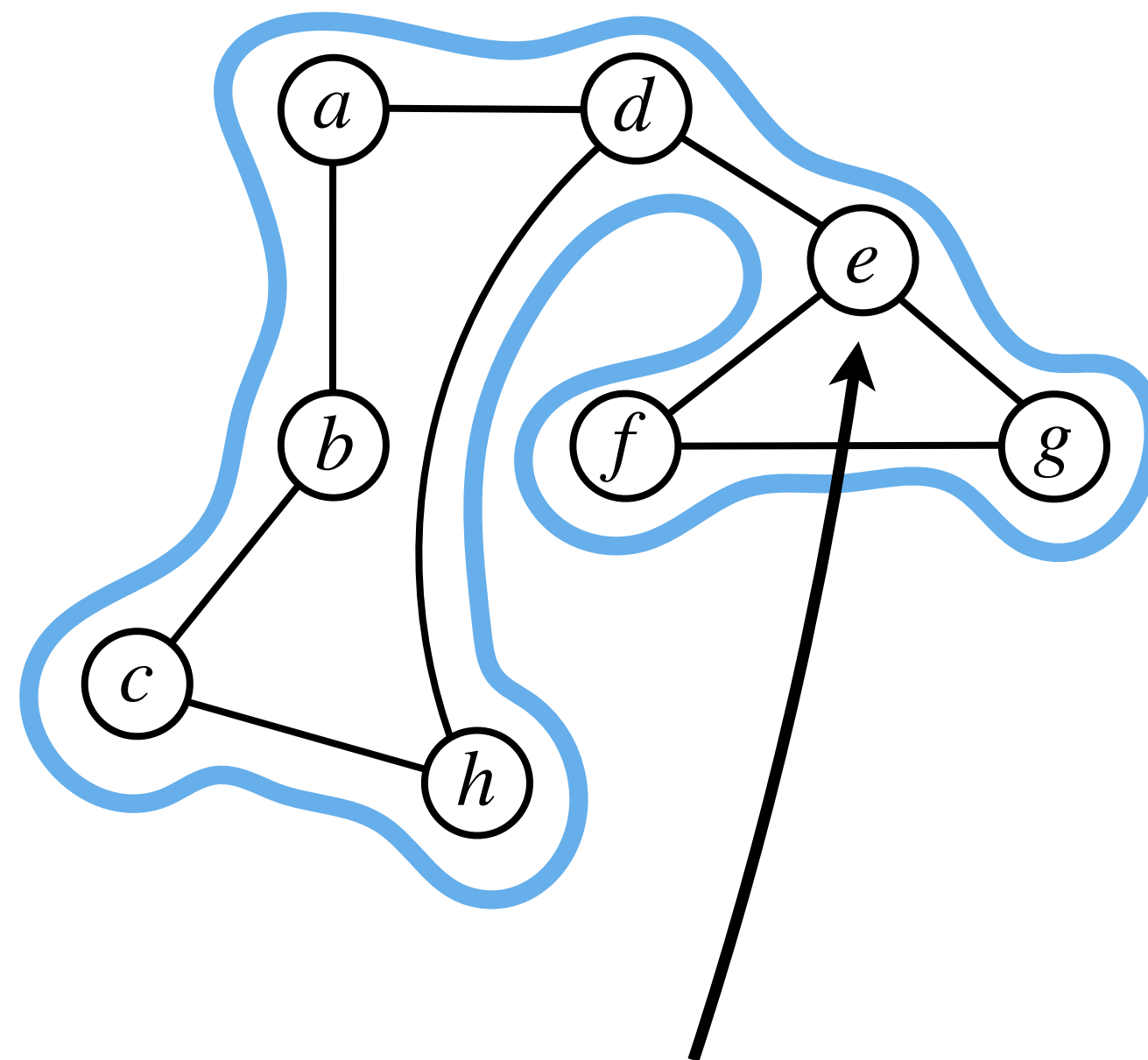
$$H = (a, b, c, \cancel{b}, h, \cancel{b}, \cancel{d}, d, e, f, e, g, e, d, a)$$

Skip e and go directly to g via edge $\{f, g\}$.

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



$$2 \cdot c(H^*) \geq c(H)$$

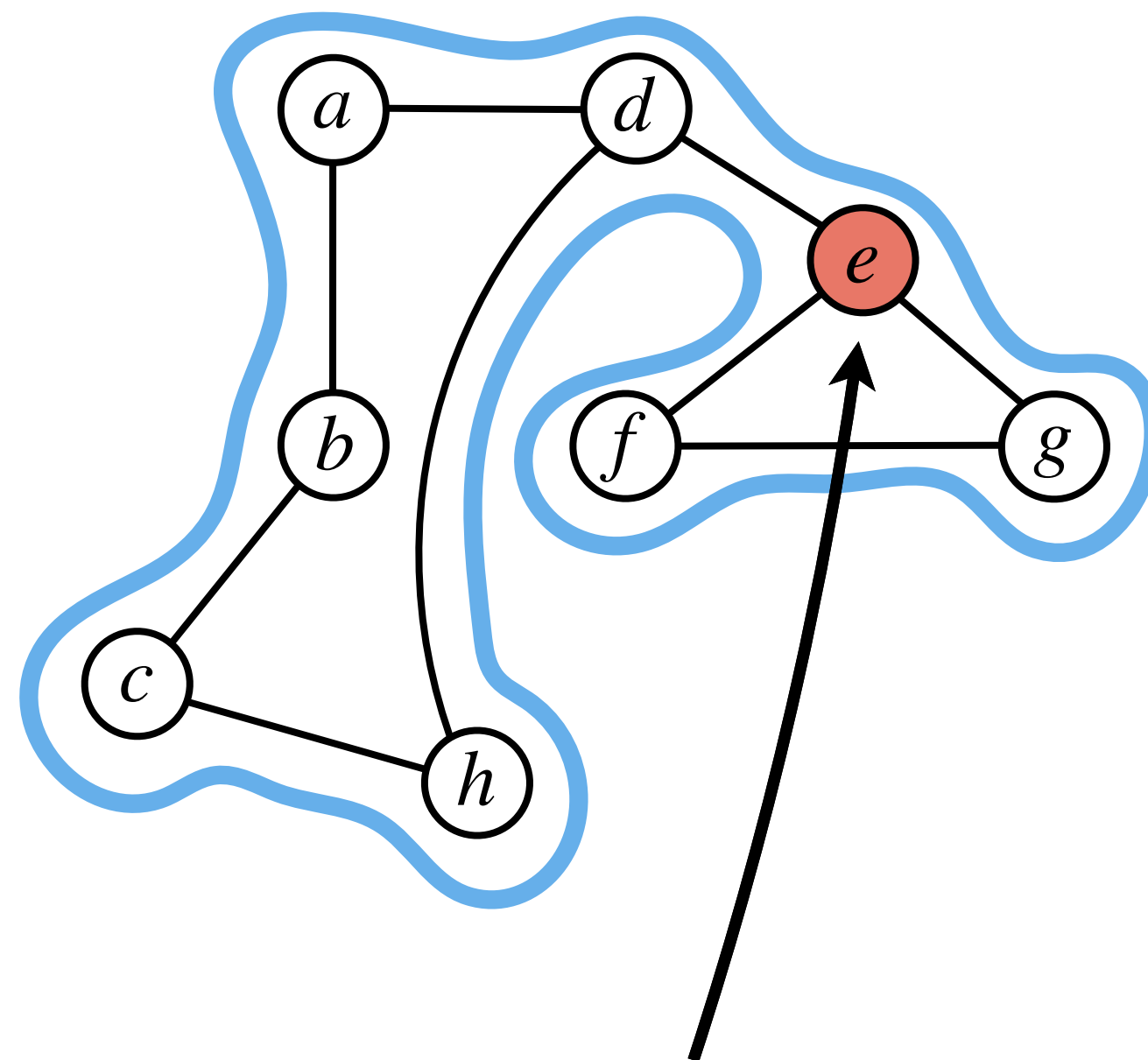
$$H = (a, b, c, \cancel{b}, h, \cancel{b}, \cancel{d}, d, e, f, \cancel{e}, g, e, d, a)$$

Skip e and go directly to g via edge $\{f, g\}$.

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



e is getting repeated again after g .

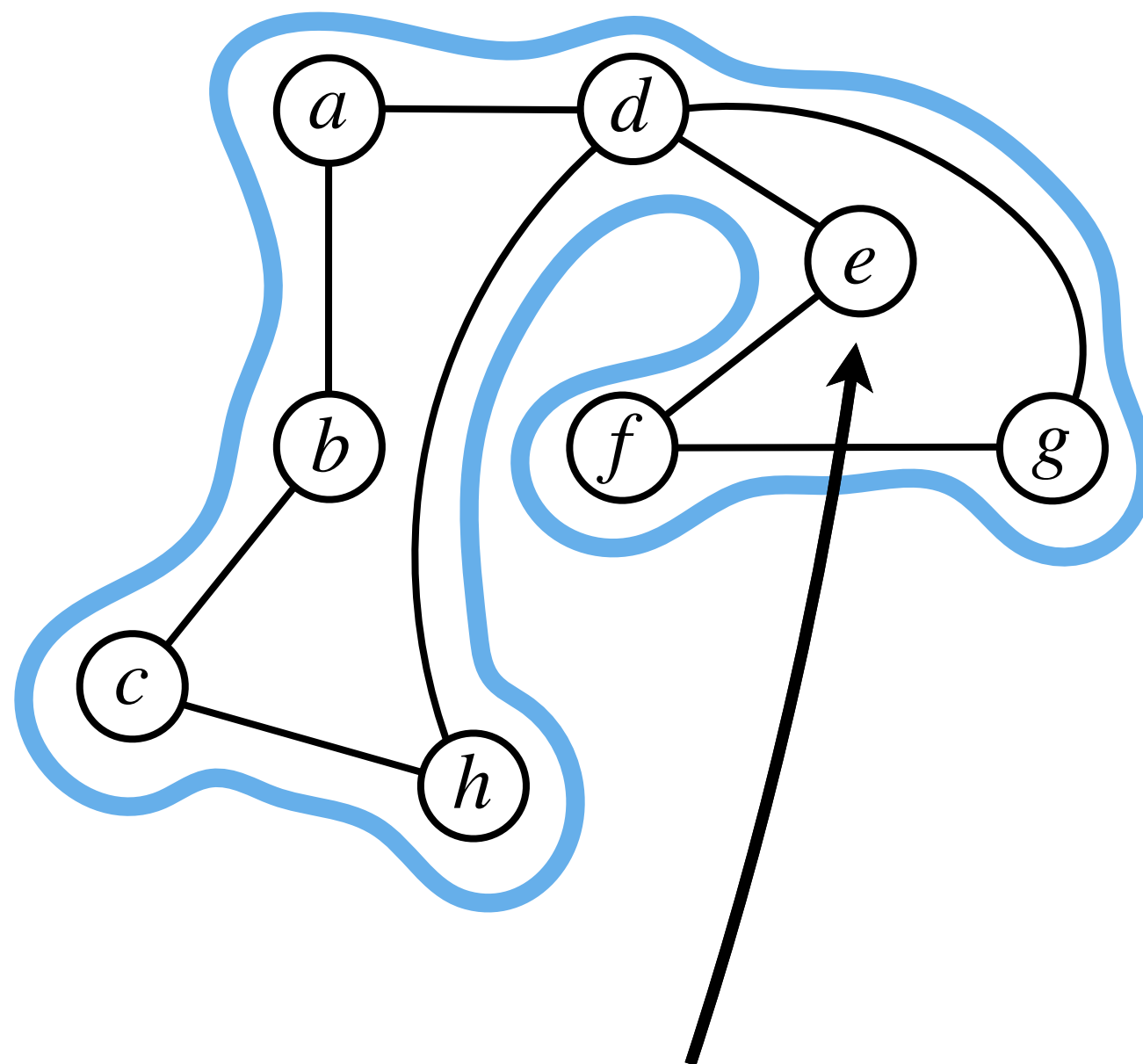
$$2 \cdot c(H^*) \geq c(H)$$

$$H = (a, b, c, \cancel{b}, h, \cancel{b}, \cancel{d}, d, e, f, \cancel{e}, g, e, d, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a TSP tour so that every change makes the cost smaller or same.

Tour around MST T



Skip e and go directly to d via edge $\{g, d\}$.

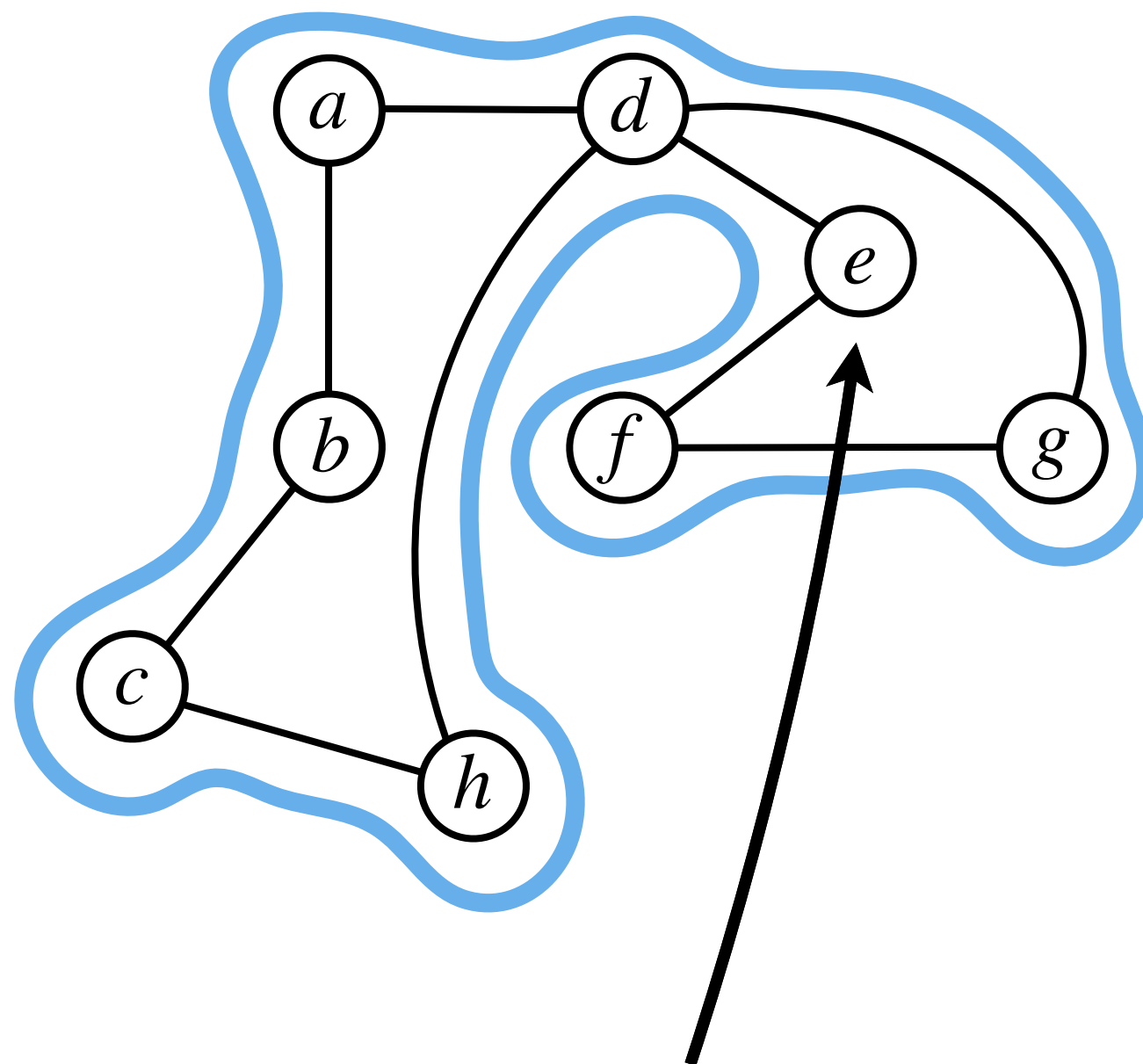
$$2.c(H^*) \geq c(H)$$

$$H = (a, b, c, \cancel{b}, h, \cancel{b}, d, d, e, \cancel{f}, g, e, d, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a TSP tour so that every change makes the cost **smaller or same**.

Tour around MST T



$$2 \cdot c(H^*) \geq c(H)$$

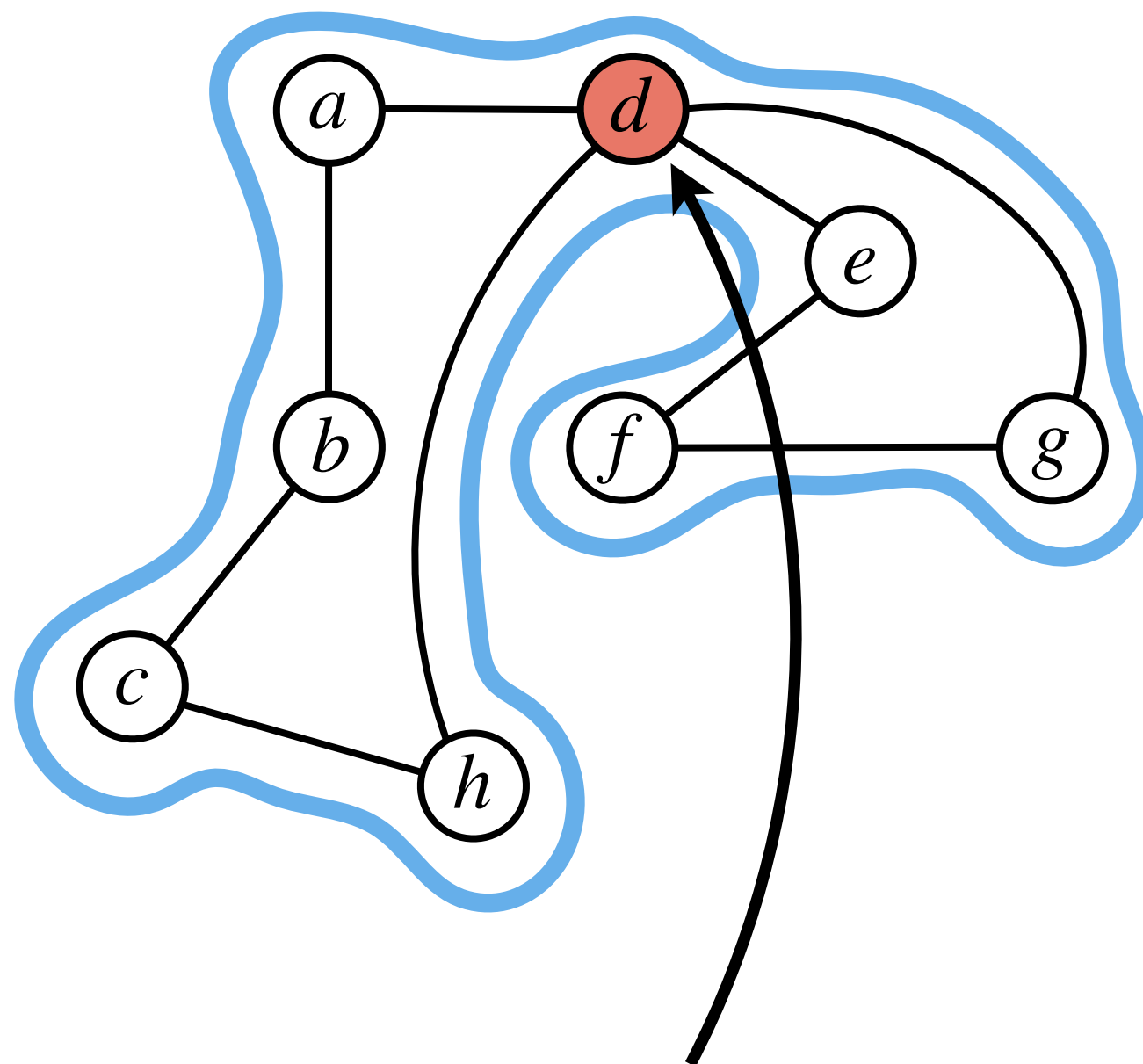
$$H = (a, b, c, \cancel{b}, h, \cancel{b}, \cancel{d}, d, e, f, \cancel{e}, g, \cancel{e}, d, a)$$

Skip e and go directly to d via edge $\{g, d\}$.

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



d is getting repeated.

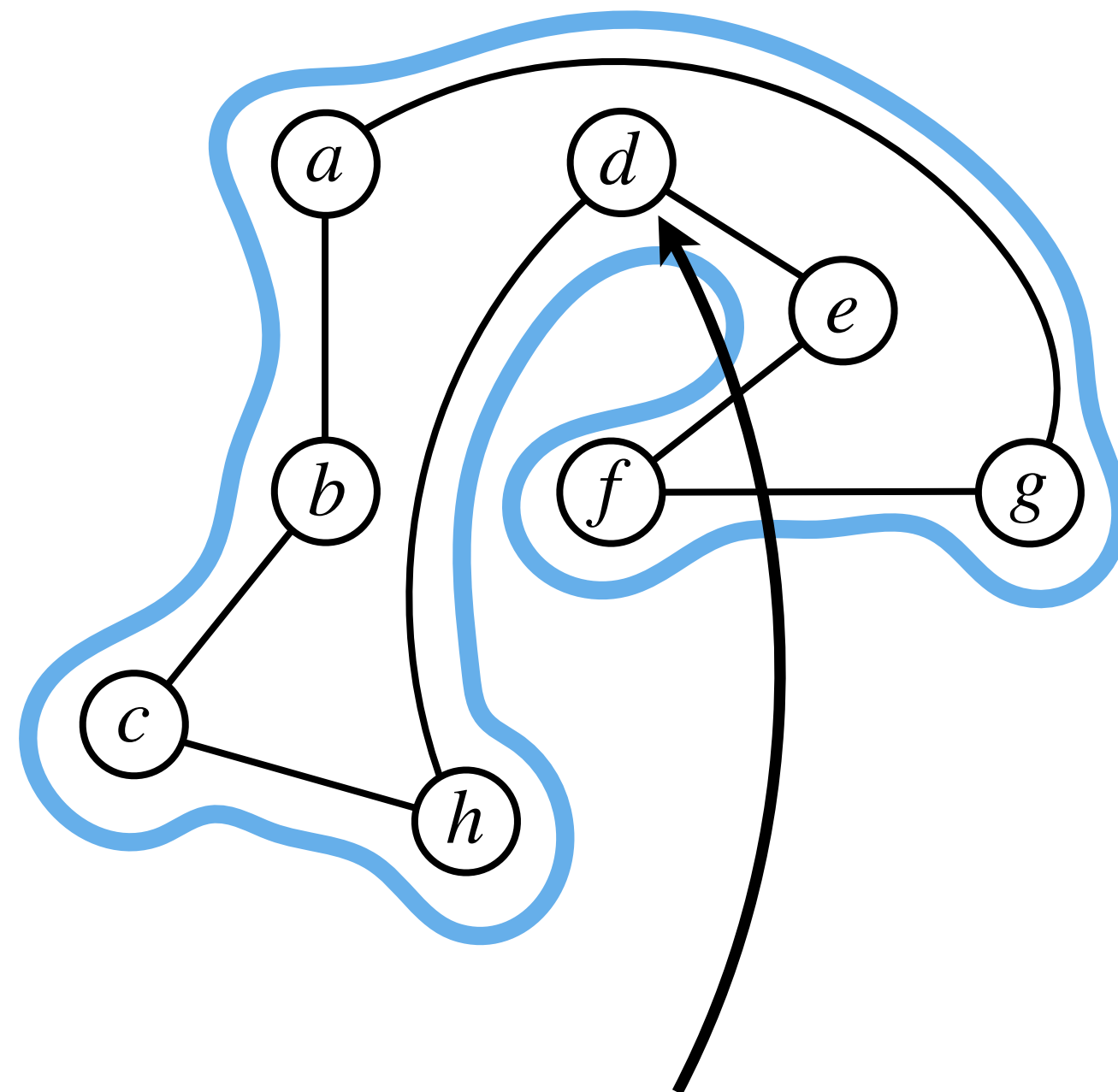
$$2 \cdot c(H^*) \geq c(H)$$

$$H = (a, b, c, \cancel{b}, h, \cancel{b}, \cancel{d}, d, e, f, \cancel{e}, g, \cancel{e}, d, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



Skip d and go directly to a via edge $\{g, a\}$.

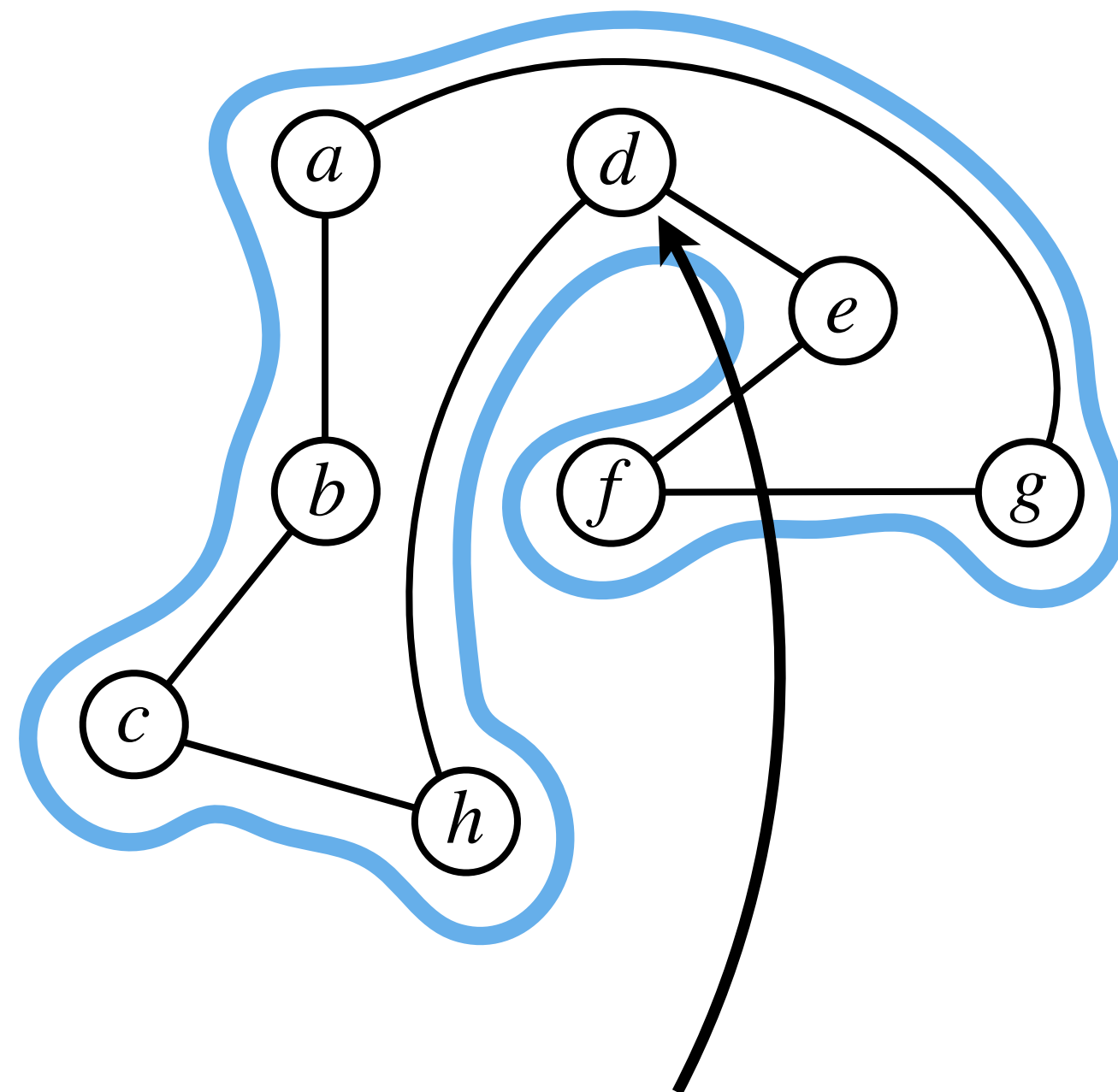
$$2 \cdot c(H^*) \geq c(H)$$

$$H = (a, b, c, \cancel{b}, h, \cancel{b}, \cancel{d}, d, e, f, \cancel{e}, g, \cancel{e}, d, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



$$2 \cdot c(H^*) \geq c(H)$$

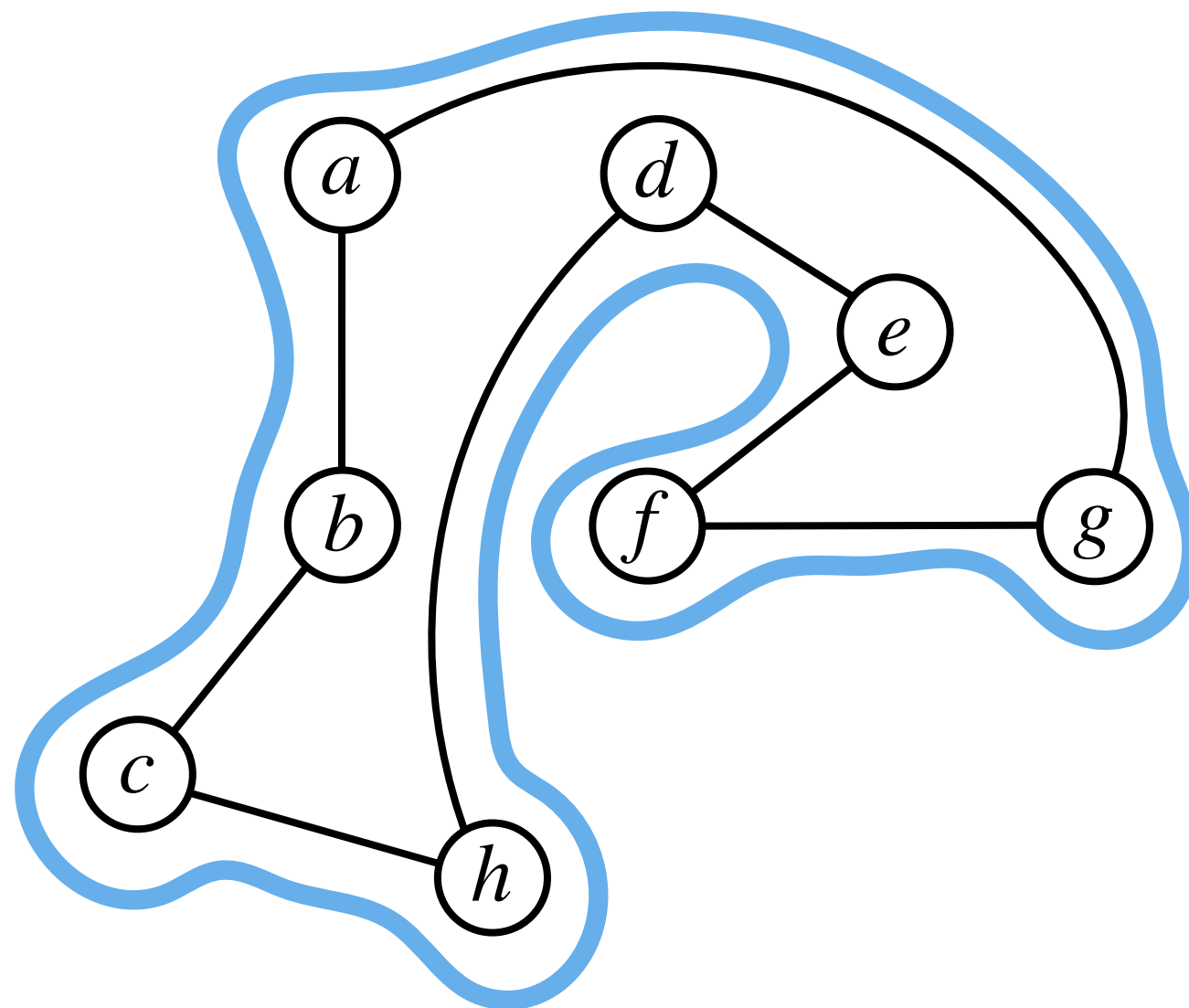
$$H = (a, b, c, \cancel{b}, h, \cancel{b}, \cancel{d}, d, e, f, \cancel{e}, g, \cancel{e}, \cancel{d}, a)$$

Skip d and go directly to a via edge $\{g, a\}$.

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



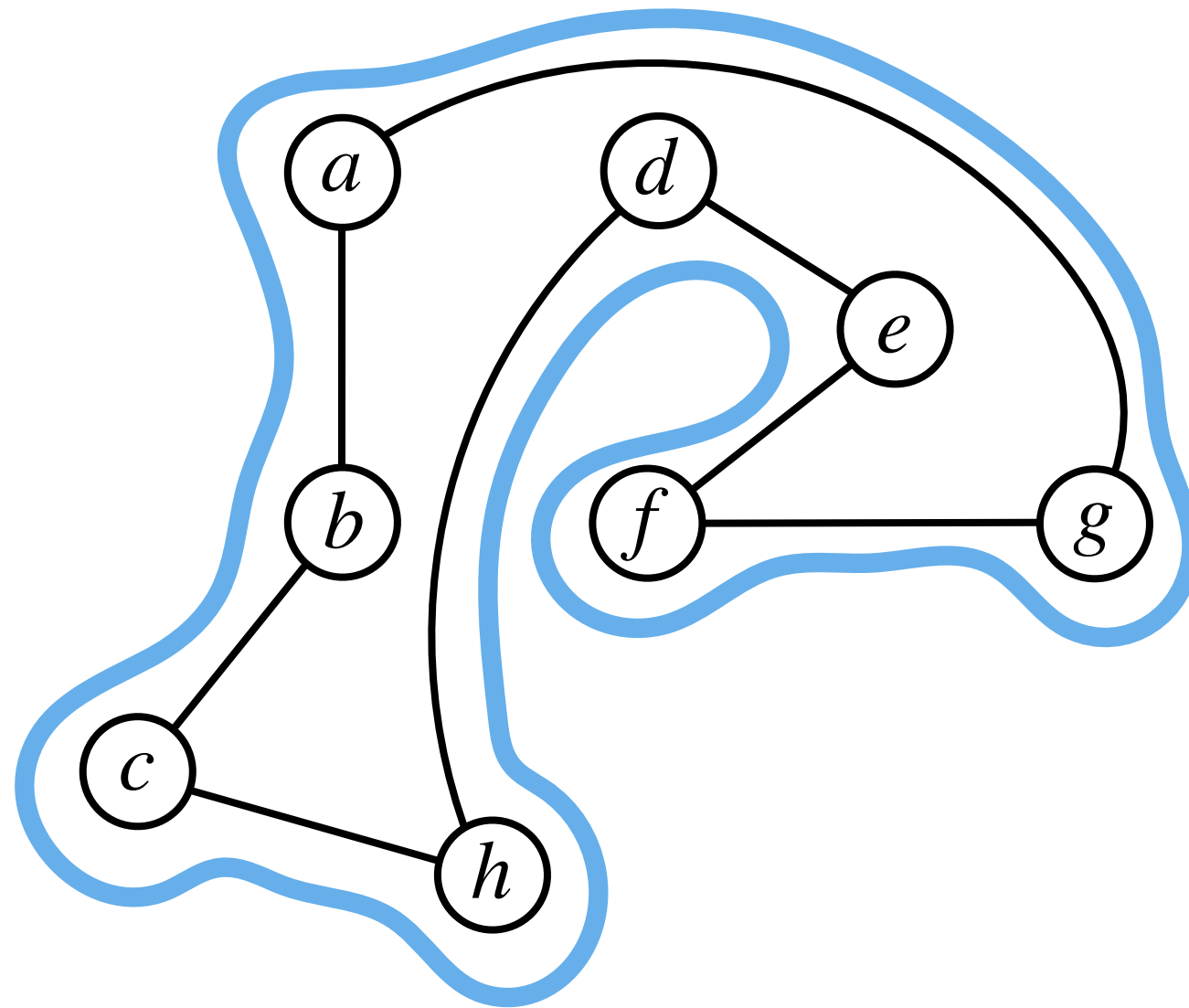
$$2 \cdot c(H^*) \geq c(H)$$

$$H = (a, b, c, \cancel{b}, h, \cancel{b}, \cancel{d}, d, e, f, \cancel{e}, g, \cancel{e}, \cancel{d}, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



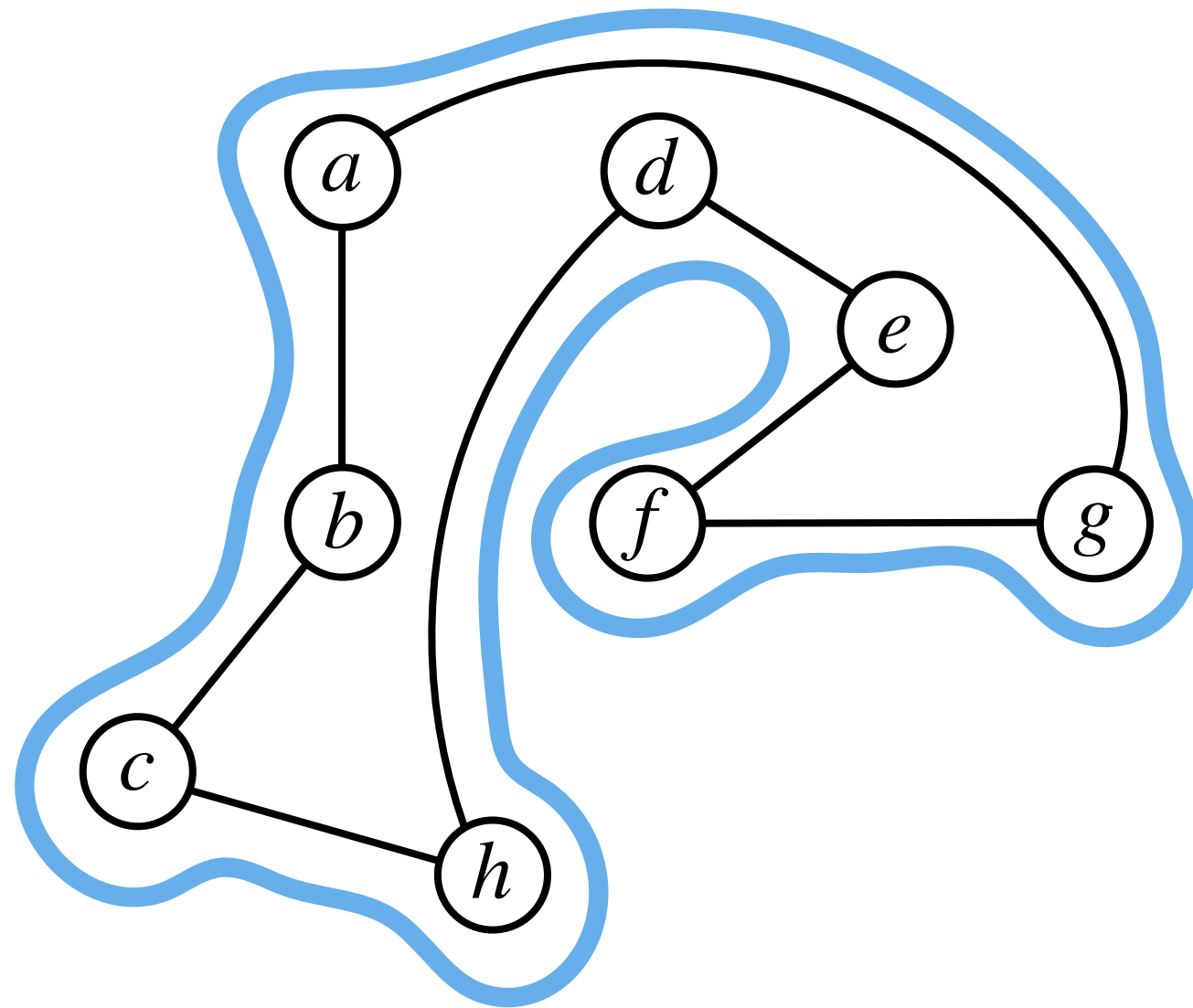
$$2 \cdot c(H^*) \geq c(H)$$

$$H = (a, b, c, h, d, e, f, g, a)$$

2-Approx. of $OptTSP$ Under Triangle Inequality

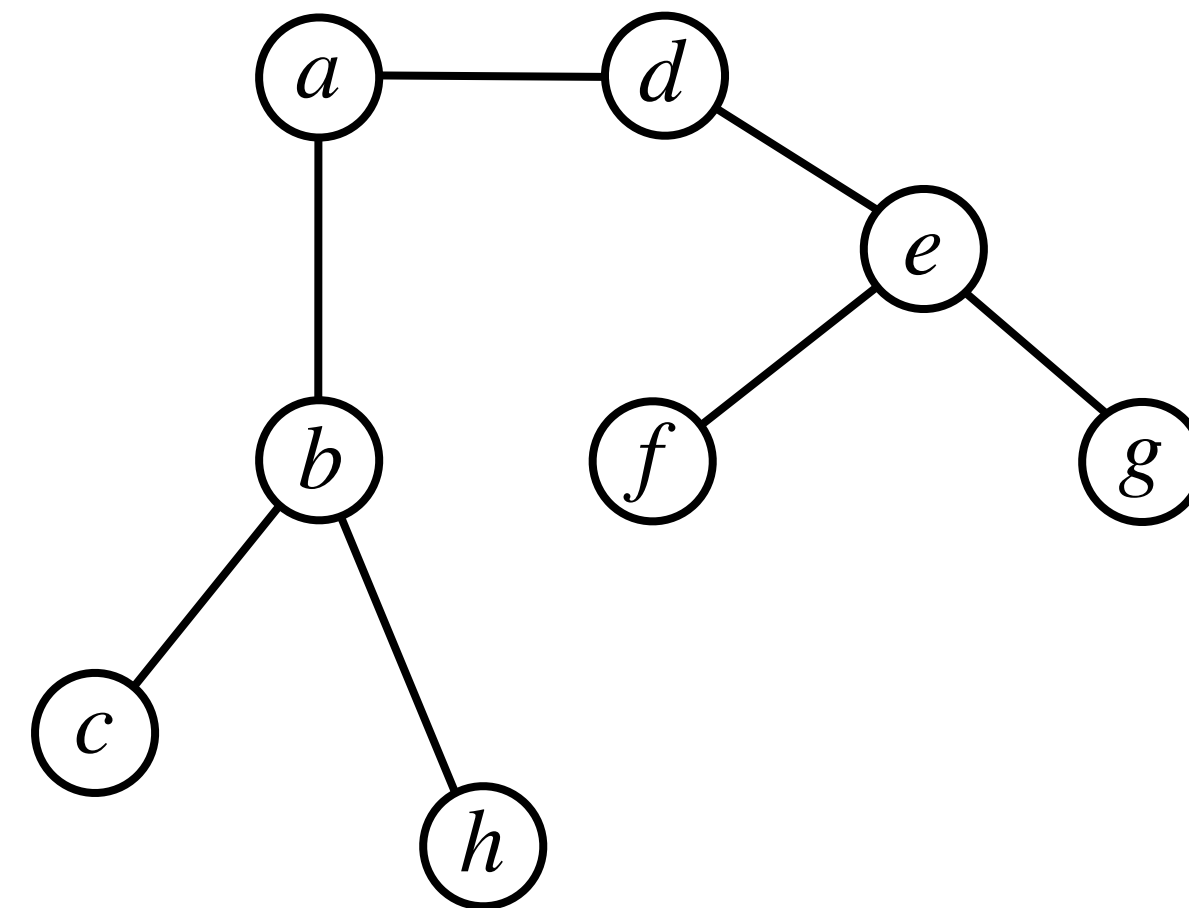
Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



$$H = (a, b, c, h, d, e, f, g, a)$$

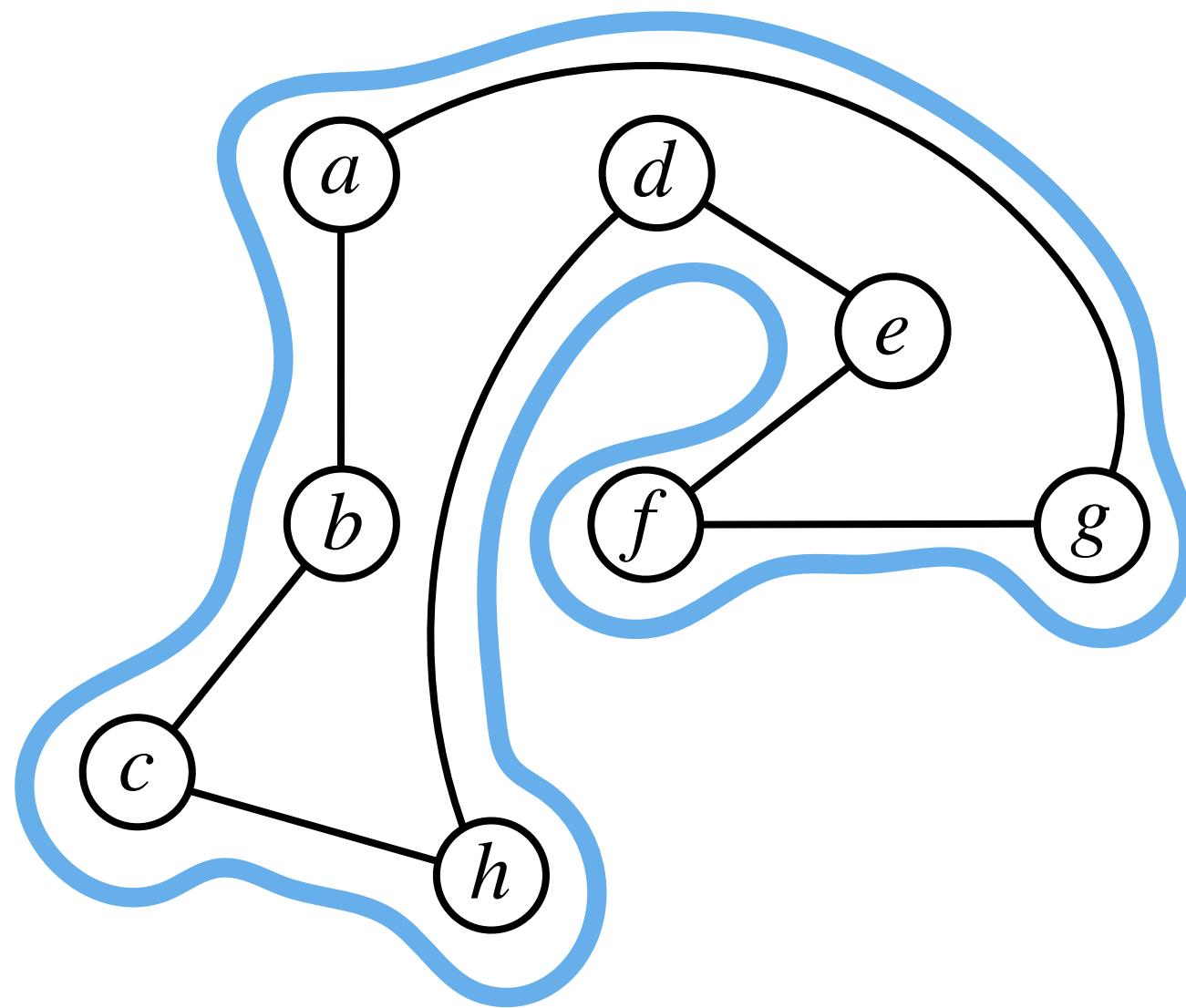
MST T of G



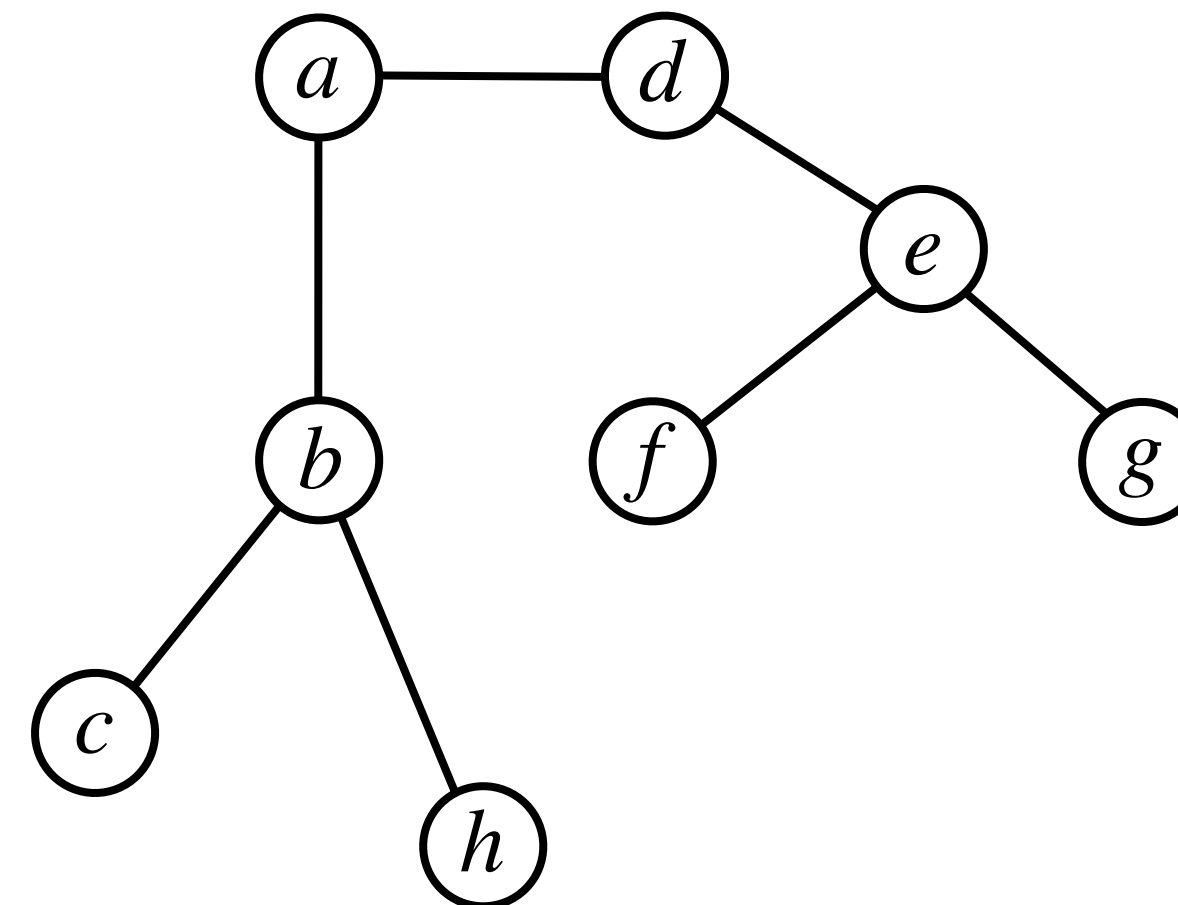
2-Approx. of $OptTSP$ Under Triangle Inequality

Plan: Gradually change H into a **TSP tour** so that every change makes the cost **smaller or same**.

Tour around MST T



MST T of G



$H = (a, b, c, h, d, e, f, g, a)$ is a preorder traversal of T .

2-Approx. of $OptTSP$ Under Triangle Inequality

2-Approx. of $OptTSP$ Under Triangle Inequality

Approx-TSP(G, c)

2-Approx. of $OptTSP$ Under Triangle Inequality

Approx-TSP(G, c)

1. Select a root vertex r of G

2-Approx. of $OptTSP$ Under Triangle Inequality

Approx-TSP(G, c)

1. Select a root vertex r of G
2. Compute a minimum spanning tree T for G from root r using Prim's algorithm

2-Approx. of $OptTSP$ Under Triangle Inequality

Approx-TSP(G, c)

1. Select a root vertex r of G
2. Compute a minimum spanning tree T for G from root r using Prim's algorithm
3. Let H be the tour obtained by visiting the vertices in the order given by the preorder walk of T

2-Approx. of $OptTSP$ Under Triangle Inequality

Approx-TSP(G, c)

1. Select a root vertex r of G
2. Compute a minimum spanning tree T for G from root r using Prim's algorithm
3. Let H be the tour obtained by visiting the vertices in the order given by the preorder walk of T
4. **return** H

2-Approx. of $OptTSP$ Under Triangle Inequality

Approx-TSP(G, c)

1. Select a root vertex r of G
2. Compute a minimum spanning tree T for G from root r using Prim's algorithm
3. Let H be the tour obtained by visiting the vertices in the order given by the preorder walk of T
4. return H

Time Complexity: Polynomial in the size of G .